Biostatistics

Descriptive Statistics & Distributions

- Data Types
    - Quantitative: data that can be measured with numbers (i.e., duration, speed)
        - Discrete: whole numbers that cannot be broken down (i.e., number of items)
        - Continuous: numbers that can be broken down (i.e., height, weight)
            - Interval: numbers with known differences between variables (i.e., time); scale in which values have order and equal intervals, without a true zero point
            - Ratio: numbers that have measurable intervals where differences can be determined (i.e., height, weight); scale in which values have order and equal intervals, with a true zero point
    - Qualitative: non-numerical data that is categorical (i.e., yes/no responses, eye color)
        - Nominal: data used for naming variables (i.e., eye color); categories with no numerical value
        - Ordinal: data used to describe the order of values (i.e., 1 = happy, 2 = neutral, 3 = unhappy); scale in which each value is rank-ordered in a way that is higher or lower than the previous value, but intervals are unknown or not equal
- Variables
    - Binary vs. non-binary
    - Independent vs. dependent
    - Predictor vs. outcome vs. confounder
    - Parametric vs. non-parametric
- Measures of Central Tendency
    - Mean: average of all numbers in a number set; influenced heavily by outliers
    - Median: the middle number when all numbers in a number set are arranged in order
    - Mode: the number that appears most often in a number set
- Measures of Dispersion
    - Range: the spread of the whole data set for any variable
    - Interquartile Range: the spread of the middle half of a data set
    - Standard Deviation: the average distance from the mean (the square root of the variance); smaller values → narrower distribution/bell curve graphs → more precise
        - 1 SD: 68% of the data
        - 2 SD: 95% of the data
        - 3 SD: 99.7% of the data
    - Variance: the average of squared distances from the mean
- Skewed Distributions
    - Skewed right → positively skewed → tail points towards the right side
    - Skewed left → negatively skewed → tail points towards the left side
- Data Visualization
    - Histograms: used for continuous & discrete data types
    - Box Plots: used for continuous & discrete data types
    - Scatter plots: used for continuous & discrete data types
    - Bar charts: used for categorical data
    - Frequency polygons: used for categorical data
    - Pie charts: used for categorical data

Box Plots

- Minimum: Q1 - 1.5xIQR
- First quartile (Q1/25th percentile): the middle number between the smallest number (not the minimum) and the median of the dataset
- Median (Q2/50th percentile): the middle value of the dataset
- Third quartile (Q3/75th percentile): the middle value between the median and the highest value (not the maximum) of the dataset
- Maximum: Q3 + 1.5xIQR
- Outliers: not contained within boxplots and are labeled outside of the minimum and maximum values

Distributions

- T-Distribution: a way of describing data that follows a bell curve when plotted on a graph, with the greatest number of observations close to the mean and fewer observations in the tails → used when the variance is unknown and there are less than 30 degrees of freedom
    - Estimated based on the degrees of freedom of a data set (total number of observations minus 1)

        - It is a more conservative form of the standard normal distribution and gives a lower probability to the center and a higher probability to the tails than the standard normal distribution
- Z-Distribution: standard normal distribution → used when there are more than 30 degrees of freedom
    - As the degrees of freedom increases, the T-distribution will get closer and closer to matching the standard normal distribution; above 30 degrees of freedom, the T-distribution will roughly match the Z-distribution
    - The Z-distribution can be used in place of the T-distribution with large sample sizes

Tests
- Chi-square test: compares two variables in a contingency table to see if they are related; it tests to see whether distributions of categorical variables differ from each other and is used to test if there is an association between two categorical variables (note: for small sample sizes, use the Fisher's exact test)
- T-test: two-sample T-tests are statistical tests used to compare the means of two groups; differences in the mean of a continuous variable between two groups → the dependent variable is continuous while the predictor is categorical
    - Paired T-tests: dependent; compares the averages/means and standard deviations of two related groups to determine if there is a significant difference between the two groups when the same group is tested twice (can be related by being the same group of people, same item, or being subjected to the same conditions); are considered more powerful than unpaired T-tests because using the same participants or item eliminates variation between the samples that could be caused by anything other than what is being tested
    - Unpaired T-tests: independent
- Analysis of Variance (ANOVA): test of hypothesis that is appropriate to compare a continuous variable in three or more independent comparison groups; the one-way ANOVA procedure is used to compare the means of the comparison groups; similar to a T-test, but is only used when there are more than 2 groups

Regression Analysis
- Regression analysis is a type of statistical evaluation that enables three things:
    - Description: relationships among the dependent variables and the independent variables can be statistically described by means of regression analysis
    - Estimation: the values of the dependent variables can be estimated from the observed values of the independent variables
    - Prognostication: risk factors that influence the outcome can be identified, and individual prognoses can be determined
- Monotonic relationship: correlation is a measure of a monotonic association between two variables
    - As the value of one variable increases, so does the value of the other variable
    - As the value of one variable increases, the value of the other variable decreases
- Linear correlation: the change in the magnitude of one variable is associated with a change in the magnitude of another variable, either in the same (positive correlation) or in the opposite (negative correlation) direction → special case of a monotonic relationship
    - Positive relationship: $b > 0$
    - Negative relationship: $b < 0$
- Covariance: the degree to which the change in one continuous variable is associated with a change in another continuous variable can mathematically be described in terms of the covariance of the variables; similar to variance, but whereas variance describes the variability of a single variable, covariance is a measure of how two variables vary together - however, covariance depends on the measurement scale of the variables, and its absolute magnitude cannot be easily interpreted or compared across studies (to facilitate interpretation, a Pearson correlation coefficient is commonly used from -1 to +1)
- Pearson's Correlation Coefficient (r): describes a linear relationship and provides information about the strength and direction of a relationship between two continuous variables; there is no distinction between the explaining variable and the variable to be explained
    - $r = +/- 1$ → perfect linear and monotonous relationship
    - $r = 0$ → no linear or monotonous relationship
    - Assumptions
        - The data are derived from a random or representative sample
        - Both variables are continuous, jointly normally distributed, random variables
- Coefficient of Determination (R^2): the proportion of variance in one variable that is accounted for by the other; this number is always positive so information on the direction of a relationship is lost

$$R^2 = (r\ x\ r);\ r = \sqrt{R^2}$$

- Simple Linear Regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Spearman's Rank Correlation (ρ): describes a monotone relationship and can be used for an analysis of the association between data that appears to be monotonic but nonlinear; calculated with the ranks of the values of each of the two variables instead of their actual values (ranking starts with the highest number); ranges from -1 to +1, has the same interpretation as r
    - Use of a Spearman coefficient is not restricted to only ordinal data; if continuous variables are ranked, then they can also use a Spearman's coefficient

Probabilities

- Rule #1: the probability of an event, which informs us of the likelihood of it occurring, can range anywhere from 0 (the event will never occur) to 1 (the event is certain to occur)
    - For any event A, $0 \leq P(A) \leq 1$
- Rule #2: the sum of the probabilities of all possible outcomes is 1
- Rule #3: the probability that an event does not occur is 1 minus the probability that is does occur
    - P(A) = 1 - P(not A)
- Rule #4: events that cannot occur at the same time are called disjoint or mutually exclusive
    - P(A or B) = P(A) + P(B)
- General Addition Rule: if A and B are not mutually exclusive events, then:
    - P(A or B) = P(A) + P(B) - P(A and B)
- Addition Rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
    - Mutually Exclusive Events: $P(A \text{ or } B) = P(A) + P(B)$
    - Non-mutually Exclusive Events: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- Multiplication Rule: a way to find the probability of two events happening at the same time
    - General rule: has to do with conditional probability
    - Specific rule: only valid if the two events, A and B, are independent (i.e., if one event does not change the probability of the other event)
    - Independent events: $P(A \text{ and } B) = P(A) \, x \, P(B)$

Power and Sample Size

- Statistical power: the probability of a hypothesis test of finding an effect if there is an effect to be found
    - The statistical power of a hypothesis test is the probability of detecting an effect if there is a true effect present to detect
    - Effect size: the quantified magnitude of a result present in the population. Effect size is calculated using a specific statistical measure, such as Pearson's correlation coefficient for the relationship between variables
    - Sample size: the number of observations in a sample
    - Significance: the significance level used in the statistical test - often set to 5% or 0.05
    - Statistical power: the probability of accepting the alternative hypothesis if it is true
        - Power = $1 - \beta$
- Power: the ability to correctly reject a null hypothesis that is indeed false
    - A power analysis can be used to estimate the minimum sample size required for an experiment, given a desired significance level, effect size, and statistical power

Central Limit Theorem

- The central limit theorem states that if you have a population with a mean (μ) and a standard deviation (σ) and take sufficiently large random samples from the population with replacement, then the distribution of the sample means will be approximately normally distributed
    - Holds true regardless of whether the source population is normal or skewed, provided that the sample size is sufficiently large ($n > 30$)
    - If the population is normal, then the theorem holds true even for samples with $n < 30$
    - Sampling with replacement: when a unit is selected at random from the population, it is returned to the population (replaced), and then a second element is selected at random
    - Assumptions
        - Samples must be selected randomly
        - Samples should be independent of each other (one sample should not influence the other samples)
        - Sample size should not be more than 10% of the population when sampling is done without replacement
        - Sample size should be sufficiently large, and larger when the population is not normal (skewed, asymmetric)
- Standard Error of the Mean (SEM)

- A measure of the precision of the study mean - the SEM is the SD of the means obtained in different hypothetical studies; indicates how different the population mean is likely to be from a sample mean and tells you how much the sample mean would vary if you were to repeat a study using new samples from within a single population
    - A large value indicates that the means in the hypothetical studies are widely scattered
    - A small value indicates that the means in the hypothetical studies are closely clustered

$$SE = \frac{\sigma}{\sqrt{n}}$$

- SEM vs. SD
    - The SEM is a measure of precision for an estimated population mean
    - It takes into account both the value of the SD and the sample size
    - SD is a measure of data variability around mean of a sample of population
    - The SD quantifies scatter - how much the values vary from one another

Confidence

- A way to describe probability (i.e., a confidence interval with a 95% confidence level → you are confident that 95 times out of 100 times, the estimate will fall between the upper and lower values specified by the confidence interval)
- Significance level (α): the maximum amount of error we allow ourselves to make (typically 0.05)
- Confidence level: $1 - \alpha$ (i.e., an α of 0.05 for statistical significance would have a confidence level of 0.95 → 95%)
- Confidence Interval (CI): the range of values that you expect your estimate to fall between a certain percentage of the time if you run your experiment again or re-sample the population in the same way (i.e., the percentage of times you expect to reproduce an estimate between the upper and lower bounds of the confidence interval, and is set by the α value)
    - An increase in sample size will decrease the length of the confidence interval without reducing the level of confidence because the standard deviation decreases as n increases

$$CI = \overline{x} \pm Z(\sigma E)$$
$$CI = \overline{x} \pm Z(\frac{\sigma}{\sqrt{n}})$$

- Margin of Error (ME): how close we can reasonably expect a sample result to fall relative to the true population value; expresses the amount of the random variation underlying a survey's results

$$ME = Z(\frac{S}{\sqrt{n}})$$

- Z-Score: describes the position of a raw score in terms of its distance (standard deviations) from the mean, when measured in standard deviation units; positive if the value lies above the mean, negative if it lies below the mean, 0 if it is on the mean

$$Z = \frac{x-\mu}{\sigma}$$

Hypothesis Testing

- Hypothesis: a supposition or proposed explanation made on the basis of limited evidence as a starting point for further investigation
    - Null hypothesis (H_0): the assumption of no difference between groups or no relationship between variables in the population → always states that the relationship between variables is equal/the same/not different
        - Rejected in favor of the alternative hypothesis if the P-value is less than α, the predetermined level of statistical significance (i.e., if the test statistic falls into the critical region)
    - Alternative hypothesis (H_1 or H_A): research prediction of an actual difference between groups or a true relationship between variables
- Significance level (α): the probability of rejecting the null hypothesis when it is true (the probability of making a Type I error) → the critical region on the normal distribution curve (right tail)
- P-value: the probability of obtaining an effect equal to, or more extreme than, the one observed - considering the null hypothesis is true (the probability of obtaining the study results by chance if the null hypothesis is true)
    - P-value < α: reject null hypothesis; conclude that there is a difference
    - P-value $\geq \alpha$: do not reject null hypothesis; conclude that there is no difference
- Type I error (α): false positive; rejecting the null hypothesis when it is true (when there is no difference)

- Type II error (β): false negative; not rejecting the null hypothesis when it is false (when there is a difference); inversely related to the statistical power of a study

	Reality: H_0 is true	**Reality: H_0 is false**
Decision: Reject H_0	**Type I error = False positive** Probability = α	**Correct decision** Probability = $(1 - \beta)$ = Power
Decision: Accept H_0	**Correct decision** Probability = $(1 - \alpha)$ = Confidence	**Type II error = False negative** Probability = β

- Trade-off between Type I and Type II errors
    - Setting a lower significance level decreases a Type I error risk, but increases a Type II error risk
    - Increasing the power of a test decreases a Type II error risk, but increases a Type I error risk
- Statistics vs. Clinical Significance

		Clinically Significant	
		Yes	No
Statistically Significant	Yes	Typically assumes the groups, outcomes, or treatments are different	Sample size may be too large
	No	Sample size may be too small	Typically assumes the groups, outcomes, or treatments are not different

Microbiology

Clinical Microbiology
- Decontamination: physical, chemical, and mechanical methods to destroy or reduce undesirable microbes
- Disinfection: the destruction or removal of vegetative pathogens, but not bacterial endospores; used on inanimate objects
- Sterilization: the complete removal or destruction of all viable microorganisms; used on inanimate objects
- Antiseptic: chemicals applied to body surfaces to destroy or inhibit vegetative pathogens; safe on tissues
- Decontaminate: sterilize equipment, disinfect abiotic surfaces, sanitize body surfaces with antiseptics
- Screening tests: highly sensitive
- Diagnostic/confirmatory tests: highly specific
- Precision: values are very close to one another without necessarily being accurate
- Accuracy: values are very close to the expected/correct value without necessary being precise

Specimen Collection
- Burn or wound specimen: swab specimens are frequently contaminated with normal flora; wound specimen should be an aspirate (pus collected by needle/syringe); do not use swabs if looking for anaerobes
- Oral-pharyngeal and throat swab specimen: high number of normal flora; used to determine if a patient has Strep throat, for example
- Sputum specimen: normal flora are low in numbers or absent; high number of squamous cells indicate saliva contamination; presence of neutrophils indicates potential infection
- Fecal specimen: high number of normal flora; MacConkey agar is used for planting - look for transparent colonies; good for 2 hours
- Urine specimen: 2 or less pathogens → infectious agent; 2+ pathogens → contamination; need clean-catch, midstream collection, good for 2 hours
    - Dipstick: screening test
    - Culture: diagnostic test
- Blood specimen: positive culture is indicative of bacteremia; 2-5 samples collected at 3 different time points and in different locations
- Cerebrospinal fluid (CSF) specimen: obtained by lumbar puncture
- Genital tract sampling
    - Females: remove mucus plug, insert cervical brush deep into cervical canal, transport within 24 hours at room temperature

- Males: metal shaft swab should be inserted approximately 2cm into the urethra and rotated 10x for a total time of more than 20 seconds

Basic Bacteriology

- Prokaryotes: includes bacteria (average of 0.2-2.0 micrometers)
    - Nucleus without a membrane
    - No specialized or membrane-bound organelles (ribosomes are 70S)
    - Partial ID done with morphology and Gram stain (crystal violet/purple), iodine mordant, decolorization (ethanol), contra-stain with safranin (pink)
        - Gram positive: thick peptidoglycan membrane (PURPLE), teichoic and lipoteichoic acid
        - Gram negative: thin peptidoglycan layer sandwiched between two membranes (PINK), porins, lipopolysaccharide (LPS) → lipid A is endotoxin, periplasm is very active
    - Peptidoglycan: unique to bacteria
        - Monomers NAM, NAG interconnected with a peptide chain
        - Monomers link with a transpeptidase or Penicillin-binding protein (PBP)
            - PBP is target for beta-lactam drugs (i.e., Penicillin) → cell wall targeting
- Metabolism
    - Medically important bacteria: chemoheterotrophs (carbon source → organic; energy source → chemical)
    - Oxygen use: anaerobic respiration, fermentation, aerobics
- Bacterial replication: binary fission
    - *E. coli*: 20 minutes
    - *S. aureus*: 28 minutes
    - *M. tuberculosis*: 360 minutes
    - *T. pallidum*: 1,980 minutes
- Growth media
    - Selective (contains components that select for the growth of specific organisms and limits the growth of others) vs. Differential (allows differentiation of growing bacteria based on different characteristics and assists with classification)
    - Complex (contains growth nutrients from a complex biological source) vs. Defined (contains very specific defined ingredients that are tailored for an organism's growth requirements)
    - Common media (routine use) vs. Specialized media (specific organisms)
        - Nutrient agar: common; rich media for non-fastidious bacteria
        - MacConkey agar: common; differential and selective for enterobacteriaceae → uses presence of bile salts to restrict growth of Gram-positive bacteria
        - Chocolate agar: common; heated blood agar releases NAD or V Factor for utilization by fastidious organisms
        - Blood agar: common; enriched differential media with TSA plus 5% sheep's blood
            - Beta hemolysis: complete lysis of red blood cells (clear zone) around each colony
            - Alpha hemolysis: partial lysis of the red blood cells (green zone) around each colony
            - Gamma hemolysis: no lysis
        - Lowenstein-Jensen agar: specialized; lipid rich to enhance slow growth, unique for *Mycobacterium tuberculosis*
        - Thayer-Martin agar: specialized; chocolate agar with antibiotics added (selective) to restrict growth of usual flora
    - Bacteria counting: loop or dilution method
        - Colony Forming Units (CFUs) / Amount Plated (mL) x Dilution Factor
- Growth Curve
    - Lag phase, exponential phase, stationary phase, and death
- Growth Conditions
    - Temperature: psychrophiles (<25 C), mesophiles (most pathogens), thermophiles (>65 C)
    - pH: acidophiles (pH 4), neutrophiles (pH 6-8 → most pathogens), alkaliphiles (pH 9-11)
    - Water activity: hypotonic, isotonic (most pathogens), hypertonic
        - Aw: index amount of water free to react, 1.0 is distilled water; absorption and solution factors reduce availability; Gram + bacteria can survive in low moisture environments longer due to their thick peptidoglycan wall
        - Salt: halophile, non-halophile (most pathogens)
    - Oxygen: obligate aerobes (SOD, catalase), facultative anaerobe (SOD, catalase), aerotolerant anaerobe (SOD), obligate anaerobe (none), microaerophiles (SOD, +/- catalase)
        - SOD = superoxide dismutase

- Disease-producing doses
  - *Salmonella typhi* → 100,000
  - *Shigella* → 10-1,000
  - *Vibrio cholerae* → 100,000,000
  - *Vibrio cholerae* + HCO_3^- → 10,000
  - *Mycobacterium tuberculosis* → 1-10
- General Terminology
  - Normal Flora: endogenous microorganisms that live on another living organism without causing disease
  - Pathogen: an organism that can cause disease
  - Opportunistic pathogen: non-pathogenic organisms that cause serious disease in immunocompromised individuals or when in other locations
  - Virulence Factors: bacterial features that contribute to infection; enhances the ability of a microorganism to cause disease beyond that intrinsic to the species (i.e., bacterial adhesins)
  - Pathogenesis: mechanism by which infections lead to symptoms
  - Dysbiosis: loss of microbiome diversity leads to altered microbiota and is associated with many health conditions
- Types of Infection
  - Primary Infection: initial infection with an organism in the host
  - Reinfection: subsequent infection by the same organism in the host after recovery
  - Secondary Infection: an infection that follows on from, or is the indirect result of, another infection
  - Superinfection: secondary infection that occurs during treatment of a primary infection
  - Opportunistic Infection: caused by normally harmless bacteria that becomes pathogenic upon favorable changes in the environment
- Bacterial adhesins (FimH adhesin): virulence factor; attached to the tip of thin pili or fimbriae; they are of medical importance because they mediate the attachment of bacteria to the mucosal surface or other surfaces
- Bacterial capsule: well-organized, gelatinous sugar material (including proteoglycans, glycoproteins, and glycolipids) that functions to help the organism resist environmental insults
  - Clinical correlate: asplenic individuals are vulnerable to sepsis because the majority of the antibodies against the capsule are produced in the spleen
- Bacterial endospores: formed during harsh environmental growth conditions and are highly resistant to desiccation, chemicals, hot/cold temperatures due to spore coat of dipicolinic acid; germinate into vegetative bacterial cells on contact with moisture and suitable temperature and pH
  - Two Gram-positive genera form spores: *Bacillus* and *Clostridium*
- Type 3 Secretion System (T3SS): acts as a needle to inject effector proteins into the host and modifies host cellular characteristics
- Exotoxins: toxins produced by Gram-positive and Gram-negative bacteria that are secreted into the environment; lysogenic conversion → spread of virulence factors
  - AB Toxins (Type III): A "active" subunit and B "binding" subunit; A-subunit dissociates from B-subunit in host cell and targets cellular function, B-subunit binds to specific host cell receptor and leads to internalization
  - Membrane-disrupting Toxins (Type II): cytotoxin secreted by bacteria from local infection; toxin acts by forming pores in host cell membrane and causes localized inflammation and damage
  - Superantigens (Type I): non-specific activation of T-cells leads to massive cytokine release, causing toxic shock
  - Enterotoxins: found in the intestinal mucosa
  - Neurotoxins: occur in nerve tissues where they cause neurological damages
  - Cytotoxins: are normally found in the general tissues of the body where they cause a series of local and systemic damages
- Endotoxins: microbial toxins which are produced only on cell lysis and produce their effect in a host by activating the complement system by the alternative pathway; they are pyrogenic in action and can cause bacterial sepsis (shock) and intravascular coagulation
  - Lipopolysaccharide (LPS) components of the outer membrane of Gram-negative bacteria (Lipid A)

<u>Antibiotics</u>

- Mechanism of Action
  - Inhibitors of Cell Wall Synthesis
    - Beta Lactams: penicillins, cephalosporins, carbapenems, monobactams
      - All contain beta lactam ring that inhibits penicillin-binding protein (PBP) for cell wall synthesis
    - Vancomycin

- Bacitracin
- Cell Membrane: polymixins
- Inhibitors of Nucleic Acid Synthesis
    - Folate Synthesis: sulfonamides, trimethoprim
    - DNA Gyrase: quinolones
    - RNA Polymerase: rifampin
- Inhibitors of Protein Synthesis
    - 30S Subunit: tetracyclines, aminoglycosides
    - 50S Subunit: macrolides, clindamycin, linezolid, chloramphenicol, streptogramins

- Mode of Action
    - Bacteriostatic: chloramphenicol, clindamycin, macrolides, sulfonamides, tetracyclines, trimethoprim
    - Bactericidal: penicillins, cephalosporins, polymixins, rifampin, vancomycin, aminoglycosides (at high doses), bacitracin
- Antibiotic Resistance
    - Cross-Resistance: resistance mechanism includes resistance to different classes of antibiotics (i.e., *Pseudomonas* efflux pump)
    - Intrinsic Resistance: all isolates are naturally resistant (i.e., *Mycoplasma*)
    - Acquired Resistance: only some isolates have become resistant (i.e., MRSA)
    - Plasmid-Mediated Resistance: antibiotic resistance genes (Abr) are highly mobile via conjugation
    - Phage-Mediated Resistance: (Abr) are highly mobile via transduction
    - Chromosomal Resistance: highly stable and can undergo spontaneous mutation; incorporation of donor DNA via transformation, incorporation of donor prophage via transduction, incorporation of plasmid via transposition
    - Sensitivity Testing: Broth Dilution Method
        - Minimum inhibitory concentration (MIC): 1st tube without visible growth
        - Minimum bactericidal concentration (MBC): 1st tube without colony-forming bacteria on agar plate
    - Horizontal Gene Transfer: virulence factors and some antimicrobial resistance genes encoded on mobile genetic elements spread through horizontal gene transfer, and can convert harmless bacteria into dangerous pathogens
        - Conjugation: the most important means of spreading antibiotic resistance and virulence factors among bacteria using sex pili
        - Transduction: bacteriophage can harbor exotoxin genes within its genome and can turn avirulent bacteria into toxin-producing virulent bacteria → lysogenic cycle
        - Transformation
            - DNA fragments: extracellular DNA fragments get uptaken by bacterial cells and are integrated into the bacterial chromosome by homologous recombination
            - DNA plasmid: extracellular DNA plasmid gets uptaken by bacterial cells and reaches a stable form
- Acid-Fast Mycolic Cell Wall (*Nocardia* and *Mycobacterium* species)
    - Mycolic acids form a thick lipid waxy layer that limits nutrient uptake and results in a slow growth rate
    - Thick lipid cell wall: interferes with normal Gram staining methods, protects the bacteria from reactive oxygen species, confers resistance to beta-lactams
    - Acid-Fast Stain = Ziehl-Neelsen Stain
- Atypical Bacteria
    - Typical features: cannot be differentiated by the Gram stain, infections are often chronic and do not respond to first-line beta-lactam antibiotics, are difficult to culture and require special media or tissue culture
    - Actinomycetes: bacteria that form long, branching filaments
        - *Actinomyces israelii*: Gram positive, anaerobic, commensal oral & GI flora, opportunistic pathogen → abscesses in anoxic tissue (periodontal disease leading to "lumpy jaw", foul-smelling draining sulfur granules, can form slow granulomatous abscesses in any tissues)
        - *Nocardia asteroides*: partially acid-fast, aerobic, transmission via inhalation & skin cuts, opportunistic pathogen (more severe in immunocompromised patients)
    - *Mycobacterium tuberculosis*: most cases are latent and non-infectious, transmission through air droplets from an infected individual; pathogen is phagocytized by alveolar macrophages but they are unable to kill and digest the bacterium due to its unique thick lipid cell wall → require cell-mediated immunity to be controlled (CD4+, Th1); granulomas are formed over weeks to months

- *Mycobacterium leprae*: obligate intracellular bacteria, replicates in macrophages and nerve cells, transmission is thought to involve coughing and sneezing (prolonged, close contact with someone with untreated leprosy over many months is needed to catch the disease)
    - Lepromatous Leprosy: more severe
    - Tuberculoid Leprosy: less severe
- *Mycobacterium marinum*: found in bodies of fresh or saltwater, skin infections of humans are relatively uncommon but are usually acquired from contact with contents of aquariums or fish and occur following skin exposure to the bacteria through a small cut or skin scrape; first signs of infection include a reddish/tan skin with a granuloma
- *Mycoplasmataceae* Family: pleomorphic bacteria that lack a cell wall → does not Gram stain well due to the lack of a cell wall and are intrinsically resistant to antibiotics that target cell walls
    - *Mycoplasma pneumoniae*: cause of atypical/walking pneumonia and other respiratory disorders; associated with a chronic dry persistent cough
    - *Mycoplasma genitalium*: believed to be involved in pelvic inflammatory disease
    - *Ureaplasma parvum & urealyticum*: can be passed through sexual contact though it is not considered a classic STD because of its low degree of pathogenicity; causes bacterial vaginosis, pregnancy complications, urethritis
- *Chlamydia* Family: obligate intracellular, lacks peptidoglycan (pleomorphic, resistant to beta-lactams), depends on host cell for ATP and NAD; complex life cycle with elementary body (infectious form) and reticulate body (replicative form)
    - *Chlamydia trachomatis*: most common bacterial STD in the US; infectious disease that affects the eyes, causes cervicitis in women & urethritis and proctitis in both men and women, although women are typically asymptomatic (untreated infection can lead to transmission to neonate, pelvic inflammatory disease, ectopic pregnancy, and sterility); can see intracytoplasmic inclusions (vacuoles) showing elementary bodies on pap smear
    - *Chlamydia pneumoniae*: respiratory tract infections including atypical pneumonia
    - *Chlamydia psittaci*: zoonotic respiratory disease from birds that can cause atypical pneumonia
- *Rickettsia* Family: obligate intracellular, pleomorphic bacteria; transmitted by arthropod bite
    - *Rickettsia rickettsii*: tick bite, onset 7-14 days → Rocky Mountain Spotted Fever (high fever, lymphadenopathy, endovasculitis → petechial rash including palms of hands and soles of feet, encephalitic signs, and gangrene of skin/tissues)
    - *Rickettsia prowazekii*: lice
    - *Rickettsia typhi*: flea
- *Spirochete* bacteria: unique morphology with endoflagella that require special microscopy (darkfield or IF)
    - *Treponema pallidum*: causative agent of Syphilis; transmitted by sexual contact or transplacental transmission
    - *Borrelia burgdorferi*: causative agent of Lyme Disease; transmitted by tick bite

Immunology

Cells and Organs of the Immune System

- Terminology
    - Immunity: a function of which an individual recognizes and excludes antigenic foreign substances; it is normally beneficial, but sometimes, it is injurious
    - Pathogen: something that causes disease
    - Antigen: any foreign substance that binds specifically to an antibody or T-cell receptor
        - Antigen factors in immunogenicity → important to select targets for vaccines
            - Foreignness: degree of foreignness relative to phylogenetic distance
            - Molecular size: generally > 100,000 Da
            - Chemical composition: greater heterogeneity = greater immunogenicity; proteins > carbohydrates > lipids; D-amino acids cannot be processed
            - Adjuvant: increases persistence of antigen and innate response (i.e., BSA in mice, alum in humans)
        - Host factors in immunogenicity
            - Genotype of host: MHC gene products, B-cell receptors, TCR
            - Dosage and administration: low doses may induce tolerance or fail to stimulate enough lymphocytes, boosters increase clonal selection
                - Route: slow release is important for immune response (subcutaneous > intraperitoneal > intravenous > intragastrico); oral route induces local mucosal immunity but not systemic immunity

- Epitope: portion of the antigen that is recognized by an antibody or T-cell receptor
- Immunogen: a substance capable of eliciting an immune response (all immunogens are antigens, but not all antigens are immunogens - i.e., haptens)
- Hapten: a small molecule that, when combined with a larger carrier such as a protein, can elicit the production of antibodies that bind specifically to it
    - Hapten molecules → no immune response
    - Carrier molecule → no immune response
    - Hapten-carrier conjugate → immune response

- Innate Immunity
    - Molecular and cellular mechanisms are performed and are fully active → immediate reaction
    - Nonspecific: distinguishes between self and pathogens but not specialized to distinguish small differences in the foreign particles
    - Components: barriers that protect the host (skin, mucous membrane, acidity of stomach, lysozymes in fluids), phagocytic cells, antimicrobial peptides (interferons, complement), temperature
- Adaptive Immunity
    - Develops in response to an infection; highly specific → adapts to recognize, eliminate, and remember the pathogen; takes several days to become fully functional
    - Components: cell-mediated immunity, antibody-mediated (humoral) immunity
    - Attributes: antigenic specificity (antibodies can distinguish between 2 proteins that differ in only 1 amino acid), diversity, immunologic memory, self/non-self recognition
- The Bloodstream
    - Hematopoietic stem cells (HSC): origin of all blood cells (lymphoid and myeloid elements), may self-renew
    - CFU-GEMM or CFU-L: can self-renew or commit to progenitor cell
    - Progenitor: committed stem cells; limited self-renewal
        - Myeloid progenitors: granulocyte, macrophage, monocyte (6%), erythroid, megakaryocyte, eosinophil (3%), basophil (1%), neutrophil (60%)
        - Lymphoid progenitors: lymphocytes (30%), natural killer (NK) cells
    - Homeostasis
        - Erythrocytes: 120-day lifespan → phagocytized by macrophages in spleen
        - Leukocytes: 1 day - 30 years lifespan
        - Apoptosis: programmed cell death
- The Mononuclear Phagocyte or Reticuloendothelial System
    - Network of connective tissue fibers and cells surrounding organs; inhabited by phagocytic cells (mononuclear phagocyte system) that are ready to attack and ingest microbes that passed the first line of defense → tissue sentinel cells
- The Lymphatic System
    - Provides an auxiliary route for return of extracellular fluid to the circulatory system (acts as a drain-off system for the inflammatory response); renders surveillance, recognition, and protection against foreign material
    - Primary lymphoid tissues and organs
        - Bone marrow (B-cells) and thymus gland (T-cells) → development and maturation of lymphocytes
            - Thymus: T-cell development and maturation
                - Surrounded by capsule and divided into lobules (outer lobule is the cortex, inner lobule is the medulla); contains network of epithelial cells, dendritic cells, and macrophages
                    - Cortex: immature thymocytes
                    - Medulla: immunocompetent T-cells
                - Induces death of T-cells that cannot recognize self-MHC molecules and those that interact with MHC molecules too strongly; function decreases with age from puberty
    - Secondary lymphoid tissues and organs
        - Spleen, adenoids, tonsils, appendix, lymph nodes, Peyer's patches, mucosa-associated lymphoid tissue (MALT) → mature lymphocytes meet pathogens
            - Lymph nodes
                - Cortex: B-cells mainly; primary follicle (unactivated lymphoid follicle), secondary follicle (activated) → germinal centers = production of plasma and memory cells (area of activated B-cells that will proliferate, go through somatic hypermutation, and class switch)

- Paracortex: T-cells, dendritic cells (APCs); activation of T-cells, interaction of T-cells and B-cells
- Medulla: plasma cells secreting antibodies and macrophages

- Spleen: filters blood, activates lymphocytes from blood-borne pathogens (important in systemic infections); surrounded by a capsule and compartments are separated by a marginal zone (antigens meet APCs, phagocytosis of antigens by innate immune system, APCs meet lymphocytes to initiate the immune response)
    - Red pulp: remove old or defective erythrocytes and platelets
    - White pulp: the periarteriolar lymphoid sheath (PALS) with T-cells, follicles with B-lymphocytes
- Mucosa-associated lymphoid tissue (MALT): organized areas along digestive, respiratory, and urogenital tracts; pathogens are directly transferred across mucosa by microfold cells (M-cells)
    - Bronchial-associated lymphoid tissue (BALT) → tonsils, adenoids
    - Gut-associated lymphoid tissue (GALT) → appendix, Peyer's patches
    - Skin-associated lymphoid tissue (SALT)

<u>Innate Immune System</u>

- Myeloid
    - Polymorphonuclear: white blood cells characterized by a lobed nucleus
        - Neutrophils (60%)
            - Multi-lobed nucleus, granulocyte; 1st to arrive at the site of inflammation (high levels is 1st indication of infection)
            - Destroys organisms via phagocytosis, neutrophil extracellular trap (NET) , release of hydrolytic enzymes
            - Short-lived (10 hours in blood, 1-3 days in tissue); come out of blood vessels ready to kill, would cause too much damage to tissues if long-lived → emphysema-elastase
        - Eosinophils (3%)
            - Bilobed nucleus, granulocyte; located mainly in tissue
            - Phagocytizing: specific granules and azurophilic granules
                - Inside specific granules: basic proteins, cationic protein, neurotoxin
                - Outside specific granules: histaminase, peroxidase, cathepsin
            - Functions: plays a role in the elimination of parasitic helminths and allergic reactions, phagocytizes antigen-antibody complexes; activated by IL5
        - Basophils (1%)
            - Bilobed nucleus, granulocyte; survives long periods of time (~2 years)
            - Granules: basophilic → heparin, histamine, chemotactic factors, peroxidase
            - Functions: precursors of mast cells, mediate inflammatory responses, plays a role in allergic reactions
        - Mast cells
            - Similar to basophils, granulocyte; plays important role in the development of allergies
            - Activated via two different pathways: innate TLRs & antibody-dependent (IgE)
    - Mononuclear: One-lobed nucleus, agranulocyte (characterized by the absence of granules in their cytoplasm); includes lymphocytes and monocytes/macrophages
        - Lymphocytes (30%)
            - Has three populations: innate lymphoid cells (natural killer cells, ILC's), B-cells, and T-cells
        - Monocytes/Macrophages (6%)
            - Monocytes circulate in blood and then migrate into tissue and differentiate into specific macrophages in different parts of the body
            - Functions of macrophages: surveys tissue compartments and discovers microbes, particulate matter, and dead or injured cells, to ingest and eliminate these materials, to extract immunogenic information (antigen presenting cell - APC)
- Myeloid/lymphoid
    - Dendritic cells (DC's): have long membranous extensions (stellar); CD14+
        - Groups: lymphoid or plasmacytoid-derived DC, monocyte-derived or myeloid DC, langerhans DC, follicular DC (involved with B-cell maturation)

        - Functions: surveillance, initiation of inflammatory response, stimulation of adaptive immune system (antigen processing and presentation; takes a snapshot of what is happening in the tissues and carries this image to the lymph nodes)
- Lymphoid: innate lymphoid cells (ILC's)
    - Killer cells: natural killer cells (NKC's)
        - Innate immune response to intracellular organisms (viruses) and tumor cells
        - Large, mononuclear, granular
        - Surface markers: CD16 and CD56
    - Helper cells: ILC1, ILC2, ILC3
        - Contribute to immunity via secretion of cytokines; derived from common lymphoid progenitors (CLP's)
        - Abundant in intestinal, pulmonary, and oropharyngeal mucosal tissue
- First line of defense: sebaceous secretions, lysozyme in tears, lactic acid in sweat, skin's acidic pH, hydrochloric acid, digestive juices, semen, vagina
- Second line of defense: epithelial layer (mechanical barrier), sentinel cells (dendritic cells, mast cells, tissue resident macrophages), recruited phagocytes (monocytes/macrophages, neutrophils), specialized lymphocytes (innate lymphoid cells → cytokine producers)
    - Plasma proteins: complement, ficolins (lectins), collectins (mannose-binding lectin), pentraxins (C-reactive protein, serum amyloid proteins)
    - Cytokines: inflammatory (IL-1, TNF, IL-6), chemokines (CXCL8, CCL2)
    - Antiviral (Type 1) interferons: IFN-α, IFN-β
    - Recognition: by pattern recognition receptors (PRRs) that recognize broad structural motifs present in the invader but absent in the host
        - Pathogen-associated molecular patterns (PAMPs): essential to microbial survival and metabolism which makes them relatively invariant and evolutionarily stable
            - Peptidoglycan, flagellin, lipopolysaccharide (LPS), nucleic acids like viral dsDNA, mannose, non-methylated CpG repeats
        - Damage-associated molecular patterns (DAMPs)
    - Toll-like receptors: TLR3, TLR7, TLR8 → viruses
    - Phagocytosis: surveys tissue compartments and discovers microbes, particulate matter, and dead/injured cells; ingests and eliminates materials; extracts immunogenic information from foreign matter
        - Professional phagocytes: neutrophils (general purpose, reacts early to bacteria and other foreign materials), macrophages (derived from monocytes, scavenge and process foreign substances to prepare them for reactions with B-cells and T-cells), dendritic cells, eosinophils
        - Opsonization: the addition of an opsonin to ensure that the cell is readily identified and more efficiently taken by phagocytes (receptors: CR1 → complement cascade for C3b fragment; Fc → antibodies for Fc region of immunoglobulins)
            - Complement molecules: opsonize the antigen
                - Opsonized agents bind to CR1 receptor on the phagocyte
                - Main opsonins: C3b, C4b, C1q
            - Antibodies: opsonize the antigen
                - Fc region of antibody binds to phagocytic cell receptors
                - Phagocytic cells do not have Fc receptors for IgM, but IgM activates complement to do the opsonization
                - Main opsonin: IgG
            - Circulating proteins: secreted pattern recognition receptors
                - Main proteins: pentraxins, ficolins, collectins (mannose-binding lectin)
        - Mechanism
            - Chemotaxis and ingestion: phagocytes migrate and recognize PAMPs that drive initiation and perpetuation of the inflammatory response
            - Phagolysosome formation: lysosome fuses with phagosome (death ~30 mins)
            - Destruction and elimination: oxygen-dependent system (respiratory burst), liberation of lactic acid, lysozyme, and nitric oxide
                - Oxygen-independent
                    - Lysozyme: split peptidoglycan
                    - Lactoferrin and reactive nitrogen intermediates deprive pathogens of iron
                    - Proteolytic enzymes degrade dead microbes
                    - Defensins, cathepsin G, and cationic proteins damage the microbial membrane

            - Oxygen-dependent (oxidative burst)
                - NADPH oxidase produces superoxide which recombines with other molecules to produce reactive free radicals
                - Superoxide anions are then used by superoxide dismutase to produce hydrogen peroxide
                - Myeloperoxidase uses the derived hydrogen peroxide to produce hypochlorite/bleach
    - Inflammatory process: localized tissue response triggered by trauma/antigenic insult; may include systemic effects and is characterized by redness (rubor), heat (calor), swelling (tumor), and pain (dolor); principal cells involved are neutrophils and macrophages
        - Injury/immediate reactions: attracts leukocytes to site of injury
        - Vascular reactions: histamine causes vasodilation and increased vessel permeability (rubor and calor); delivers leukocytes, fluid, and clotting factors to site of injury
        - Edema and pus formation: edema (tumor) helps contain the infection and attracts neutrophils, debris and leukocytes form pus, bradykinin stimulates pain receptors (dolor)
        - Resolution/scar formation: macrophages clean the area, fibroblasts form granulation tissue; lymphocytes mediate long-term immunity
        - Diapedesis: migration of cells out of blood vessels into the tissues
        - Chemotaxis: migration in response to specific chemicals at the site of injury/infection
        - Cell adhesion molecules: selectins, mucins, integrins ($\alpha L\beta 2$, $\alpha M\beta 2$, $\alpha X\beta 2$), ICAMs
        - Leukocyte extravasation
            - Rolling: mediated by binding of selectins (endothelium) to mucin-like CAMs (leukocytes)
            - Activation: chemokine binding induces conformational change in integrins (leukocyte)
            - Arrest/adhesion: integrins now capable of binding Ig-superfamily CAMs (endothelium)
            - Transendothelial migration: migration through tight junctions of inflamed endothelium
        - Systemic acute phase response
            - Mediated by IL-1, IL-6, TNF-α
                - Acts on the hypothalamus → fever
                - Acts on liver → increased acute phase proteins
                    - C-reactive protein (increases x1,000), fibrinogen, haptoglobin, amyloid, ceruloplasmin; activates complement
                - Acts on bone marrow → increased leukocytosis
                    - Left-shift, increase in immature cells (bands)
        - Chronic inflammatory response
            - Inflammation lasting 2 weeks or longer that is often related to an unsuccessful acute inflammatory response; IFN-γ and TNF-α play a major role
            - May result in disease state: allergies, autoimmune disease, microbial infections, transplants
            - Other causes: high lipid and wax content of microorganisms (*Mycobacteria*), ability to survive inside the macrophage, toxins, chemicals/particulate matter/physical irritants
- NADPH Oxidase Deficiency: X-linked recessive condition; absence of NADPH oxidase will prevent the formation of reactive oxygen species and will result in chronic granulomatous disease (CGD)
    - Associated with catalase + infections (C-shapes → *Candida, Serratia, H. pylori, Actinomyces, Pseudomonas, E. coli, S. aureus*)
    - Healthy patient: production of hydrogen peroxide via respiratory burst is greater than the catalase produced by the microbes → microbes are overwhelmed and die
    - Patient with CGD: production of hydrogen peroxide via respiratory burst is less than the catalase produced by the microbes → microbes tolerate the phagocytic environment and survive
- Chronic Granulomatous Disease (CGD): suffer from recurrent bouts of infection due to the decreased capacity of their immune system to fight off disease-causing organisms
    - Presentations: pneumonia, abscess, suppurative adenitis, osteomyelitis, bacteremia/fungemia, cellulitis, meningitis
        - Aphthous ulcers (mouth ulcers): neutropenia frequently presents as mouth ulcers (although there isn't neutropenia, neutrophil function is impaired)
    - Treatment: prophylactic antibiotic treatment, IFN-γ, bone marrow transplants)
    - Testing: nitroblue-tetrazolium test (negative test → yellow), dihydrorhodamine 123 (positive test → both peaks overlap)

- Chediak-Higashi Syndrome (CHS): rare autosomal recessive disorder that arises from a mutation of a lysosomal trafficking regulator protein (microtubule dysfunction) leading to a lack of phagolysosome formation
    - Characterized by large lysosomal vesicles in phagocytes with poor killing action leading to: susceptibility to infections, abnormalities in nuclear structure of leukocytes, anemias, hepatomegaly, Dohle bodies (cytoplasmic inclusions of neutrophils)
    - Clinical presentation: albinism, neutropenia, periodontal disease, recurrent pyogenic infections
- Leukocyte Adhesion Defect (LAD): deficiency of the β-2 integrin subunit (CD18) that is involved in the formation of the β-2 integrins (i.e., LFA-1) by dimerization with different CD11 subunits (a, b, c)
    - Neutrophils cannot extravasate and fight against bacteria in tissues
    - Clinical presentations: recurrent skin infections, delayed umbilical separation, absence of pus
- Antibacterial Serum Agents
    - Lactoferrin: iron-binding protein that competes with pathogens for iron, an essential metabolite
    - C-reactive protein (CRP): an acute phase protein that binds to phosphocholine in bacteria membranes and functions in opsonization. It activates complement via the classic pathway
    - Mannose-binding lectin (MBL): recognizes carbohydrate patterns, found on bacteria, viruses, protozoa, and fungi, resulting in activation of the lectin pathway of the complement system
    - Serum amyloid A protein (SAP): binds to bacterial cell wall lipopolysaccharide and serves as a receptor for phagocyte attachment. Implicated in several chronic inflammatory diseases, such as amyloidosis, atherosclerosis, and rheumatoid arthritis
- Interferons
    - Small proteins produced by certain white blood cells and virally-infected cells
    - Produced in response to viruses, RNA, immune products, and various antigens
    - Types
        - Interferon-α and interferon-β (Type I)
            - Type I interferons can be produced by almost any cell upon stimulation by a virus
            - Act on neighboring uninfected cells
            - Inhibit transcription and translation of viral proteins
        - Interferon-γ (Type II)
            - Produced by Th1, CTLs (CD8+), NK cells
            - Promotes NK cell and macrophage activity
            - Activates inducible nitric oxide synthase (iNOS) → relaxes smooth muscle & vasodilation
            - Induces the production of antibodies
            - Causes normal cells to increase expression of MHC I & MHC II
            - Promotes adhesion and binding required for leukocyte migration
            - Induces the expression of intrinsic defense factors with direct antiviral effects
            - Responds to cancerous growth
            - Granuloma: body's way of dealing with a substance that it cannot remove
                - Infectious causes: tuberculosis, leprosy, histoplasmosis, cryptococcosis, blastomycosis, coccidioidomycosis, and cat scratch disease
                - Activation of Th1 helper cells by macrophages (IL-1 and IL-12 by macrophages)
                - Th1 helper cells aggregate around the macrophages (IFN-γ release activates macrophages)
                - Macrophages surround the Th1 helper cell and becomes fibroblast-like cells walling off the infection
- Complement System: important for the recruitment of inflammatory cells and the killing/opsonization of pathogens
    - Classical Pathway
        - Adaptive immunity
        - C1 binds to Ag-Ab complex → activated C1 cleaves C4 & C2 → formation of C3 convertase → C3 is cleaved → C5 is cleaved → sequential binding of C5b, C6-C9
        - Assays
            - Hemolytic titration (CH50) assay: amount of patient serum required to lyse 50% of the antibody-sensitized sheep erythrocytes (needs all complement molecules C1-C9 to function)
            - ELISA
    - Alternative Pathway
        - Innate immunity
        - Spontaneous cleavage of C3 based on multiple initiators (requires Factors B & D) → formation of C3 convertase (less stable; requires Properdin)

        - Assays
            - AH50
    - Mannose-Binding Lectin Pathway
        - Innate immunity
        - Mannose-binding lectin (MBL), an acute phase protein, binds to carbohydrate residues on cells or pathogen surfaces
        - MBL-associated serine protease (MASP) binds to MBL → cleaves C4 & C2 (similar to classical pathway)
    - Membrane-Attack Complex
        - Three above pathways converge with the activation of C5 convertase → C5b binds antigenic surface, MAC is formed on the surface
            - C5 components: C5b, C6-C9
            - Poly-C9: perforin-like molecule
    - Functions
        - C5b-C9 → cell lysis
        - C3b, C4b, C1q → opsonization
        - C3a, C5a → inflammation & anaphylactic reactions
        - C3b → clearance of immune complexes
        - C3b, C5b-C9 → viral neutralization
    - Inhibitors
        - Inhibit C1: C1 inhibitor (classical pathway)
        - Prevent assembly of C3 convertase: CR1 (CD35), MCP (CD46), C4b-binding protein, Factor H
        - Accelerate decay of C3 convertase: DAF (CD55)
        - Block MAC: CD59 (protectin), S protein, clusterin
        - Inactivates anaphylatoxin C3a & C5a: anaphylatoxin inactivator
    - Genetics
        - Most complement deficiencies are autosomal recessive
        - Properdin deficiency may be inherited as an X-linked trait
        - MBL deficiency can be both autosomal dominant and recessive
        - Complement deficiencies can also be acquired (post-infection)
    - Deficiencies
        - Activator disorders: under-reactive system; more susceptible to infections
        - Inhibitor disorders: over-reactive system
            - Hereditary Angioedema: overproduction of C2b (vasodilator)
                - Rare syndrome characterized by episodic, nonpruritic, localized subcutaneous and submucosal swelling → laryngeal edema can precipitate fatal respiratory obstruction
                - Dysregulation of three different systems
                    - Complement: classic (C1) and MBL (MASP-2) pathways
                        - Excessive activation of the complement due to mutation on C1inh results in high concentration of anaphylatoxins: C3a & C5a
                        - Reduced C4 on serum test
                    - Coagulation: factor XI & XII leading to fibrin buildup
                    - Contact cascade: results in increased bradykinin levels leading to inflammation, coagulation, and other body effects
                - Absence of urticaria (rush) that distinguishes this disease from an allergic reaction

<u>Adaptive Immune System</u>

- The adaptive immune response is initiated by antigen recognition, which is accomplished by antigen receptors of lymphocytes
- Lymphoid (mononuclear)
    - B-cells: protein, polysaccharides, lipids, nucleic acids (soluble form or cell surface-associated antigens)
        - Site of maturation: bone marrow in mammals; displays membrane-bound immunoglobulin (antibody)
        - Once antigen is encountered, undergoes differentiation
            - Plasma cells: antibody-secreting cells, die within 1-2 weeks
            - Memory B-cells: same membrane-bound antibody as parent B-cell with a longer lifespan

- T-cells: peptide fragments of protein antigens (only when presented on specialized molecules); do not recognize native antigens - antigens must be processed in the APC in order to be recognized by T-cells
    - Site of maturation: thymus
    - T-cell receptor: only recognizes antigen (peptides) that is bound to cell membrane proteins called major histocompatibility complex (MHC); once antigen is encountered with MHC, differentiation occurs → effector T-cells & memory T-cells
        - Effector T-cells: T-helper, T-cytotoxic, T-regulator/T-suppressor
            - T-helper cells (Th)
                - Mature Th cells express the surface glycoprotein CD4 and are referred to as CD4+ T-cells
                - Recognizes antigen presented in MHC Class II (extracellular/phagocytized); MHC II only present in antigen-presenting cells
                    - MHC II: $\alpha_1, \alpha_2, \beta_1, \beta_2$
                - Helps with the activation of B-cells, Tc cells, and macrophages in the immune response
            - T-cytotoxic cells (Tc)
                - Mature Tc cells express the surface protein CD8 and are referred to as CD8+ T-cells
                - Recognizes antigen presented in MHC Class I (eliminates virus/intracellular microbes infected cells or cancerous cells)
                    - MHC I: $\alpha_1, \alpha_2, \alpha_3, \beta_2$
            - T-regulatory cells (Tr)
                - Mature T-regulatory (suppressor) cells express the surface glycoprotein CD4, CD25, and FOXP3
                - Cytokines produced: IL4, IL10, TNF-α, TGF-β
                - Helps suppress the immune system after its been upregulated and the infection has cleared
                - Maintains tolerance to self-antigens, and prevents autoimmune diseases
        - Memory T-cells
            - T-cells that have previously encountered and responded to their cognate antigen
            - During a second encounter with an invader, memory T-cells can reproduce to mount a faster and stronger immune response
            - May be either CD4+ or CD8+
    - T-cell antigen presentation by dendritic cells
        - Dendritic cell in epithelium: capture microbial antigen
        - Activation of dendritic cells
        - Migration to draining lymph node (chemokines)
        - Maturation: due to cytokines and TLR-signaling
        - Expression of MHC II and co-stimulators for stimulation of T-cells
        - Most important antigen-presenting cell: dendritic cells (are in the same region as T-cells in the lymph node)
- Major Histocompatibility Complex (MHC)
    - Cell surface molecules that are required for: antigen processing, antigen presentation, and graft rejection
    - Cluster of genes: HLA (human leukocyte antigen), chromosome 6
    - Diversity
        - Generation of B-cell receptors (antibodies) and T-cell receptors is dynamic
            - Gene rearrangement and others
        - In contrast, MHC molecules are fixed in the genes
            - Differences in population due to the large number of alleles (370 A alleles, 660 B alleles, 190 C alleles)
    - MHC Class I: heterodimeric protein (Ig superfamily); expressed on the majority of nucleated cells; A, B, C genes
        - Components
            - Alpha-1, alpha-2, alpha-3
            - Beta-2 microglobulin
        - Alpha chain

            - Transmembrane
            - Encoded by A, B, and C regions in the human MHC complex
            - Alpha-1/alpha-2: highly polymorphic peptide binding region
            - Alpha-3: binds to CD8 on cytotoxic T-cells
            - Binds short peptides, usually between 8-10 amino acids, presents an endogenously processed antigens
        - Beta chain
            - Beta-2 microglobulin
            - Induced by highly conserved gene on different chromosome
        - CD8+ T-cytotoxic cells are MHC Class I restricted
            - Can only recognize antigens presented by MHC Class I molecules, therefore, cells with MHC Class I are called “target cells”, killed by cytotoxic T-cells
        - Fate: exported to the cell surface; sent to lysosomes for degradation
        - Evasion of immunity via interference with cytosolic antigen processing
            - Many viruses produce immuno-evasins that interfere with antigen presentation by MHC Class I molecules
                - Viruses block MHC Class I from properly presenting them to CD8 T-cells
                    - Inhibit peptide transport
                    - Inhibit peptide loading
                    - Cause MHC Class I degradation
    - MHC Class II: heterodimeric protein (Ig superfamily); expressed on antigen-presenting cells (APC’s); DP, DQ, DR genes
        - Components
            - Alpha-1 & alpha-2 chain: transmembrane
            - Beta-1 & beta-2 chain: transmembrane
        - Alpha-1/beta-1 chain: highly polymorphic peptide (binding region)
        - Beta-2: highly conserved, binds to CD4 on helper T-cells
        - Alpha-2: highly conserved
        - Binds long peptides, usually between 13-25 amino acids
        - Presents antigens that are in the phagosome
        - CD4+ T-helper cells are MHC Class II restricted
            - Cells with MHC Class II are called antigen-presenting cells (APC’s)
        - In the endoplasmic reticulum
            - Need to prevent newly synthesized, unfolded, self-proteins from binding to immature MHC
            - Invariant chain stabilizes MHC Class II by non-covalently binding to the immature MHC Class II molecule
        - Class II associated invariant chain peptide (CLIP)
            - $(\alpha\beta inv)3$ complexes directed towards endosomes by invariant chain → cathepsin L degrades invariant chain → CLIP blocks groove in MHC molecule → MHC Class II containing vesicles fuse with antigen-containing vesicles
            - HLA-DM catalyzes the removal of CLIP
                - Replaces CLIP with a peptide antigen using a catalytic mechanism
                - HLA-DO may play a role in blocking HLA-DM
        - Bacterial evasion of MHC II antigen presentation
            - Escape endosomes, neutralize endosome acidification, block fusion with lysosome, sequesters MHC Class II molecules after vesicle fusion: preventing from going to the surface
- Antigens generated by endogenous and endocytic antigen processing activate different effector functions
    - Exogenous pathogens
        - Eliminated with the help of T-helper cells that use antigens generated by endocytic processing; antigens presented to T-helper cells by MHC Class II
        - Exogenous antigen processing takes place in lysosomes and uses invariant chain and HLA-DM
    - Endogenous pathogens

- Endogenous processing is non-lysosomal and uses proteasomes and peptide transporters in antigen processing (pathogens can evade immunity by disrupting this antigen processing)
- Eliminated by the killing of infected cells by cytotoxic T-lymphocytes (CTL) that use antigens generated by cytosolic processing; antigens presented to T-cytotoxic cells by MHC Class I
    - Inactive virus raises a weak CTL response
        - Antigens from inactive viruses are processed via the exogenous pathway (MHC Class II presentation)
    - Infectious virus raises a strong CTL response
        - Protein synthesis is required for non-lysosomal antigen processing
        - Antigens from infectious viruses are processed via the endogenous pathway (MHC Class I presentation)
- Peptide generation
    - Proteins targeted for lysis combine with a small protein → ubiquitin
    - Ubiquitin-protein complex is degraded by a proteasome
    - Specific proteasomes generate peptides which can bind to MHC I
    - Peptide antigens produced in the cytoplasm are physically separated from the newly formed MHC Class I that are produced in the endoplasmic reticulum
        - Peptides need access to the endoplasmic reticulum in order to be loaded onto the MHC Class I molecules
- Proteasome degradation
    - Cytoplasmic cellular proteins, including non-self proteins are degraded continuously by a multi-catalytic protease of 28 subunits
        - The components of the proteasome are induced by IFN-γ

- Exceptions
    - Cross-presentation
        - Exogenous antigens normally process by phagolysosomes (MHC II) are presented by MHC I (exception to the MHC Class I rule)
    - Cross-priming
        - Presented to, and activates, CD8+ T-cell
    - Example
        - *Rickettsia rickettsia*
            - Endocytosed but rapidly lyses endosomal membrane and escapes prior to phagolysosome formation
            - Degraded by proteasome and presented on MHC I

- Antigen Presentation & Receptors
    - Antigen receptors are critical in the maturation of lymphocytes and in all adaptive immune responses
        - B-cells express antibodies (B-cell receptors/BCR's)
            - Recognize a broad range of chemical structures: proteins, lipids, carbohydrates, and nucleic acids; can recognize microbes and toxins in their native forms
            - Antibodies exist as membrane-bound antigen receptors on B-cells & as secreted proteins
            - Antigens bind to antibody by non-covalent interactions (affinity → affinity maturation)
                - Avidity: strength of multiple affinities of individual non-covalent binding interactions
                - Cross-reactivity: one antibody, different epitopes
            - Monoclonal antibodies: the realization that one clone of B-cells makes an antibody of only one specificity gave rise to the production of monoclonal antibodies
                - CD20: depletion of B-cells; seen in rheumatoid arthritis, multiple sclerosis, other autoimmune cells, and B-cell lymphoma
            - Recognizes antigen, but cannot transmit signals on its own → signaling functions are mediated by Igα and Igβ (2 heavy chains + 2 light chains → ends are antigen binding site/Fab fragment)
                - Light chain
                    - Amino (N) terminal → variable, site of antigen binding
                    - Carboxy (C) terminal → constant, can be κ or λ

- Heavy chain: determines the 5 classes of Ig: G, A, M, E, D
    - Variable region
        - Isotypes: sequence variability of H/L chain constant regions (5 heavy chain isotypes & 2 light chain isotypes)
        - Allotype: allelic variants; affects constant region
        - Idiotype: sequence variability of H/L chain variable regions (individual, clone-specific)
    - Constant region: nearly invariant → not involved in antigen binding; amino acid sequence defines the five heavy chain classes (γ, α, μ, ε, δ)
- Goal: production of plasma and memory cells via generation of mature immunocompetent cells, activation of mature cells, and differentiation of active B-cells
    - B-cell maturation: antigen independent phase
        - Pro B-cells: expression of CD45R and CD19; requires bone marrow stromal cells to develop into a pre-B cells → IL-7 supports this process
            - Rearrangement of Ig heavy chain genes: RAG1 & RAG2 are expressed → drives recombination
            - Terminal deoxyribonucleotidyl transferase (TdT) is expressed → catalyzes insertion of N-nucleotides at the heavy chain
        - Pre B-cells: presence of Igμ heavy chain in cytoplasm, few expressed on cell surface associated with a surrogate light chain and Igα/Igβ
            - Signaling: induces recombination of Igκ light chain locus; RAG1 & RAG2 are expressed
        - Immature B-cells: IgM expressed on surface → negative selection occurs
            - Further matures in bone marrow and the spleen → expression of IgM and IgD = mature B-cell
        - Naïve B-cells: migrate out of bone marrow; quiescent/non-dividing, has not encountered antigen
        - Treatment targeting: CD20 is expressed on all stages of B-cell development except Pro B-cells and plasma cells
            - CD20 is the target of monoclonal antibodies for treatment of all B-cell lymphomas and leukemias → leaves Pro B-cells as a bank (no transplantation needed)
    - B-cell development
        - Bruton's tyrosine kinase (BTK) helps with B-cell developmental mutations in this gene are implicated in X-linked agammaglobulinemia (XLA)
            - Patients have normal Pre B-cell populations but they fail to mature and enter circulation
- T-cells express T-cell receptors (TCR's) → heterodimer recognizes peptides displayed by MHC molecules; not produced in a secreted form and do not undergo class switching
    - Composed of two chains: α & β, each with a variable and constant region
        - CDR3 is the most variable
    - Recognizes antigen, but cannot transmit signals on its own
    - Associated with CD3: essential for TCR membrane expression and post-antigen recognition signal transduction
        - Complex of 3 dimers
            - γε dimer
            - εδ dimer
            - ζζ dimer or ζη dimer
        - Cytoplasmic tails contain immunoreceptor tyrosine-based activation motif (ITAM) → signal transduction
    - Thymocyte development: hematopoietic stem cells in bone marrow → common lymphoid progenitor → migrates to thymus (site of maturation) → proliferation via IL-7

- Precursors: Pro T-cell; no TCR or CD3 expressed
- Double-negative cells: Pre T-cells → no expression of CD4 or CD8 → rearrangement of TCR-β genes followed by α genes in the cortex of the thymus
  - Outcome: expression of CD4 & CD8
- Double-positive cells (CD4+ & CD8+) → positive selection (self-restriction)
  - Cells that do not bind with MHC die via apoptosis
  - Cells that bind moderately to MHC molecules become single positives (CD4 or CD8)
- Negative selection (self-tolerance)
  - Elimination of any CD4 or CD8 cells that have a high-affinity to self-MHC complexes → apoptosis

- Clonally distributed: each clone has a unique receptor that is specific for an antigen
- Antigen recognition site: variable regions (V)
  - Variability is concentrated - hypervariable regions or complementary-determining regions (CDR)
- Conservative regions: constant region (C)
- Lymphopoiesis
  - B-cell production occurs throughout life and does not wane as T-cell production does
    - Naïve B-cells survive about 1 week and undergo negative selection
- Autoimmune regulator (AIRE): transcription factor expressed in the medulla of the thymus
  - AIRE drives negative selection of self-recognizing T-cells
    - Medullary thymic epithelial cells (mTEC) expresses major proteins from elsewhere in the body (tissue-restricted antigens → TRA) and T-cells that respond to those proteins are eliminated through apoptosis
  - When AIRE is defective, this can result in a variety of autoimmune diseases (hypoparathyroidism, primary adrenocortical failure, chronic mucocutaneous candidiasis)

- Classes of Immunoglobulins
  - IgG: monomer; produced by plasma cells (primary response) and memory cells (secondary response), most prevalent; can cross placenta
  - IgA: monomer; circulates in blood, dimer in mucous and serous secretions; breast milk
  - IgM: pentamer; first class synthesized following antigen encounter
  - IgD: monomer; serves as a receptor for antigen on B-cells
  - IgE: monomer; involved in allergic responses and parasitic worm infections

Features of the Pathways of Antigen Processing

Feature	Class I MHC Pathway	Class II MHC Pathway
Composition of stable peptide-MHC complex	Polymorphic α-chain of MHC, β 2-microglobulin, peptide	Polymorphic α and β chains of MHC, peptide
Cells that express that MHC class	All nucleated cells	Dendritic cells, mononuclear phagocytes, B-lymphocytes, endothelial cells, thymic epithelium
Responsive T-cells	CD8+ T-cells	CD4+ cells, T-cells
Source of protein antigens	Cytosolic proteins (mostly synthesized in the cell; may enter cytosol from phagosomes)	Endosomal/lysosomal proteins (mostly internalized from extracellular environment)
Enzymes responsible for peptide generation	Enzymatic components of cytosolic proteasome	Endosomal and lysosomal proteases (i.e., cathepsins)
Site of peptide loading of MHC	Endoplasmic reticulum	Late endosomes and lysosomes
Molecules involved in transport of peptides and loading of MHC molecules	Transporters associated with antigen processing (TAPs I & II)	Invariant chain, DM

Professional APC's

Cell Type	Expression of:		Principal Function
	Class II MHC	Co-stimulators	
Dendritic cells	Constitutive; increases with maturation; increased by IFN-γ	Constitutive; increases with maturation; increased by TLR ligands, IFN-γ, and T-cells (CD40-CD40L interactions)	Antigen presentation to naive T-cells in the initiation of T-cell responses to protein antigens (priming)
Macrophages	Low or negative; inducible by IFN-γ	Low, inducible by TLR ligands, IFN-γ, and T-cells (CD40-CD40L interactions)	Antigen presentation to CD4+ effector T-cells in the effector phase of cell-mediated immune responses
B-lymphocytes	Constitutive; increased by cytokines (i.e., IL-4)	Induced by T-cells (CD40-CD40L interactions), antigen receptor cross-linking	Antigen presentation to CD4+ helper T-cells in humoral immune responses (T-cell/B-cell interactions)

- Cluster of Differentiation (CD)
    - Protocol for identification and investigation of cell-surface molecules
    - CD number assigned on basis of 1 cell surface molecule recognized by 2 specific monoclonal antibodies
        - CD34: all stem cells
        - CD45: marker of all white blood cells
        - CD3: marker of all T-cells
            - CD4: marker for T-helper cells
            - CD8: marker for T-cytotoxic cells
        - CD19, CD20, CD21: markers of B-cells
    - Leukocyte markers
        - Granulocytes → CD45+, CD15+
        - Monocytes → CD45+, CD14+
        - T-cells → CD45+, CD3+
            - T-helper cells → CD45+, CD3+, CD4+
            - T-cytotoxic cells → CD45+, CD3+, CD8+
            - T-regulatory cells → CD45+, CD3+, CD4+, CD25+, FOXP3
        - B-cells → CD45+, CD19+, CD20+, CD21+
        - Natural killer cells → CD45+, CD16+, CD56+
        - Dendritic cells → CD45+, CD14+
- Lymphocyte Diversity
    - Before interacting with the antigen:
        - Somatic recombination: variable region of receptors → VDJ joining for heavy chain. VJ joining for light chain; antigen-independent phase
            - Important enzyme: RAG1/RAG2 (expresses from pro stage to immature stage)
            - Allelic exclusion: recombine only one of your parent's genes
        - Junctional diversity: VDJ joining after somatic recombination; antigen-independent phase
            - P-nucleotide addition: repair enzymes fix broken pieces
            - N-nucleotide addition: tDt add nucleotides during the joining (only heavy chain) → tDt is expressed from pro stage to pre stage of development
    - After interacting with the antigen:
        - Somatic hypermutation: refinement of variable region of receptors to make it more compatible with the antigen; antigen-dependent phase
            - Important enzyme: AID
        - Class switching: constant region of antibodies is changed; antigen-dependent phase
            - Important enzyme: AID
            - Only occurs if T-cell activates B-cell through CD40-CD40L
            - Switch depends on cytokines present (IFN-gamma, IgG, IL-4, IgE)

Overall Summary
- Innate Immune System: non-specific, no memory
  - Recognize antigens by PAMPS & DAMPS using PRRs
  - Inflammatory signs: acute phase proteins, cytokines, erythrocyte sedimentation rate (ESR)
  - APCs: B-cells, dendritic cells, macrophages
    - Presentation of antigen
      - Cytosolic pathway: proteasome to ER by TAP 1 + 2 to MHC I
        - Presented to T-cytotoxic → CD8
      - Endocytic pathway: endosome/lysosome/endosome-containing MHC II
        - Involved with invariant chain, CLIP, HLA-DM
        - Presented to T-helper → CD4
  - Diseases
    - Chronic granulomatous disease (CGD): associated with oxidative burst (NADPH oxidase)
    - Leukocyte adhesion deficiency (LAD): involved in adhesin mutation (CD18)/diapedesis → neutrophils cannot cross to site of inflammation
    - Chediak higashi: no fusion between lysosome and phagosome
    - Hereditary angioedema: lack of C1-inhibitor → over-active cytokine cascade
- Adaptive Immune System: specific, memory
  - Cells involved: B-cells, T-cells
  - Specificity given by their receptors which undergo development and affinity maturation
    - Pro cells: rearrangement of heavy chains starts → CD45R & CD19
    - Pre cells: end of arrangement of heavy chains → μ receptors in cytoplasm (Igμ); start of arrangement of light chain
    - Immature cells: end of arrangement of light chains → presence of IgM on surface
    - Mature cells: presence of IgM & IgD
- Cytokines
  - IL-1, IL-6, TNF-α: acute inflammation (TNF-α important in neutrophil activation)
  - IFN-α, IFN-β: antiviral
  - IFN-γ: activation of macrophages, increase of MHC expression → chronic inflammation
  - IL-8: chemokine → chemotaxis
  - IL-5: eosinophil activation
  - IL-4: class switch to IgE
  - IL-7: development of lymphocytes from pro cells → pre cells
  - IL-3: development of myelocytes
  - IL-10, TGF-β: downregulation of immune response
  - IL-12: produced by macrophages to activate T-helper 1 cells → important for granuloma formation

Epidemiology

Introduction to Epidemiology
- Terminology
  - Epidemiology: the study of the distribution and determinants of health-related states or events (including disease), and the application of this study to the control of diseases and other health problems
  - Incidence: the number of individuals who develop a specific disease or experience a specific health-related event during a particular time period (i.e., the number of new cases only that developed during a time period in a population free from the disease or condition)

$$\frac{New\ Cases}{Population\ at\ Risk}$$

  - Incidence Rate/Density: a true rate; number of new cases in a specified period divided by the total person-time at risk during this period
  - Risk factor: a patient characteristic associated with the risk of contracting a specific disease
  - Prognostic factor: a patient characteristic that identifies subgroups of untreated patients having different outcomes
  - Factor predictive of treatment effect: a patient characteristic that identifies subgroups of treated patients having different (as a consequence of treatment) outcomes
  - Endemic: refers to the constant presence and/or unusual prevalence of a disease or infectious agent in a population within a geographic area
  - Hyperendemic: refers to persistent, high levels of disease occurrence
  - Sporadic: refers to a disease that occurs infrequently and irregularly
  - Epidemic: refers to an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area

    - Pandemic: refers to an epidemic that has spread over several countries or continents, usually affecting a large number of people
- Distributions
    - Negatively skewed = left skewed = right modal (i.e., mode is closer to the right)
        - Mean < median < mode
    - Positively skewed = right skewed = left modal (i.e., mode is closer to the left)
        - Mode < median < mean
- Levels of Disease Prevention
    - Primordial: population prevention (i.e., government policy/legislation, taxes on cigarettes, seatbelt laws, etc.)
        - Illness: what patient experiences
        - Disease: what the physician diagnoses
    - Primary: prevent disease from occurring (i.e., not smoking, wearing seatbelts, vaccines, exercise, diet, sleeping well, etc.)
        - Illness absent; disease absent
    - Secondary: screening (i.e., PFR, roadblocks, pap smears, mammography, colonoscopy, annual physical, etc.)
        - Illness absent; disease present
    - Tertiary: treatment, prevent sequelae (i.e., PT, diabetic foot care, cardiac rehabilitation, incentive spirometry, etc.)
        - Illness present; disease present
    - Quaternary: prevent over-treatment (i.e., preventing all types of harm associated with medical interventions, etc.)
        - Illness present; disease absent

<u>Diagnostic & Screening Tests</u>

		Disease	
		Present	**Absent**
Test	**Positive**	True Positive TP	False Positive FP
	Negative	False Negative FN	True Negative TN

- Sensitivity: true positive rate (TPR): the probability that a patient with the disease will have a positive test result; used to rule out disease (snOUT)
    - Increases while moving down/left on a distribution curve (at the expense of specificity)

$$\frac{TP}{(TP+FN)}$$

- 1-Sensitivity: false-negative rate (FNR): the probability that a patient with the disease will have a negative test result

$$\frac{FN}{(TP+FN)}$$

- Specificity: true negative rate (TNR): the probability that a patient without the disease will have a negative test result; used to rule in disease (spIN)
    - Increases while moving up/right on a distribution curve (at the expense of sensitivity)

$$\frac{TN}{(TN+FP)}$$

- 1-Specificity: false positive rate (FPR): the probability that a patient without the disease will have a positive test result

$$\frac{FP}{(TN+FP)}$$

- Positive predictive value (PPV): the probability that a patient with a positive test result will not have the disease

$$\frac{TP}{(TP+FP)}$$

- Negative predictive value (NPV): the probability that a patient with a negative test result will not have the disease

$$\frac{TN}{(TN+FN)}$$

- Accuracy: the probability that the results of a test will accurately predict the presence or absence of disease

$$\frac{(TP+TN)}{(TP+TN+FP+FN)}$$

- Prevalence: a proportion; the probability that a portion of a particular population will be found to be affected by a specific disease at a given time; also known as the "pre-test probability"

$$\frac{(TP+FN)}{(TP+TN+FP+FN)}$$

$$\frac{All\ Cases}{Population}$$

- Regression of the mean
    - The phenomenon that occurs when clinicians encounter an unexpectedly abnormal test result and repeat the test only to find that the second test result is closer to normal (i.e., patients selected because they represent an extreme value in a distribution can be expected, on average, to have less extreme values on subsequent measurements); this is an application of the Central Limit Theorem
- The Receiver Operator Characteristic (ROC) Curve
    - A graphical representation of the relationship between sensitivity and specificity for a given test; constructed by plotting the true positive rate (sensitivity) against the false positive rate (1-specificity)
        - Accuracy: the area under the curve (an area of 1 represents a perfect test → right angle curve)
        - Likelihood ratio: the slope of the tangent line at a cut-off point gives this value
- Likelihood Ratio: the likelihood that a given test result would be expected in a patient with the target disorder compared to the likelihood that the same result would be expected in a patient without the target disorder → a proportion of probability
    - LR+: the increase in the odds of having the disease after a positive test result
        - >10 → strong evidence to rule in disease
        - <0.1 → strong evidence to rule out disease

$$\frac{Sensitivity}{1-Specificity}$$

    - LR-: the decrease in the odds of having the disease after a negative test result

$$\frac{1-Sensitivity}{Specificity}$$

- Bayes' Theorem: the odds of having, or not having, a disease after testing

$$Posttest\ Odds = Pretest\ Odds\ x\ Likelihood\ Ratio$$

- Parallel Testing: increases the sensitivity and negative predictive value for a given disease prevalence above those of each individual test; on the other hand, it decreases the specificity and positive predictive value for a given disease prevalence below those of each individual test
    - Useful when rapid assessment is necessary (i.e., hospitalized or emergency patients)
- Serial/Sequential Testing: increases the specificity and positive predictive value but decreases sensitivity and the negative predictive value
    - Useful when none of the individual tests available are highly specific

Measures of Disease Association

- Risk: the chances of something happening divided by the chances of all things happening

$$\frac{a}{a+b}$$

- Odds: the chances of something happening divided by the chances of something not happening

$$\frac{a}{b}$$

- Odds Ratio (OR): a measure of association between an exposure and an outcome; represents the odds that an outcome will occur given a particular exposure, compared to the odds of the control having that exposure
    - Ratio of = 1.0: means that the odds of exposure among cases is the same as the odds of exposure among controls; the exposure is not associated with the disease
    - Ratio of > 1.0: means that the odds of exposure among cases is greater than the odds of exposure among controls; the exposure may be a risk factor for the disease

- Ratio of < 1.0: means that the odds of exposure among cases is lower than the odds of exposure among controls; the exposure may be protective against the disease

	Disease	
	Case	Control
Exposed	A	B
Unexposed	C	D

$$OR = \frac{odds\ that\ a\ case\ was\ exposed\ (\frac{A}{C})}{odds\ that\ a\ control\ was\ exposed\ (\frac{B}{D})} = \frac{AD}{BC}$$

- Incidence Rate (IR): number of individuals that have a disease out of the total population (can be with exposure or without exposure)

$$IR\ exposed = \frac{have\ a\ disease}{total\ population} = \frac{A}{A+B}$$
$$IR\ unexposed = \frac{have\ a\ disease}{total\ population} = \frac{C}{C+D}$$

- Incidence Rate Ratio (IRR): calculated by taking the incidence rate among exposed patients and dividing it by the incidence rate among unexposed patients

$$IRR = \frac{IR\ exposed}{IR\ unexposed} = \frac{\frac{A}{A+B}}{\frac{C}{C+D}}$$

- Hazard Ratio (HR): ratio of hazard in the treatment group over the hazard in the control group
    - HR = 0.5: at any particular time, half as many patients in the treatment group are experiencing an event compared to the control group
    - HR = 1.0: at any particular time, event rates are the same in both groups
    - HR = 2: at any particular time, twice as many patients in the treatment group are experiencing an event compared to the control group

$$HR = \frac{hazard\ in\ treatment\ group}{hazard\ in\ control\ group}$$

- Relative Risk (RR): measure of the strength of an association; can be used interchangeably with the Odds Ratio only if the outcome if interest is rare (i.e., rare disease) → less than 10%; for common events, the two calculations are very different
    - RR > 1: exposure is a risk factor for disease
    - RR < 1: exposure is a protective factor for disease

$$RR\ exposed = \frac{exposed + has\ disease}{exposed\ population} = \frac{A}{A+B}$$
$$RR\ unexposed = \frac{unexposed + has\ disease}{unexposed\ population} = \frac{C}{C+D}$$
$$RR = \frac{RR\ exposed}{RR\ unexposed} = \frac{\frac{A}{A+B}}{\frac{C}{C+D}}$$

- Risk Difference (RD)/Absolute Risk Reduction (ARR): the excess risk involved with exposure and disease
    - Example: 140 smokers die of lung cancer per year & 10 non-smokers die of lung cancer per year → RD = 130
    - RD < 0: risk is lower in the treatment group than the control group
    - RD = 0: risk is the same for both groups
    - RD > 0: risk is higher in the treatment group than the control group

- Risk Reduction: the reduction in risk between a treatment and control/placebo group
    - Example: 210 individuals receiving a placebo experience heart attacks per year & 120 individuals receiving treatment experience heart attacks per year → Risk Reduction = -90
- Attributable Risk (AR): the amount/proportion of disease incidence (or disease risk) that can be attributed to a specific exposure

$$AR = incidence\ in\ the\ exposed\ group - incidence\ in\ the\ unexposed\ group$$

- Odds Ratio & Relative Risk
    - OR/RR < 1: odds/risk of an event are lower in the treatment group than in the control group
    - OR/RR = 1: odds/risk of an event are the same for both groups
    - OR/RR > 1: odds/risk of an event are greater in the treatment group than in the control group
- Other Essential Formulas
    - Experimental Event Rate (EER): probability of outcome occurring in experimental group

$$EER = \frac{A}{A+B}$$

    - Control Event Rate (CER): probability of outcome occurring in control group

$$CER = \frac{C}{C+D}$$

    - Absolute Risk Reduction (ARR): difference in the event rate between control and treatment groups

$$ARR = |CER - EER|$$

    - Number Needed to Treat (NNT): average number of patients who need to be treated to prevent one additional bad outcome (i.e., the number of patients that need to be treated for one of them to benefit compared with a control in a clinical trial)

$$NNT = \frac{1}{ARR}$$

    - Number Needed to Harm (NNH): indicates how many persons on average need to be exposed to a risk factor over a specific period to cause harm in an average of one person who would not otherwise have been harmed

$$NNH = \frac{1}{AR}$$

Study Designs & Measures of Effect

Main Types of Epidemiological Studies

Type of Study	Characteristics
Experimental	Studies preventions and treatments for diseases; investigator actively manipulates which groups receive the agent under study Examples: randomized controlled trials (RCT's), non-randomized controlled trials (NRCT's)
Observational	Studies causes, preventions, and treatments for diseases; investigator passively observes as nature takes its course Examples: descriptive, analytical, ecological
Descriptive	Examples: case report, case series, case study, cross-sectional

Analytical	Examples: cross-sectional, case-control, cohort
Cohort	Typically examines multiple health effects of an exposure; subjects are defined according to their exposure levels and followed for disease occurrence Use Relative Risk calculation, not Odds Ratio (unless outcome is rare; less than 10%)
Prospective	Observing participants as time passes to study if exposure affects outcome Example: 100 patients were exposed to asbestos → follow for 10 years to see how many develop mesothelioma
Retrospective	Observing participants with an outcome and reviewing their past exposure to see if there is a correlation Example: 100 patients have mesothelioma → retrospectively study how many were exposed to asbestos in the past
Case-Control	Typically examines multiple exposures in relation to a disease; subjects are defined as cases and controls, and exposure histories are compared; subjects are also identified by outcome status Use Odds Ratio calculation
Cross-Sectional	Typically examines the relationship between exposure and disease prevalence in a defined population at a single point in time (vs. Longitudinal study → data collected repeatedly over time with the same group of study individuals)
Ecological	Examines the relationship between exposure and disease with population-level rather than individual-level data

- Case Report: a detailed report of the diagnosis, treatment, response to treatment, and follow-up after treatment of an individual patient; contains demographic information about the patient
- Case Series: a group of case reports involving patients who were given similar treatments; contains demographic information about the patient(s)
- Ecological Fallacy: a type of confounding specific to ecological studies that occurs when relationships which exist for groups are assumed to also be true for individuals
- Principle of Equipoise: a trial is ethical in as much as the researcher/clinical community is genuinely uncertain of the relative aggregate effectiveness of the treatments being tested → at the outset of the trial, no patient is knowingly disadvantaged by being randomized
- Efficacy: can treatment work under ideal circumstances?
- Effectiveness: can treatment work under ordinary circumstances?
- Equivalence: new treatment is not any better or any worse than the standard treatment (-M → +M)
- Superiority: new treatment is better than the standard treatment (0 → infinity)
- Non-inferiority: new treatment is not worse than the standard treatment (-M → infinity)

<u>Types of Clinical Trials</u>
- Treatment trials
    - Tests: new treatments, new combinations of drugs, new approaches to surgery or radiotherapy
- Screening trials
    - Identifies: best way to detect diseases, best way to detect health conditions
- Diagnostic trials
    - Identifies: better procedures for diagnosing diseases or conditions
- Prevention trials
    - Tests: alternative ways of preventing disease/recurring disease via medicines, immunizations, vitamins, and lifestyle changes
- Health economic evaluation trials
    - Compares: cost data between similar treatments
- Health-related quality of life trials

        - Tests: approaches aimed at improving comfort and quality of life via providing pain relief, improving patient functioning, and boosting well-being
- Phases
    - Phase 0: preclinical, no testing on human subjects
    - Phase I: evaluate safety, determine safe dosage, identify side effects
    - Phase II: test effectiveness, further evaluate safety
    - Phase III: confirm effectiveness, monitor side effects, compare to other treatments, collect information
    - Phase IV: provide additional information after approval including risk, benefits, and best use

Types of Experimental Designs
- Non-experimental: measures outcomes before and after program for participants only; no comparison group
- Quasi-experimental: measures outcomes for program participants and non-participants without random assignment; control for bias, comparison group
- Experimental/randomized controlled trials (RTC's): randomizes participants to treatment or control groups and measure outcomes for both groups; explicit comparison group
- Cross-over trial: patients are randomized into two different treatment groups and followed for a certain period of time followed by a washout period equivalent to the half-life of the treatment drugs, and then are crossed over to the other treatment (i.e., treatment A → washout → treatment B & treatment B → washout → treatment A)
- Blinding
    - Unblinded: all participants were aware of treatment assignment
    - Single-blind: only patients are unaware of treatment assignment
    - Double-blind: only patients, investigators, and outcome assessors are unaware of treatment assignment
    - Triple-blind: all participants (including data managers and biostatisticians) are unaware of treatment assignment

Summarizing Evidence & Meta-Analysis
- Systematic Review: high-level overview of primary research on a focused question that identifies, selects, synthesizes, and appraises all high-quality research evidence relevant to that question
- Literature Review: qualitatively summarizes evidence on a topic using informal or subjective methods to collect and interpret studies
- Hierarchy of Evidence (from highest reliability → lowest reliability)
    - Speed Knowledge Creation + Critical Appraisal
        - IPD Meta-Analysis
    - Critical Appraisal
        - Meta-Analysis
        - Systematic Review
    - Interventional Studies
        - Randomized Controlled Trial
        - Non-Randomized Controlled Trial
    - Observational Studies
        - Cohort Study
        - Case-Controlled Study
        - Case Reports
    - Animal Studies
    - In-Vitro Studies
- Formulating a research question: PICO
    - P) patient, population, or problem: how would you describe a group of patients similar to your own?
    - I) intervention, prognostic factor, or exposure: what do you plan to do for your patient/group?
    - C) comparison: what is the main alternative to compare with the intervention
    - O) outcome: what do you hope to accomplish, measure, improve, or affect?
- Forest Plot
    - Solid line: line of no effect
    - Dashed line: meta-analysis weighted mean
    - Pooled estimate: diamond
        - Left point of diamond: lower confidence interval
        - Right point of diamond: upper confidence interval
        - Size of diamond: weight (%)
- Fixed-Effects vs. Random Effects

        - A random-effects model assumes each study estimates a different underlying true effect, and these effects have a normal distribution (multiple true treatment effects). Fixed-effects model should be used only if it is reasonable to assume that all studies share the same, common effect (single true treatment effect)
    - Funnel Plot: used to determine the presence of publication bias
        - Center line: overall effect
            - Left of line: favors treatment
            - Right of line: favors control
        - Strong triangle shape → no publication bias
        - Weak triangle shape → potential publication bias

Outcomes & Causation
- Bradford Hill's Criteria of Causality
    - Strength: a strong effect size for an association
    - Consistency: repeated observations of an association in different populations/circumstances
    - Specificity: a risk factor leads to a single outcome; the weakest of the criteria & should be removed
    - Temporality: the cause should precede the effect; mandatory for all studies to be valid
    - Biological Gradient: a dose-response relationship
    - Plausibility: the hypothesis is biologically plausible
    - Coherence: the interpretation does not conflict with the known biology of the disease
    - Experimental Evidence: evidence from interventional research
    - Analogy: similar associations in other fields

Types of Bias
- Hawthorne effect: the phenomenon that subject behavior changes by the mere fact that they are being observed
- Lead-time bias: overestimation of survival duration due to earlier detection by screening than clinical presentation
- Length-time bias: overestimation of survival duration due to the relative excess of cases detected that are slowly progressing
- Overdiagnosis bias: occurs when diseases are diagnosed in individuals who would not have presented with clinical symptoms during their lifetimes in the absence of screening; for example, many people diagnosed with very small thyroid cancers are over diagnosed. While these tiny thyroid tumors may at first look like dangerous cancers under the microscope, almost all will never actually progress to cause harm
- Recall bias: a systematic error caused by differences in the accuracy or completeness of the recollections retrieved by study participants regarding events or experiences from the past
- Selection bias: bias introduced by the selection of individuals, groups, or data for analysis in such a way that proper randomization is not achieved, thereby failing to ensure that the sample obtained is representative of the population intended to be analyzed
- Publication bias: the potential for studies to favor one outcome/conclusion over another (i.e., favoring treatment over favoring placebo)
- Berkson's bias: a form of selection bias that causes hospital cases and controls in a case control study to be systematically different from one another because the combination of exposure to risk and occurrence of disease increases the likelihood of being admitted to the hospital. This produces a systematically higher exposure rate among hospital patients, so it distorts the odds ratio
- Healthy worker effect: a special type of selection bias, typically seen in observational studies of occupational exposures with improper choice of comparison group (usually general population); example → sample that has more medical screening or greater physical requirements for a job → typically results in better health status initially than the general population or certain control populations
- Healthy smoker effect: a type of selection bias that masks the true severity of the topic being studied; example → lung function in adolescent smokers > lung function in elderly non-smokers
- Pygmalion effect (i.e., self-fulfilling prophecy or observer expectancy bias): a foretelling of a subject's potential and performance brought about by a teacher's expectations; example → teacher's beliefs about their students → teachers action towards their students → students' beliefs about their own capabilities → students' actions in school and life → students' outcomes in school and life → teacher's beliefs about their students
- Confounding: a variable that influences both the dependent variable and independent variable, causing a spurious association; for example, a study looking at the association between obesity and heart disease might be confounded by age, diet, smoking status, and a variety of other risk factors that might be unevenly distributed between the groups being compared

Preventing Heart Disease & Diabetes Type II

- Heart Disease
    - Leading cause of morbidity and mortality globally
    - Contributes to over 30% of all disability cases and deterioration of the quality of life
    - Novel Risk Factors
        - Excess homocysteine in blood
        - Inflammatory markers (C-reactive protein)
        - Abnormal blood coagulation (elevated blood levels of fibrinogen)
    - Modifiable Risk Factors
        - High blood pressure
        - Abnormal blood lipids
        - Tobacco use
        - Physical inactivity
        - Obesity
        - Unhealthy diet
        - Diabetes mellitus
        - Heavy alcohol use
        - Lipoprotein A
- Diabetes Type II
    - Diagnosed if overnight fasting blood glucose level is > 126 mg/dL OR two hour oral glucose tolerance test (OGTT) level is > 200 mg/dL OR glycosylated hemoglobin (HbA1C) level is ≥ 6.5%
    - Risk Factors
        - Family background
        - History of gestational diabetes/macrosomia
        - Dyslipidemia (HDL-C < 35 mg/dL OR triglycerides > 250 mg/dL)
        - Overweight/obese BMI
        - Poor diet and physical inactivity
- Pre-Diabetes Type II (impaired fasting glucose - IFG/impaired glucose tolerance - IGT)
    - Diagnosed if the overnight fasting blood glucose level is between 100-125 mg/dL OR two hour oral glucose tolerance test (OGTT) level is between 140-199 mg/dL
    - After 10 years, 50% of patients reach full diabetes type II, 25% remain pre-diabetic, and 25% revert to normal
        - Lifestyle changes and medication have potential to increase the probability of reverting from IGT to normal glucose tolerance
- Note: obesity and physical inactivity are the most preventable risk factors for heart disease and diabetes type II, and could potentially lead to an over 50% reduction in prevalence of those diseases

Public Health

Health Systems: US Organization and Delivery
- Models of Organizing Care
    - Primary Care: common health problems, minor interventions, preventative measures; can be given by a general practitioner or another caregiver (accounts for 80-90% of health care visits)
    - Secondary Care: problems that require more specialized care and clinical expertise; usually involves hospital setting for care and specific specialties such as general surgery
    - Tertiary Care: more complex and often rare disorders that require extensive clinical expertise and hospital care; involves subspecialties such as cardiac surgeons, immunologists, hematologists, etc.
- The Dispersed Model: multiple access points, more fluid roles for providers, less distinction in hospital care, higher value on tertiary care
- Health Maintenance Organization (HMO)
    - You receive most or all of your healthcare from a network provider
    - You choose a primary care physician responsible for managing and coordinating care
    - Specialist care diagnostic services require an approval referral
- Independent Practice Associations (IPA)
    - A loose collection of private doctors who own their own practices
    - Contracts with HMO on behalf of the doctors
        - The IPA receives a capitation payment from the HMO and pays its doctors with through capitation or fee-for-service
        - Both usually involve fee-for-service referrals with bonus arrangements
- Preferred Provider Organization (PPO)
    - Emerged in the 1980s for Medicare patients
    - The PPO payer receives monthly premiums from subscribers and employers
    - Patients are required to select physicians and hospitals approved (preferred) by the payer

            - Providers discount their fees or allow payer to manage the care they give
- Accountable Care Organizations (ACO)
    - Affordable Care Act authorized Medicare to initiate ACO program; authorized to improve the safety and quality of care while reducing Medicare costs
        - Medicare Shared Savings Program includes 404 ACOs in 49 states
        - Consists of groups of doctors, hospitals, and other healthcare providers that work together to give coordinated, high-quality care to their Medicare patients
        - Goal: avoid unnecessary duplication of services and prevent medical errors
    - Private insurance and employers plan have also developed ACOs
        - 744 ACOs covering 24 million in 2015
        - Emphasis on regionalized, integrated care

<u>Basics of Healthcare</u>

- Affordable Care Act (ACA - 2010): comprehensive health care reform law with the following three goals
    - Make affordable health insurance available to more people
    - Expand medicaid to cover all adults with income below 133% of the federal poverty line (FPL) → not all states have expanded Medicaid
    - Key changes
        - No denial of coverage based on pre-existing conditions
            - For children <19 as of September 2010
            - For everyone as of 2014
        - No cancellation of coverage based on illness
        - No lifetime monetary caps on insurance coverage
        - Private health insurance plans required to include young adults up to age 26 under their parents policies
        - Individual mandate (tax penalty now eliminated)
- Terminology
    - Premiums: basic fees collected by the insurer on a monthly or an annual basis
        - Community rating: a health insurance mechanism to distribute health care based on human need versus ability to pay; insurer sets the premiums evenly based on the entire community of its subscribers (i.e., all individuals, regardless of risk, pay the same monthly premium)
            - Within each group: people who become ill receive benefits in excess of the premium they pay, while those who remain healthy receive fewer benefits than paid for
            - Among different groups: groups who use less health care than their premiums are worth help pay for those who use more health care than their premiums could buy
        - Experience rating: insurer sets premiums for each group based on the experience of each group's use of health services; it is less redistributive than community rating (i.e., monthly premiums are adjusted based on the relative risk of each individual)
            - Within each group: those who become ill subsidized by those who remain well
            - Among different groups: healthier groups do not subsidize high-risk groups
    - Deductible: amount of money to be paid out-of-pocket before the insurer will begin to pay
    - Co-insurance: percentage of costs of a covered health care service that one pays after they've paid their deductible
    - Co-payments: a fixed amount one pays for a covered health care service after they've paid their deductible
- General Coverage & Financing
    - U.S. Healthcare Coverage: government financing is responsible for the largest portion
    - Healthcare Financing: employment-based private insurance is responsible for the largest portion
- Medicare: federal insurance program
    - Part A: hospital coverage
        - Includes: hospitalization, skilled nursing facilities, home health care, hospice care
            - Does not cover unskilled (custodial) nursing home care
        - Eligibility
            - Americans ≥65 years old and eligible for social security
                - Retirement status is irrelevant
                - Must have paid social security for 10 years minimum
                - Covers spouse once they turn 65
            - Americans <65 years old with permanent disability

- Waiting period: must have received social security disability for 24 months
  - Exceptions: end-stage renal disease/dialysis, ALS
- Financing: social security tax system
- Part B: medical coverage
  - Includes: medical expenses, physical therapy, occupational therapy, speech therapy, physician services, medical equipment, diagnostic tests, preventative care
    - Does not cover eye refractions, hearing aids, dental services
  - Eligibility
    - People who are eligible for Medicare Part A & elect to pay the Medicare Part B monthly premium (variable for individuals based on income levels)
  - Financing: general federal revenues (personal income & other federal taxes) & payment of monthly premiums
- Part C: Medicare Advantage
  - Includes: bundled plan that incorporates services from Medicare Part(s) A + B, and often Part D as well
  - Financing: Medicare subsidizes the Medicare Advantage Plan's premium, rather than direct reimbursement to providers
  - Note: beneficiaries sacrifice some freedom of choice of healthcare providers for lower out-of-pocket payments
- Part D: prescription coverage
  - Includes: assistance for paying for prescription drugs (optional)
  - Eligibility
    - Purchase Part D once you already have either Part A or Part B
    - Purchase Part D once enrolled in Medicare Part C (Medicare Advantage Plan)
    - Purchase Part D as a standalone plan
  - Financing: through monthly premiums + yearly deductible
  - Donut hole: medicare stops paying for prescriptions once you reach your drug coverage limit
    - 1st stage: starts when the year begins and an individual reaches their deductible
    - Initial coverage stage: patient is only responsible for copayments → after the cost of drugs reaches $4,430, they fall into the donut hole
    - In the donut hole, patients pay 25% of the cost of generic and brand-name drugs until they reach $6,550
    - Catastrophic coverage kicks in when the patient is out of the gap and requires them to be responsible for 5% coinsurance of drug costs → lasts until the end of the year when their coverage ends and their plan restarts

- Medicaid: federal & state assistance program
  - Includes: inpatient & outpatient hospital services, physician services, laboratory and X-ray services, home health services
    - Optional benefits: prescription drugs and physical therapy
  - Eligibility: low-income families, qualified pregnant women, children, and those with disabilities
  - Financing: federal (50-76%), state, and local taxes
- Children's Health Insurance Program (CHIP): companion program to Medicaid
  - Includes: doctors visits, dental & vision, laboratory services and X-ray services, prescriptions, immunizations, routine checkups, inpatient & outpatient hospital care, emergency services
    - Routine doctor and dental visits → free
    - May have copayments
    - Some states charge a monthly premium
  - 1997: covers children in families with incomes ≤ 200% of the federal poverty level, but above the Medicaid income eligibility level
  - 2018: covers children in families with incomes ≤ 300% of the federal poverty level

<u>Reimbursing Healthcare Providers</u>

- Four Types of Physician Payments
  - Fee-for-Service: the physician is paid a fee for each office visit, ECG, IV fluid, or other service/supply provided; only form of payment based on the individual components of health care
    - Financial risk: the party paying the bill (insurance company, government agency, or patient) absorbs all the risk
  - Payment-per-Episode of Illness: single bundling of payments for multiple services (i.e., surgery); for example: surgeons have an economic incentive to limit the number of postoperative visits and perform more surgeries (similar to the fee-for-service system)

- Financial risk: bundling of services transfers a portion of the risk from the payer to the physician
- Capitation: monthly payments made to a physician for each patient signed up to receive care from them (usually for primary care physicians); requires patients to register with a physician/group of physicians
  - Financial risk: shifts financial risk from insurers to providers; methods to decrease risk include carve-outs and risk-adjusted capitation
    - Carve-outs: specific services not covered by capitation and paid separately as fee-for-service
    - Risk-adjusted capitation: patients with greater health needs need more time with their physician (thus higher monthly payments for elderly patients and those with chronic illness)
  - Two-tiered (UK): under the National Health Service (NHS), each patient enrolls with a PCP. For each patient, a PCP receives a monthly capitation payment (more patients = more money for the PCP). All non-emergency medical needs go through the PCP who makes referrals for specialist services and hospital care
  - Three-tiered: Health Maintenance Organizations (HMOs) do not pay capitation fees directly to individual physicians/small group practices → pay to an intermediary administrative structure (i.e., IPAs/Independent Practice Associations: physicians remain in their own private offices but join into physician groups)
    - Financial risk: borne by the IPA organization and split among its participating physicians
      - Capitation-plus-bonus payment: financial incentives IPAs give to PCPs to limit their use of diagnostic and specialist services → any surplus funds that remain at the end of the year
      - Issue: a conflict of interest for PCPs as their income increases by denying diagnostic and specialty services to patients
- Payment-per-Time (Salary): physicians in the public sector and in community clinics are usually paid by salary
  - Financial risk: physicians bear little if any financial risk → the employer is at financial risk; to manage risk, administrators may schedule their physicians to a high volume of patient visits → now physician is at risk
    - Without overtime pay, high volume of complex patient visits may turn an 8-hour day into a 12-hour day with no increase in income

- Four Types of Hospital Payments
  - Per Procedure: insurance companies make fee-for-service payments to private hospitals based on reasonable cost → hospitals influence payment amount
  - Per Diem (Day): bundles all services provided for a patient on one day into one payment; removes the hospital's financial incentive as it loses money by performing expensive studies
    - Financial risk: the insurer is at risk for the number of days a patient may have to stay in the hospital; the hospital is at risk for the number of services performed per day because it incurs more costs without additional payment
      - Insurers conduct utilization reviews to reduce the number of hospital stays (less concern with number of services performed/day)
  - Per Episode of Hospitalization: bundles together all services performed during one acute care hospital episode
    - Financial risk: the Medicare program is at risk for the number of admissions but not the length of stay as it pays the same amount per illness; the hospital is at risk for the length of hospital stay and the number of expensive procedures performed
      - Hospitals have started to use internal utilization reviews to reduce the costs of Medicare patients
  - Per Institution (Global Budget): a fixed payment made for all services performed on every patient for one year (used in VA and DOD hospitals in the US); the most extensive bundling of services
    - Financial risk: hospital is entirely at risk no matter the number of patients admitted and expensive services done → hospital must stay within its fixed budget
- Three Cost-Control Mechanisms in Healthcare Delivery Systems
  - Financing Controls: limits how much money flows from individuals and employers to health care plans
  - Reimbursement Controls: controls how much money flows from insurance plans to doctors/hospitals/etc.
  - Utilization Controls: restricts consumption by patients and avoids excessive care by physicians
- Payment Reform

- Value-Based Payment (VBP) and Payment Reform: the US is currently undergoing payment reform towards a value-based payment method, including eliminating the fee-for-service payment for payments that reward value rather than volume
    - Pay-for-Performance: paying for units of service and how well physicians/other providers deliver those services
        - Health payers give bonus payments to physicians/hospitals that achieve a specified high level of performance on certain measures → moving from volume-driven to value-driven

<u>Biomedical Informatics</u>

- Informatics: the science of information management and usage
    - Helps us use data, information, and knowledge improves areas of interest; it is interdisciplinary and it drives innovation
- Biomedical Informatics: uses data to help clinicians, researchers, and scientists improve human health and provide healthcare
    - Uses big data and works across disciplines to provide clinical insight
- Electronic Health Record (EHR) System: capture, store, and share comprehensive patient information across the entire continuum of care in health organizations; a digitized record system of all types of health information (diagnoses, immunization calendar, lab reports, imaging records, allergy records, demographic data, insurance records, etc.)
    - Easily accessible and can be shared with healthcare providers and practitioners across health organizations; clinical data can be shared on a real-time basis
        - The ability to integrate and share health data depends on interoperability, privacy, and security
    - Functional components
        - Integrated view of patient data: single longitudinal medical record of lab tests, medications, notes, etc.
        - Clinical Decision Support: computerized physician order entry (CPOE), reminders and alerts (transcription errors, drug-drug interactions, drug allergies), treatment and follow-up suggestions, identify patients' compliance and adherence
        - Access to knowledge resources: proactive presentation of short informational nuggets → pulls library resources relevant to a particular clinical situation
        - Integrating communication and reporting
- Health Level-7 (HL-7): data transmission and sharing standard; allows data to be exchanged between providers' systems
- International Classification Disease v10 (ICD-10): data definition standard; transforms descriptions of medical data into universal medical codes → describes diseases, morbidity, mortality, etc.; identifies health trends and statistics
- Health Information Exchange (HIE): allows various organizations across the entire continuum of care to share electronic health information locally, regionally, and nationally; the ultimate objective is to improve the safety, quality, and efficiency of healthcare
    - The organizations that support and govern HIE is called the Health Information Exchange Organization (HIEO) → Regional Health Information Organization (RHIO) & Statewide Health Information Network
- Benefits of biomedical informatics: reduction in medical errors, facilitation of patient record retrieval, improves accuracy for health insurance administration, improved protection of patient information

<u>Medical Errors</u>

- Error of Commission: doing something wrong (action)
- Error of Omission: failing to do something right (inaction)
- Adverse Drug Events (ADE)
    - Potential ADE: error that is intercepted before reaching the patient
    - Preventable ADE: error that reaches the patient and causes some degree of harm
    - Non-preventable ADE: adverse outcome even though medications are prescribed and administered appropriately
- Diagnostic Errors
    - No-fault errors: results from factors outside the control of the clinician or the health care system
        - Atypical disease presentation
        - Patient providing misleading information; uncooperative patient
    - Systems-related errors: the result of technical or organizational flaws
        - Inadequate communication and care coordination, inefficient processes, technical failures, and equipment problems

- Cognitive errors: diagnoses that are wrong, missed, or unintentionally delayed due to clinician error
- Diagnostic Cognitive Errors
  - Anchoring bias: when the clinician maintains initial impressions when making a diagnosis and becomes dismissive to signs and symptoms that point to another diagnosis
  - Confirmation bias: looking for evidence and interpretation of information to fit a preconceived diagnosis rather than the converse
  - Availability bias: more recent and cognitively available answers and solutions are preferentially favored because of ease of recall and incorrectly perceived importance
  - Diagnosis momentum: when the diagnosis considered by one clinician becomes the definitive diagnosis as it passes from one clinician to the next and becomes accepted without question by clinicians down the line
  - Framing effect: diagnostic decision-making unduly biased by extraneous and collateral information
- Treatment Medical Errors
  - Slips: actions not carried out as intended or planned
    - Example: injecting a medication intravenously when you meant to give it subcutaneously
  - Lapses: missed actions and omissions
    - Example: forgetting to monitor and replace serum potassium in a patient treated with furosemide (Lasix) for acute congestive heart failure
  - Mistakes: a wrong intended action, faulty plan, or incorrect intention
    - Example: extubating a patient prematurely based on misapplication of guidelines
- Not Medical Errors
  - Violations: deliberate actions whereby someone does something and knows it to be against the rules; even if the action was well-intentioned (quicker, easier, etc.), it is still considered a violation
    - Example: deliberately failing to follow proper procedures
- Outcomes of Medical Errors
  - Near-misses: errors that occur but do not result in injury or harm to patients because they are caught in time, or simply because of luck
  - Adverse events: harm or injury that results directly from the management of a patient's disease or condition by health care professionals rather than by the underlying disease or condition itself
    - Sentinel events: an adverse event in which death or permanent/severe temporary harm to a patient has occurred
    - Never event/serious reportable events: adverse events in medical care that are clearly identifiable, preventable, and serious in their consequences for patients, and that indicate a real problem in the safety and credibility of a healthcare facility
- Fatigue: the inability to continue effective performance, caused by excessive workload, stress, sleep loss, and circadian disruption resulting in suboptimal patient care and medical errors → can lead to burnout
- Burnout: the state of emotional exhaustion, cynicism, depersonalization, and decreased sense of personal accomplishment that can result in suboptimal patient care and medical errors
- Communication Improvement
  - SBAR: situation, background, assessment, recommendation
    - Situation
      - Who are you?
      - Who are you calling about?
      - What is happening with the patient?
    - Background
      - Why is the patient in the hospital?
      - What was previously done?
    - Assessment
      - What is your assessment based on available medical findings?
    - Recommendation
      - What do you recommend?
  - Call-out: inform all team members simultaneously during team events and emergent situations
  - Read-back: closed-loop communication where receiver repeats message and sender confirms it is correct
  - Critical language: assertive structured communication that provides key uniformly understood words
- Reporting
  - Near misses: not disclosed to patients or families; however, should be reported to the hospital so that the error can be studied to learn how to prevent it in the future
  - Adverse & Sentinel events: should be disclosed to patient and family, and reported to the hospital
- James Reason's Swiss-Cheese Model: used to put up as many barriers to error as possible to reduce the possibility of negative events occurring

- Latent factors → error-producing factors → active failures → defenses
    - Latent factors: errors in a system or process design; present but may go unnoticed for a long time with no ill effect (i.e., faulty equipment, lack of staff training, ineffective organizational structure → faulty IV pump)
    - Active failures: errors taking place between a person and an aspect of a larger system at the point of contact (i.e., slips, mistakes → wrong IV pump dose programmed)
- Analyzing Medical Errors
  - Root Cause Analysis (RCA): a retrospective approach used to identify the causative factors that underlie variations in performance; first done to analyze and identify the underlying causes and potential interventions
    - Typically evaluates sentinel events that result in physical or psychological injury or death
    - Identifies the cause and improves the systems and processes to decrease the odds of a repeat event
  - Fishbone (Ishikawa) Cause-and-Effect Diagram: identifies system causes of error and develops action plans that include strategies to reduce the risk of future similar events
  - Failure Mode and Effects Analysis (FMEA): a systematic and proactive process of anticipating failures, determining the impact of those failures and determining the likelihood of that failure being detected before it occurs
  - Risk Priority Number Assignment: a numeric assessment of risk assigned to a process, or steps in a process
    - RPN = severity of effect X probability of occurrence of cause X probability of detection
  - Human Factors in Design (Ergonomics): application of psychological and physiological principles to the engineering and design of products, processes, and systems to reduce human error, increase productivity, and enhance safety and comfort with a specific focus on the interaction between the human and the thing of interest
    - Forcing functions: prevents the user from taking an action without consciously considering information relevant to that action
    - Standardization: implementation and use of guidelines/checklists
    - Simplification: reduces wasteful activities, increases efficiency, reduces error-prone situations → EMR's
  - Plan-Do-Study-Act (PDSA) Cycle: four step cycle that allows one to quickly implement relatively small-scale change, solve problems, and iteratively or continuously improve processes
    - Plan: define problem and plan a solution
    - Do: test new process
    - Study: measure and analyze data
    - Act: integrate new process into regular workflow
  - Six-Sigma Model: a data-driven approach to solve problems requiring large-scale transformation
    - Define the problem in detail, measures defects, analyze conditions under which defects occur, improve current process to reduce defects, control processes to maintain performance

<u>Vaccinations</u>

- Herd Immunity
  - Requirements
    - Disease agent must be restricted to a single host species within which transmission occurs
    - Transmission must be relatively direct from one member of the host species to another
    - Operates optimally when there is random mixing in the population or community
    - Infections must induce solid immunity
  - Works when:
    - Probability of an infected person encountering every other individual (i.e., random mixing) is the same
    - Critical percentage of the population is immune
- Vaccination Recommendations
  - Healthcare Workers
    - Hepatitis B, influenza (annually), MMR (if no evidence of immunity), varicella, Tdap (tetanus, diphtheria, pertussis), meningococcal (for those routinely exposed to N. meningitidis)
  - Travelers
    - Japanese encephalitis, meningococcal, rabies, typhoid, yellow fever
- Poliomyelitis Vaccine
  - Inactivated Polio Vaccine (IPV): can be used for immunocompromised individuals

- Live Attenuated Oral Polio Vaccine (OPV): longer-lasting immunity than IPV
    - Live attenuated vaccines induce both humoral and cell-mediated immunity and usually requires fewer doses to achieve strong immunity
- Vaccine Schedule
    - Hepatitis B
        - Dose 1 → birth
        - Dose 2 → 1-2 months
        - Dose 3 → 6-18 months
    - DTaP
        - Dose 1 → 2 months
        - Dose 2 → 4 months
        - Dose 3 → 6 months
        - Dose 4 → 15-18 months
        - Dose 5 → 4-6 years
    - Tdap
        - Dose 1 → 11-12 years
            - Follow-up: Td/Tdap booster every 10 years
    - MMR
        - Dose 1 → $\geq$ 19 years
        - Dose 2 → $\geq$ 19 years
- Types of Vaccines
    - Live Attenuated
        - Tuberculosis, oral polio vaccine (OPV), measles, rotavirus, yellow fever, varicella-zoster virus, seasonal influenza (nasal spray)
    - Inactivated
        - Whole-cell pertussis, inactivated polio vaccine (IPV), hepatitis A, seasonal influenza (injectable)
    - Subunit
        - Acellular pertussis, haemophilus influenzae type b, pneumococcal, hepatitis B, human papillomavirus
    - Toxoid
        - Tetanus toxin, diphtheria toxoid
    - Viral vector
        - Zaire Ebola virus
- Notes
    - DTaP (diphtheria, tetanus, acellular pertussis) → children $<$ 7 years old
    - Tdap (tetanus, diphtheria, pertussis) → children $\geq$ 7 years old
    - Hepatitis B vaccine is given to newborns before being discharged from the hospital
    - The U.S. Preventive Services Task Force (USPSTF) uses an evidence-based approach to reviewing literature and developing recommendations

Microbiology

Viruses

- General Overview
  - Viruses are non-cellular obligate intracellular parasitic organisms that can infect any living organism and consist of either DNA or RNA that is surrounded by a protein coat; requires a host cell for replication
  - They are invisible via light microscopy; range in size from 20-400 nm
  - Individual viral components self-assemble into a virion (a complete infectious form of a virus outside of the host cell → extracellular)
- Structural Components
  - Genome: nucleic acids → either DNA or RNA
  - Capsomere: viral protein subunits that assemble into a capsid
  - Nucleocapsid: genome assembled into the capsid
  - Virus-specific glycoproteins
  - Envelope: outer lipoprotein layer that originates from the host membranes and covers the capsid to maintain aqueous solution; poor in host cell proteins but is rich in virus-specific glycoproteins → viral attachment protein (VAP)
    - Forms
      - Cell plasma membrane: HIV, most of the enveloped viruses
      - Nuclear or other internal membranes → endoplasmic reticulum: Herpesvirus
    - Tegument: cluster of proteins that line the space between the envelope and the nucleocapsid
    - Enveloped Viruses: viral nucleocapsid is surrounded by a lipoprotein membrane derived from the host cell membrane (i.e., Measles), nuclear membrane (i.e., Herpes), or other internal membranes and contains virus-specific glycoproteins
      - Sensitive to inactivation by organic solvents (i.e., alcohol, chloroform, ether, etc.), detergents, drying, acid, and heat
      - Transmitted by secretions, large droplets, blood, or sexual contact (i.e. Measles)
      - Typically spherical or pleomorphic in shape except Rhabdovirus (Rabies → bullet-shaped) and Poxvirus (Smallpox → complex)
    - Non-Enveloped Viruses: viral nucleocapsid is naked
      - Tough and relatively resistant to inactivation by organic solvents (i.e., alcohol, chloroform, ether, etc.), detergents, drying, acid, and heat
      - Transmitted by fecal/oral route, fomites, or small droplets
      - Released from infected cells by lysis (i.e. Poliovirus, Adenovirus)
  - Viral Proteins
    - Viral attachment proteins (VAP) or peplomers: facilitate host cell entry
      - Hemagglutinin (HA): binds erythrocytes on Influenza
      - VAP: binds C3d receptor (CR2) on B-cells on the Epstein-Barr Virus
    - Viral polymerase: present within the genome of viruses and is involved in the transcription and replication of the viral genome
    - Matrix proteins: present between the nucleocapsid and the envelope; stabilizes the organization of viral glycoproteins, directs the viral genome to intracellular sites of viral assembly, and facilitates virus assembly and budding
    - Note: depending on the virus and its expression pattern, viral proteins are divided into immediate early phase proteins, early phase proteins, and late phase proteins
  - Capsid: protects the nucleic acid genome that is held together by non-covalent, reversible hydrophobic or hydrogen bonds; consists of single or several different subunits (capsomeres)
    - Forms
      - Icosahedral (spherical)
        - Capsids form independently of the genome
        - Few proteins assemble into basic protomers → protomers assemble into pentamers
        - Complex 5-3-2 axes of symmetry (i.e., Poliovirus)
      - Helical (rod-like)
        - Capsomers vind to the viral genome in a regular fashion
        - Capsids are open-ended and form around the genome (no empty helical capsids may form)
        - Note: all known examples of animal viruses with helical symmetry contain RNA genomes and have flexible nucleocapsids wound into a ball and is surrounded by an envelope - except Rhabdovirus
      - Complex (non-symmetrical)

                - Bacteriophages or phages exhibit complex symmetry (i.e., Poxvirus → Vaccinia Virus)
                    - Head: nucleic acid and protein
                    - Tail and contractive sheath
                    - Tail fibers and tail pins
                    - Base/end plate
            - Other Forms
                - Cone-shaped capsid structure: Retrovirus (i.e., HIV)
                - 3-layer capsid with 11 dsDNA segments: Rotavirus
        - Functions
            - Defines tissue or species-specific transmission by interaction with host receptors to facilitate the host cell entry
            - Interacts with the viral nucleic acid for packaging/assembly
            - Assists in viral and/or host gene regulation
            - Evades/blocks host immune system
- Viral Nomenclature and Classification
    - Order → family (*-idae*) → subfamily (*-inae*) → genus (-virus) → species → [isolates, strains]
    - Group Classification
        - Group I (DNA +/-): dsDNA viruses
            - Adenovirus, Herpesvirus, Poxvirus
            - DNA +/- → mRNA → proteins
            - Properties
                - Non-enveloped or enveloped
                - Replication occurs in the nucleus (except Poxvirus → cytoplasm), early proteins take over the cell command and target genome replication (like viral DNA polymerase), late proteins are structural and translocate back into the nucleus to form a capsid
        - Group II (DNA +): ssDNA viruses
            - Parvovirus
            - DNA + → DNA +/- → mRNA → proteins
            - Properties
                - Non-enveloped
                - Replication occurs in the nucleus
        - Group III (RNA +/-): dsRNA viruses
            - Reovirus, Rotavirus
            - RNA +/- → mRNA → proteins
            - Properties
                - Virus entry into target cells leads to loss of VP4 and VP7 proteins from the triple-layered particle (TLP) resulting in the formation of a double-layered particle (DLP)
                - Ends with the re-formation of DLP which will acquire VP4 and VP7 proteins to form the TLP that is ready for release
        - Group IV (RNA +): ssRNA viruses
            - Picornavirus, Togavirus, Poliovirus
            - RNA + → mRNA → proteins
            - RNA + → RNA - → mRNA → proteins
            - Properties
                - Non-enveloped or enveloped
                - Replication occurs in the cytoplasm; positive sense ssRNA genome does not have a CAP but contains an IRES element
        - Group V (RNA -): ssRNA viruses
            - Orthomyxovirus, Rhabdovirus, Influenza, Hepatitis D Virus
            - RNA - → mRNA → proteins
            - Properties
                - Enveloped
                - Non-segmented (Rhabdovirus, Paramyxovirus) or segmented (Orthomyxovirus, Arenavirus, Bunyavirus)
        - Group VI (RNA +): ssRNA reverse transcription viruses (RNA with DNA intermediate)
            - Retrovirus (i.e., HIV)
            - RNA + → reverse transcription → DNA +/- → mRNA → proteins
            - Properties
                - Enveloped

- Replication occurs in the nucleus and requires RNA-dependent DNA polymerase (reverse transcriptase)
            - Viral genome becomes permanently integrated randomly into the host genome → provirus (similar to bacteriophages without the excision part; can be tumorigenic → 8% of human genome is of retroviral origin)
    - Group VII (DNA +/-): dsDNA reverse transcription viruses (DNA with RNA intermediate)
        - Hepadnavirus, Hepatitis B Virus
        - DNA +/- → RNA + → reverse transcription → DNA +/- → mRNA → proteins
        - Properties
            - Enveloped
            - The pregenomic (positive sense) ssRNA intermediate is first encapsidated prior to reverse transcription by RNA-dependent DNA polymerase (reverse transcriptase)
        - Example: Hepatitis B Virus → may integrate into the host genome and cause chronic infection leading to hepatocellular carcinoma (HCC)
- General Classification
    - DNA Viruses
        - Enveloped
            - Double-stranded linear: Herpesvirus, Poxvirus
            - Double-stranded circular: Hepadnavirus
        - Non-Enveloped
            - Single-stranded linear: Parvovirus
            - Double-stranded linear: Adenovirus
            - Double-stranded circular: Papillomavirus, Polyomavirus
    - RNA Viruses
        - Enveloped
            - Single-stranded positive sense: Togavirus, Flavivirus, Coronavirus
            - Single-stranded negative sense
                - Linear: Rhabdovirus, Paramyxovirus, Filovirus, Bornavirus
                - Segmented: Arenavirus, Bunyavirus, Orthomyxovirus
            - Retroviruses: Lentivirus, Oncovirus
        - Non-Enveloped
            - Single-stranded positive sense: Astrovirus, Calicivirus, Picornavirus
            - Double-stranded: Reovirus, Rotavirus

- Viral Properties
    - General Properties
        - Parvovirus: requires cells undergoing DNA synthesis to replicate
        - Papovavirus: stimulates cell growth and DNA synthesis
        - Hepadnavirus: stimulates cell growth, cell makes RNA intermediate, encodes a reverse transcriptase
        - Adenovirus: stimulates cellular DNA synthesis and encodes its own polymerase
        - Herpesvirus: stimulates cell growth, encodes its own polymerase and enzymes to provide deoxyribonucleotides for DNA synthesis, establishes latent infection in the host
        - Poxvirus: encodes its own polymerases and enzymes to provide deoxyribonucleotides for DNA synthesis, replication machinery, and transcription machinery in the cytoplasm
    - Properties of RNA Viruses
        - Picornavirus, Togavirus, Flavivirus, Calicivirus, Coronavirus
            - Positive RNA genome resembles mRNA and is translated into a polyprotein which is proteolyzed. A negative RNA template is used for replication. For Togavirus, Coronavirus, and Calicivirus, early proteins are transcribed from the genome and late proteins are transcribed from the template
        - Orthomyxovirus, Paramyxovirus, Rhabdovirus, Filovirus, Bunyavirus
            - Negative RNA genome is a template for individual mRNAs, but full-length positive RNA template is required for replication. Orthomyxovirus replicate and transcribe in the nucleus and each segment of the genome encodes one mRNA and template
        - Reovirus
            - Positive/negative segmented RNA genome is a template for mRNA
                - Positive RNA may also be encapsulated to generate the positive/negative RNA → more mRNA
        - Retrovirus

            - Positive retrovirus RNA genome is converted into DNA which is integrated into the host chromatin and is transcribed as a cellular gene
- Viral Synthesis and Growth
    - Initial infection is followed by a disappearance of all viral particles (eclipse phase)
    - Viral genome takes over the control of the host cell and directs production of early enzymes, nucleic acids, and protein coats as a part of the eclipse phase during the latent period
        - Note: latent period is different from latent infection
    - Once the new viral particles are assembled, the eclipse phase ends and the maturation phase begins which deals with assembly and release
- Stages of Viral Replication
    - General Principles
        - Replication is an error-prone process; DNA viruses are inherently more stable than RNA viruses
        - Recombination: occurs between two viruses that are infecting the same host; very infrequent (i.e., HSV-1, HSV-2)
        - Complementation: occurs when one or two viruses which are simultaneously infecting the host cell have defective functions but at different gene locations; both viruses can successfully propagate because of complementing and rescuing each others' defective functions (i.e., relationship between HBV and HDV)
        - Antigenic drift: evolution due to small changes in the genome sequence within a defined strain which usually occurs at a slower rate compared to antigenic shift (i.e., Influenza, Rhinovirus)
        - Antigenic shift: occurs in segmented viruses (i.e., Influenza) which results in exchanges (reassortment) of RNA segments between two different viruses that subsequently give rise to a highly pathogenic strain
        - Phenotypic mixing: association of a genotype with a heterologous phenotype
            - Transcapsidation: non-genetic interaction in which viral particles are released from a cell that is infected with two different viruses and has components from both viruses, but only has a genome from one of the viruses (i.e., Poliovirus, Myxovirus)
            - Pseudotypes: non-genetic interaction in which the nucleocapsid of one virus acquires the envelope from another type of virus
    - Attachment to specific host cell receptors → this binding determines what cells can be infected (tropism); requires viral attachment protein (VAP) and cellular receptors

Common Viral Attachment Proteins & Receptors

Virus	Viral Attachment Protein (VAP)	Target Cell	Viral Receptor
Rhinovirus	VP1-VP2-VP3 Complex	Epithelial Cells	ICAM-1
Influenza A	HA gp	Epithelial Cells	Sialic Acid
HIV	gp 120	Helper T-Cells	CD4+ Chemokine Receptors

        - Tropism: might be limited to a single organ, tissue, specialized cell type, or range of different organs and tissues
            - Susceptibility: viral glycoproteins (VAP) integrated into the outer coat → either the capsid or the envelope that target receptors are acting as doors on the surface of the host cells
            - Presence of transcription factors allows expression of viral genes
            - Permissivity: presence of cell enzyme pathways to produce viral proteins
    - Penetration
        - Enveloped viruses: fusion
            - Viral glycoproteins attach to host cell receptors → envelope-membrane fusion occurs → capsid enters → capsid is uncoated and virus is released (i.e., Herpesvirus, Paramyxovirus, HIV)
        - Non-enveloped viruses: direct entry across plasma membrane
            - Virus attaches to host cell receptors → sinks into cell membrane → injects its genome through a pore into the cell (i.e., Poliovirus)
    - Uncoating: release of nucleic acids

- Enveloped viruses: endocytosis & acidification
    - Host cell cytoplasmic membrane wraps around virus and brings it inside → capsid is uncoated and the viral genome is released into the host cell (i.e., Influenza)
- Non-enveloped viruses: endocytosis
    - Host cell cytoplasmic membrane wraps around virus and brings it inside → capsid is uncoated and the viral genome is released into the host cell (i.e., Parvovirus)
- Macromolecular synthesis
  - Early mRNA and protein synthesis: proteins shut off host cell and replicate viral genome
    - Early proteins → includes synthesis of viral DNA or RNA polymerase and other proteins which play an important role in viral replication
  - Replication of the genome: leads to complementary strand synthesis and additional templates using nucleic acids
  - Late mRNA and protein synthesis: structural proteins
    - Late proteins → structural proteins which participate in the formation of the viral capsomeres (some viruses use chaperone proteins to assist in capsomere folding)
- Post-translational modification of proteins
- Assembly of new virus particles
- Release: lysis of the cell or budding out
  - Exocytosis: causes the viral capsid to grab cellular membrane in a form of an envelope which is laced with viral proteins → most commonly observed in enveloped viruses
  - Cell lyses: most commonly observed in non-enveloped viruses
- Bacteriophage Life Cycle
  - Virulent (lytic) phages: kills the host following infection
  - Lysogenic (temperate) phages: undergo lysogeny wherein the host is not immediately killed and the phage genome becomes a prophage (provirus) either by integration into the host chromosome or exists as an independent entity but replicating with the rate equal to the host genome multiplication
- Consequences of Viral Infection
  - Viral cell → transformation into tumor cell → tumor cell division → transformation of normal cells to tumor cells
  - Viral cell → viral multiplication → lytic infection (death of cell and release of virus)
  - Viral cell → viral multiplication → persistent infection (slow release of virus without cell death)
  - Viral cell → viral multiplication → latent infection (virus present but not causing harm to cell; later emerges in lytic infection)
- Viral Genome
  - Convergence point: all viruses must go through mRNA positive strand synthesis to produce proteins
    - Positive strand nucleic acid (DNA or RNA) is the gene coding strand (5' → 3')
    - Negative strand nucleic acid (DNA or RNA) is the complementary strand (3' → 5')
      - Transcription is done off of the negative strand nucleic acid to create the positive mRNA strand (5' → 3')
      - Translation is done off of the positive mRNA strand
  - All viruses are haploid except Retrovirus (i.e., HIV)
  - DNA-based genomes are always presented by a single molecule, either dsDNA or ssDNA
  - RNA-based genomes can be presented by either single molecules (non-segmented) or several molecules (segmented)
    - Examples:
      - dsRNA: Reovirus (segmented)
      - ssRNA: Retrovirus (positive), Paramyxovirus (non-segmented; negative), Orthomyxovirus (segmented; negative), Arenavirus (Group V negative; ambisense genome), Bunyavirus (Group V negative; ambisense genome)
  - Specific viral enzymes
    - RNA-dependent DNA polymerase (reverse transcriptase)
    - RNA-dependent RNA polymerase (RNA replicase)
  - Viral mRNA components
    - 5'-5' N^7-methylguanosine-triphosphate CAP
    - Poly-A tail (100-200 adenosine residues)
    - Note: when viruses replicate in the cytoplasm, they can either 1) make their own 5'-CAP or 2) possess a 3D RNA structure known as an internal ribosomal entry site element (IRES)

- Three methods of viral protein assembly
    - Transcription of individual monocistronic mRNA molecules from the genome
    - Segmented genome where each molecule gives single monocistronic mRNA
    - Production of a single long polyprotein that is later cleaved into individual functional peptides
    - Note: some viral genomes can be transcribed by shifting the reading frame with one or two nucleotides and thus produce different mRNA and proteins (i.e., Rous Sarcoma Virus → encodes Gag and Pol proteins that overlap in a -1 reading frame)
- Atypical Viruses and Virus-Like Agents
    - Defective (satellite) viruses: cannot replicate without a helper virus (up to x100 more defective viruses are produced than normal due to genome mutations)
        - Hepatitis D Virus (HDV) is a defective virus that replicates in the nucleus and that can only infect in the presence of Hepatitis B Virus (HBV) → helper virus. HDV utilizes HBV surface antigens/envelope proteins (HBsAg) for entry into host cells and produces delta antigens (HDsAg). The HDV genome is translocated into the nucleolus where it is transcribed by cellular RNA polymerases and the mRNA is translated in the cytoplasm. The HDV antigen is then translocated into the nucleolus and then the HDV nucleocapsid associates with the HBsAg in the cytoplasm during assembly and is released from infected cells
    - Viroids: infectious agents that contain RNA with some double-stranded regions → associated with certain plant diseases
    - Prions: infectious agents that are devoid of any nucleic acids (DNA or RNA) → composed solely of proteins that are associated with diseases in humans and animals; can be acquired through diet, transfusion, surgical procedures, corneal transplants, etc.
        - Hereditary: autosomal dominant mutation of PrP on chromosome 20
        - Spontaneous: about 1/1,000,000 people usually late in life, or 1-2 million infected worldwide
            - Conversion from the normal/innocuous protein (PrPc) to the harmful/disease-causing protein (PrPSc); this conversion proceeds via a chain reaction
                - Forms long, filamentous aggregates that gradually damage neuronal tissue (vacuolization within neurons and neuropils)
        - Transmissible Spongiform Encephalopathies: characterized by loss of motor control, dementia, paralysis, encephalitis, and widespread neuronal loss

<u>Parasites</u>
- General Features
    - Protozoan Parasites
        - Unicellular organisms that are physiologically simple; acquires nutrition via pinocytosis or phagocytosis and are typically facultative anaerobes
    - Metazoan Parasites
        - Multicellular organisms that have digestive, circulatory, nervous, excretory, and reproductive systems; possesses bilateral symmetry (head and tail) and has tissue differentiation (endo/meso/ectoderm); spends most/all of their life in the host and are generally classified as Helminthes. Acquire nutrition from the host by active ingestion or passive absorption; reproduce sexually and/or asexually (no multiplication of adult form in humans) → produces > 200,000 eggs/day
        - Nemathelminthes (roundworms)
            - Class → Nematoda
                - Cylindrical bodies that are covered with a cuticle, oral opening may have hooks/suckers to attach to mucosal epithelium
                - Classifications
                    - Intestinal: *Ascaris* (roundworm), *Enterobius* (pinworm), *Trichuris* (whipworm), *Ancylostoma* (hookworm), *Necator* (hookworm), *Strongyloides* (threadworm), *Trichinella* (pork roundworm), *Toxocara* (dog/cat roundworm)
                    - Tissue (lymphatic filariae): *Wuchereria*, *Brugia*
                    - Tissue (cutaneous filariae): *Onchocerca*, *Loa loa*, *Dracunculus*
        - Platyhelminthes (flatworms)
            - Class → Trematoda (flukes)
                - Have a dead-end digestive system where nutrients are taken in, digested, and then expelled through the same opening; life cycles involve more than one host

- Classifications
  - Intestinal: *Fasciolopsis*
  - Lungs: *Paragonimus*
  - Liver: *Fasciola*, *Clonorchis* (Chinese liver fluke)
- Class → Cestoda (tapeworms)
  - Simple anatomy from the flat segments (proglottids) that develop sequentially from the head (scolex); scolex contains hooks/suckers to attach to the intestinal epithelium; has no digestive tract but absorbs nutrients directly
  - Hermaphroditic (each proglottid produces sperm and egg and engages in self-fertilization); humans are the definitive hosts (when they are intermediate hosts, more serious diseases occur)
  - Classifications
    - Intestinal: *Taenia*
    - Systemic: *Echinococcus*

- Terminology
  - Medical parasitology: the study of the parasites of man and their medical consequences
  - Parasite: living organism requiring intimate prolonged contact with another living organism to meet some of its basic nutritional needs
  - Host: organism harboring a parasite
  - Definitive host: host harboring a parasite in the adult/sexually mature stage
  - Intermediate/transient host: host harboring a parasite in which development occurs but in which adult/sexually mature stages are not reached
  - Reservoir host: host harboring a parasite that can be transmitted to another host to cause infection
  - Carrier: host harboring a parasite but exhibits no clinical signs or symptoms
  - Life cycle: the sequence of morphologic and environmental stages in the lifespan of a parasite
  - Parasitic infection: invasion by endoparasites (i.e., protozoa and helminths)
  - Parasitic disease: invasion and pathology produced by endoparasites (i.e., protozoa and helminths)
  - Parasitic infestation: external parasitism by ectoparasites (i.e., arthropods)
  - Commensalisms: the association of two different species or organisms in which one is benefited and the other is neither benefited nor harmed (i.e., non-pathogenic intestinal protozoa)
  - Vector: a living carrier that transports a parasite from an infected to a non-infected host
  - Zoonosis: disease involving a parasite for which the normal host is an animal but humans are able to be infected
- Life Cycles
  - Protozoa
    - Trophozoite → motile, metabolically active, multiplies by replication
    - Cyst/Oocyst → non-motile, passed in feces, resistant to hostile environment, does not multiply
    - Sexual multiplication → zygote
    - Asexual → schizonts, sporogons
  - Helminths
    - Egg → larval stage(s) → adult → eggs
- Infective Forms
  - Intestinal (fecal-oral)
    - Cysts, Trophozoite, Oocyst
  - Urogenital (sexual transmission)
    - Trophozoite
  - Blood & Tissue (insect vectors)
    - Cyst, Trophozoite, Oocyst, Pyriform Body, Promastigote, Trypomastigote
  - Nematodes
    - Egg, Filariform Larvae, 3rd-stage Larvae
  - Trematodes
    - Cercaria, Metacercaria
  - Cestodes
    - Proglottid, Embryonated Egg, Cysticercus, Cysticercoid
- Transmission Routes
  - Contact and Penetration of Eyes
    - *Acanthamoeba* → multiple
      - *Acanthomoeba*: ubiquitous amoeba of soil and water; often contaminates fluid used to treat contact lenses but infection may also come from dust or swimming

→ infection of the eye causes keratitis which results in eye pain, redness, tearing, and potential vision loss

- Inhalation
  - *Acanthamoeba* → multiple
  - *Enterobius* → multiple
    - *Enterobius*: pinworm; transmitted via fecal-oral route and is prevalent worldwide but is mainly found in children
  - *Naegleria*
    - *Spp. fowleri*: found in freshwater bodies such as lakes and streams; invasion from nasal mucosa into the brain and causes rapidly-progressing primary acute meningoencephalitis
- Fecal-Oral, Ingestion
  - *Ascaris*
    - *Spp. lumbricoides*: common roundworm → lives in small intestine; infects > 25% of the human population via fecal-oral transmission; co-occurs with 47 other species of helminths or protozoa (co-infection can enhance morbidity for other infectious diseases)
    - Ascariasis: mostly asymptomatic; infects the lungs (larvae) and the intestines (adult worm)
      - Lungs: asthma or pneumonia-like; cough, shortness of breath, wheezing
      - Intestines: diarrhea/bloody stools, nausea, vomiting, abdominal pain; severe infections → malnutrition, weight loss
  - *Balantidium*
    - *Spp. coli*: ciliate protozoan, lives in the colon of pigs, humans, and rodents; can lead to colonic ulceration & reproduces asexually
  - *Clonorchis*
    - *Spp. sinensis*: Chinese liver fluke; acquired by ingestion of infective metacercariae in raw, pickled, smoked fish
  - *Cryptosporidium*
    - *Spp. parvum*: non-motile; more prevalent in immunocompromised patients & reproduces asexually and sexually
  - *Cyclospora*
    - *Spp. cyatenensis*: non-motile; parasitizes the small intestinal mucosa and may cause diarrhea for several weeks & reproduces asexually and sexually
  - *Echinococcus*
    - *Spp. granulosus*: dog tapeworm; hydatid disease occurs when the larval stages of these organisms are ingested; larvae may develop in the human host and cause lesions in several organs
    - *Spp. multilocularis*: rodent tapeworm; hydatid disease occurs when the larval stages of these organisms are ingested; larvae may develop in the human host and cause lesions in several organs
  - *Entamoeba* → multiple
    - *Spp. histolytica*: uses pseudopodia for locomotion, may invade the colon and cause bloody diarrhea → amoebic dysentery; also causes amoebic liver abscess & reproduces asexually
  - *Enterobius* → multiple
  - *Fasciola* → multiple
    - *Spp. hepatica*: a parasite of sheep that infects humans when they ingest freshwater plants; parasite lives in the intrahepatic bile ducts of the liver; fascioliasis → severe anemia
  - *Fasciolopsis*
    - *Spp. buski*: parasite of humans and pigs that lives in the upper intestine; chronic infection causes inflammation, ulceration, and small intestine hemorrhages
  - *Giardia* → multiple
    - *Spp. lamblia*: flagellate protozoan with world-wide distribution, lives in the small intestine and results in malabsorption & reproduces asexually
  - *Taenia* → multiple
    - *Spp. saginata*: beef tapeworm; causes cysticercosis worldwide; acquired by ingestion of uncooked beef that contains cysticerci
    - *Spp. solium*: pork tapeworm; causes cysticercosis worldwide; acquired by ingestion of uncooked pork that contains cysticerci; can cause neurocysticercosis
  - *Toxocara*

- *Toxocara*: worldwide infection of dogs and cats; human infection occurs when embryonated eggs are ingested from dog or cat feces present in the soil; it is common in children and can cause visceral larva migrans (VLM)
    - *Toxoplasma*
    - *Trichinella*
        - *Trichinella*: pork roundworm; associated with consumption of infected and unfrozen or poorly-cooked meat; encysted larvae occur in striated muscle and symptoms occur based on location
    - *Trichuris*
        - *Trichuris*: whipworm; prevalent in warm and humid conditions; can cause diarrhea, rectal prolapse, and anemia in heavily-infected individuals
- Sexual Contact
    - *Entamoeba* → multiple
    - *Giardia* → multiple
    - *Trichomonas*
        - *Spp. vaginalis*: flagellate, urogenital parasite; sexually transmitted & reproduces asexually
- Contact and Penetration of Skin
    - *Ancylostoma*
        - *Ancylostoma*: hookworm; major cause of anemia in the tropics
    - *Necator*
        - *Necator*: hookworm; major cause of anemia in the tropics
    - *Schistosoma*
    - *Strongyloides*
        - *Strongyloides*: threadworm; inhabits the small bowel; infection is more severe in immunocompromised individuals (i.e., HIV/AIDS, malnutrition, intercurrent disease)
- Vector-Borne
    - Black Fly → *Onchocerca*
    - Kissing Bug → *Trypanosoma* → multiple
        - *Trypanosoma*: flagellate protozoan; transmitted by Reduviid/Triatomine "kissing bug" → causative agent of Chagas Disease
    - Mosquito → *Plasmodium*, *Wuchereria*, *Brugia*, *Onchocerca*, *Loa*
        - *Plasmodium*: non-motile; transmitted by Anopheles mosquito → causative agent of Malaria; four species infect humans → *P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae*
        - *Wuchereria bancrofti*: microfilarial nematodes transmitted by mosquitoes and flies → causative agents of lymphatic filariasis
        - *Brugia malayi*: microfilarial nematodes transmitted by mosquitoes and flies → causative agents of lymphatic filariasis
        - *Onchocerca volvulus*: microfilarial nematodes transmitted by mosquitoes and flies → causes visual impairment, blindness, and severe itching of the skin in infected individuals
        - *Loa loa*: microfilarial nematodes transmitted by mosquitoes and flies → causes visual impairment, blindness, and severe itching of the skin in infected individuals
    - Pigs (one intermediate host) → *Taenia solium* → multiple
        - *Spp. saginata*: beef tapeworm; causes cysticercosis worldwide; acquired by ingestion of uncooked beef that contains cysticerci
        - *Spp. solium*: pork tapeworm; causes cysticercosis worldwide; acquired by ingestion of uncooked pork that contains cysticerci; can cause neurocysticercosis
    - Reduviid Bug (one intermediate host) → *Trypanosoma cruzi* → multiple
    - Sand Fly → *Leishmania*
        - *Leishmania*: flagellate protozoan; associated with visceral, cutaneous, and mucocutaneous Leishmaniasis
    - Snails/crustaceans (two intermediate hosts) → *Paragonimus westermani*
        - *Paragonimus westermani*: acquired by ingestion of infective metacercariae in raw or pickled crustaceans
    - Snails/plants (two intermediate hosts) → *Fasciola* → multiple
        - *Fasciola hepatica*: a parasite of sheep that infects humans when they ingest freshwater plants; parasite lives in the intrahepatic bile ducts of the liver; fascioliasis → severe anemia
    - Tsetse Fly → *Trypanosoma brucei* → multiple

            - *Trypanosoma*: flagellate protozoan; causative agent of African Sleeping Sickness
        - Tick → *Babesia*

Fungi

- General Features
    - Fungi are eukaryotes and possess a chitinous cell wall and produce filamentous structures and spores; they are heterotrophs and consume decomposing organic material
        - Saprophytes decompose dead material and parasitic fungi invade and decompose living material
        - Cell wall structure: chitin, β(1,3) & β(1,6) glucans, mannose-modified proteins, and glycosylphosphatidylinositol (GPI) anchored proteins; has ergosterol instead of cholesterol
            - Control of fungal infections: inhibition of ergosterol, β-glucan, chitin biosynthetic pathways, DNA synthesis via depletion of thymidine pools
    - Morphology: unicellular (yeast) or multicellular (mold)
        - Multicellular fungi connect together in long strands → hyphae; tangled masses of hyphae → mycelia; hyphae may also form asexual spores (conidia)
            - Septate: perforated → mitochondria or nuclei can migrate along a hyphal strand
            - Aseptate: coenocytic (single plasma membrane surrounds many nuclei to create a multinucleated cell)
        - Fungi are haploid except during the formation of a zygote → dikaryon (not diploid)
    - Reproduction: asexually (via mitosis) or sexually (spore-forming)
- Pathogenesis
    - Primary mycoses: respiratory portal (inhaled spores)
    - Cutaneous & superficial mycoses: contamination of skin surface
        - Cutaneous: macular, papular, or pustular (herpetiform) lesions that itch and are spread by scratching; infection is confined to cutaneous tissue and is rarely spread systemically and is caused by soil organisms introduced into the extremities by trauma
            - Tinea Barbae: beard
            - Tinea Pedis: foot
            - Tinea Cruris: groin
            - Tinea Capitis: scalp
            - Tinea Corporis: trunk
        - Superficial: colonization of the outer layers of skin, hair, and nails; rarely invades deeper tissues
            - Pityriasis (Tinea) Versicolor: caused by dimorphic *Malassezia furfur* which infects skin and alters color; fluoresces under UV light
            - Tinea Nigra: caused by *Hortaea werneckii* which causes skin to darken
            - Black Piedra: caused by *Piedra hortae* which causes a superficial infection of the hair shaft
                - Note: White Piedra also exists
            - Onychomycosis: caused by *Trichophyton rubrum* and *Trichophyton mentagrophytes* that causes an infection of the fingernails and toenails
            - Note: dermatophytes typically do not penetrate deeper and acquire nutrition because they are keratinolytic
    - Subcutaneous mycoses: inoculated skin (trauma)
        - Subcutaneous: involves skin and deep viscera and may become widely disseminated
            - Sporotrichosis: subacute or chronic granulomatous infection that follows lymphatics; caused by the soil fungus *Sporothrix schenckii*
            - Mycetoma/eumycetoma: granulomatous inflammation that may extend beneath the subcutaneous region to the bone; pigmented nodules may drain through sinuses and produce colored grains which are helpful in identification
            - Chromoblastomycosis: infection that forms warty pigmented lesions which grow outwards from the site of introduction
    - Opportunistic mycoses: do not normally cause disease in healthy people, but is more common in immunocompromised people
        - Common examples: Candidiasis, Aspergillosis, Cryptococcosis
    - Virulence factors: thermal dimorphism, toxin production, capsules, adhesion factors, hydrolytic enzymes, inflammatory stimulants
- Diseases
    - Hypersensitivity: allergic reaction to molds and spores (i.e., Farmer's Lung, Tea Picker's Lung, Bark Stripper's Disease)

- Mycotoxicoses: poisoning of man/animal by feeds/products contaminated by toxin-producing fungi that colonize crops such as grains, corn, peanuts → lethal to poultry and livestock (i.e., aflatoxin, ergot alkaloids)
- Mycetismus: ingestion of preformed toxins (i.e., mushroom poisoning)
- Infections: mycoses

- Classifications
  - *Ascomycota*: sexual reproduction in a sack (ascus) with the production of ascopspores (i.e., *Candida albicans*, *Pneumocystis jirovecii*) → Note: largest fungal phylum; includes most of the human fungal pathogens, typically septate
    - *Candida albicans* → Candidiasis: superficial skin infections, oral cavity, genitalia, and large intestine that forms off-white, pasty colonies with a yeasty odor; causative agent of thrush, vulvovaginal yeast infection, and cutaneous candidiasis; has pseudohyphae
    - *Pneumocystis jirovecii* → Pneumocystis: small, unicellular fungus that causes pneumocystis pneumonia (PCP) → the most prominent opportunistic infection in AIDS patients; can be rapidly fatal if not controlled with medication
  - *Basidiomycota*: sexual reproduction in a sack (basidium) with the production of basidiospores (i.e., *Cryptococcus neoformans*) → Note: common mushrooms
    - *Cryptococcus neoformans* → Cryptococcosis: inhabits soil around pigeon roosts and usually infects lungs, the brain, and other organs; common infection of AIDS, cancer, or diabetes patients
  - *Zygomycota*: sexual reproduction by gametes and asexual reproduction with the formation of zygospores (i.e., *Mucor*, *Rhizopus*, *Absidia*) → Note: most primitive fungal phylum, typically aseptate and has right-angle branching
    - *Mucor*, *Rhizopus*, *Absidia* → Zygomycosis/Mucormycosis: saprobic fungi found in soil, water, organic debris, and food; usually harmless but can invade the nose, eyes, heart, and brain of people with diabetes and malnutrition
  - *Deuteromycota*: mitosporic fungi/fungi imperfecti → no recognizable form of sexual reproduction; includes most pathogenic fungi (i.e., *Penicillium*, *Trichophyton*, *Candida*, *Aspergillus*) → Note: septate mycelium
    - *Aspergillus* → Aspergillosis: very common airborne soil fungus that infects the lungs and presents a serious opportunistic threat to AIDS, leukemia, and transplant patients; invasive aspergillosis can involve many organs; low-angle branching
- Laboratory Testing
  - Culture Mediums
    - Most commonly used solid medium is Emmons Modification of Sabouraud Dextrose Agar (SDA) → pH: 5.4 (acidity inhibits bacterial growth); can be supplemented with antibiotics like gentamicin and chloramphenicol to minimize bacterial contamination, and with cycloheximide to minimize saprophytic fungi
    - Special media may be used for isolation and to help rapid identification when the identity of a particular pathogen is strongly suspected (i.e., Bird Seed Agar → *Cryptococcus neoformans*)
    - Other media types: Potato Dextrose Agar, Brain Heart Infusion (BHI) Agar, Czapek Dox Agar
  - Staining
    - Sputum, washings, and aspirates may be prepared with 10% KOH; only the chitinous hyphae survive this treatment and are visible as hyaline shadows under the light microscope
    - Calcofluor white and/or Gram stain are also used
      - Calcofluor white: binds chitin and fluoresces under UV
      - Gram stain: *Candida albicans* stains Gram +
    - Gomori methenamine silver staining is used to demonstrate hyphae, spores, and conidiophores; detects *Pneumocystis jirovecii*

Immunology

Cell-Mediated Immunity

- General Principles
  - Cell-mediated immunity is responsible for immune responses to intracellular pathogens
  - Cells Involved
    - Antigen-specific cells: T-helper CD4 cells & T-cytotoxic CD8 cells
    - Antigen non-specific cells: NK cells, macrophages, neutrophils, and eosinophils
      - NK Cells
        - Innate immunity

- Lymphoid cells
  - ~5% of lymphocytes
  - Shares early lineage to T-cells
  - Do not develop exclusively in the thymus
- Defends against viruses, tumors, and other intracellular pathogens
  - Nonspecific cytotoxicity, no TCR/CD3, not MHC-restricted, no memory
- Recognizes non-peptides (glycolipid & CD1d), altered cell-surface molecules (lowered MHC I), and tumor or virus-infected cells
  - Cytotoxic granule-mediated cell apoptosis (perforin, granzymes, α-defensins)
  - Induces infected macrophages to kill (responds to IL-12 with IFN-γ production)
  - Antibody-dependent cell-mediated cytotoxicity (ADCC)
- Produces important cytokines
- Killing of target cells is similar to cytotoxic T-cells
- Activation is determined by a balance between activating and inhibitory receptors
  - Inhibitory → no killing
  - Activating + inhibitory → no killing
  - Activating → killing

- Functions
  - Activation of M1 macrophages (via IFN-γ or INF-γ)
    - Functions: pro-inflammatory, microbicidal, tumoricidal, Th1 response, APC capacity, intracellular pathogen destruction, tissue damage, etc.
  - Induce B-cells (IFN-γ) to class switch (IgG) to opsonize
  - Direct killing (T-cytotoxic cells)
    - Granules present: perforin (action similar to C9), granzymes (act as nucleases)
    - Fas/FasL → lymphocytosis/apoptosis → splenomegaly (in ALPS)
      - Note: Autoimmune Lymphoproliferative Syndrome (ALPS)

- Antigen Recognition
  - Phagocytosed microbes (exogenous microbes) → intracellular bacteria, fungi, protozoa
    - Pathway: Dendritic cells (APCs) → Naïve CD4+ T-cells → production of IL-2 cytokines → Effector CD4+ T-cells (activation of macrophages, B-cells, and other cells → inflammation) & Memory CD4+ T-cells
  - Non-phagocytosed microbes (cytosolic microbes) → viruses, rickettsiae, protozoa
    - Pathway: Dendritic cells (APCs) → Naïve CD8+ T-cells → production of IL-2 cytokines → Effector CD8+ T-cells (i.e., cytotoxic T-cells; killing of infected target cells, macrophage activation) & Memory CD8+ T-cells
  - Superantigens
    - Cause pathology through massive load of cytokines (i.e., Toxic Shock Syndrome); T-cell mitogens → induce cell proliferation
      - Examples: Staphylococcal enterotoxins, Staphylococcal toxic shock toxin, Staphylococcal exfoliating toxin, Streptococcal pyrogenic exotoxins
  - Adhesion Molecules (CAMs)
    - CAMs: selectins, mucins, integrins, and immunoglobulin superfamily
      - Integrins: $\alpha L\beta 2$ (LFA-1) → T-cell
      - Immunoglobulin superfamily: ICAM-1, ICAM-2, ICAM-3 → APC
    - Function: stabilization of binding between ligand on APCs and T-cells (overcome moderate affinity between TCR and peptide/MHC)
  - Inhibitors
    - Block B7/CD28: used in treatment of rheumatoid arthritis, graft rejection, etc.
    - Block CTLA-4 & PD1: enhance immune response in cancer patients
  - T-Cell Responses
    - Secretion of cytokines and expression of cytokine receptors
      - First cytokine to be produced by CD4+ T-cells → IL-2 (around 1-2 hours after activation); gives autocrine signal → survival, proliferation, differentiation
    - Note: X-Linked Severe Combined Immunodeficiency (SCID): defect in common γ-chain

- Associated Diseases
  - DiGeorge Syndrome
    - Microdeletion on chromosome 22 → born without thymus → lack T-cell components

- Can fight extracellular pathogens, but have issues with intracellular pathogens (viruses, intracellular bacteria)
- Symptoms: cardiac abnormalities, abnormal facies, thymic aplasia, cleft palate, hypocalcemia

Surface Molecules of T-Cells	Function	Ligands of APCs	
		Name	Expression
CD3	Signal transduction by TCR complex	None	None
ζ	Signal transduction by TCR complex	None	None
CD4	Signal transduction	MHC II	APCs
CD8	Signal transduction	MHC I	All nucleated cells
CD28	Signal transduction Co-stimulation (guarantees response to foreign antigen & is essential for naïve T-cell response)	B7-1 (CD80) B7-2 (CD86)	APCs
CD40L (CD154)	Activate APCs to express more B7 co-stimulators and secrete cytokines (IL-12) to enhance T-cell differentiation	CD40	APCs
CTLA-4	Inhibition	B7-1 (CD80) B7-2 (CD86)	APCs
PD-1	Inhibition	PD-L1 PD-L2	APCs, tissue cells, tumor cells
LFA-1 (αLβ2)	Signal transduction, adhesion	ICAM-1	APCs, endothelium

<u>CD4+ T-Cell Subunits</u>

- General Principles
    - Effector cells of the CD4+ helper lineage functions:
        - Cell-mediated immunity: activate phagocytes (Th1) → intracellular organisms
        - Humoral immunity: extracellular organisms
    - Effector function mechanisms: secretion of cytokines
- Maturation
    - Naïve T-cells are stimulated by microbial antigens presented by dendritic cells in the lymph nodes and the spleen
        - Express adhesion molecule: L-selectin
        - Express chemokine receptor: CCR7 (selective migration into lymph node)
    - Gives rise to effector cell → differentiated effector T-cell migrates to the site of infection
        - Express: E-selectin, P-selectin, integrin LFA-1, VLA-4
        - Changes chemokine from CCR7 to CXCR3 (among others)
- Th1 Cells
    - Production of IFN-γ: stimulates phagocyte-mediated ingestion
        - Expression of lysosomal proteases
        - Synthesis of reactive oxygen species and nitric oxide (deals with organisms that escape the phagosome)
        - Stimulates production of antibodies that promote phagocytosis (IgG)
        - Stimulates expression of MHC II and B7 on APCs → amplify T-cell response
        - Activated macrophages produce IL-12 → drives CD4+ T-cells to Th1 cells (amplification)
    - Delayed-type hypersensitivity reaction
    - Balance
        - High Th1: autoimmune problem
        - Low Th1: cancer, hepatitis, HIV, shingles, TB, etc.

- Th2 Cells
    - Stimulates phagocyte-independent eosinophil-mediated immunity → against helminthic parasites
    - Produces IL-4
        - Stimulates production of IgE antibodies
        - IgE coats parasites and eosinophils/mast cells release granule enzymes that kill them
    - Produces IL-5
        - Activates eosinophils
    - IL4 and IL-13 leads to IL-10 production by M2 macrophages (alternative macrophage activation)
        - Synthesis of extracellular matrix proteins for repair
        - Inhibits microbicidal activity of macrophages → suppresses Th1 cell-mediated immunity
    - Balance
        - High Th2: allergic reaction
        - Low Th2: infection by extracellular organisms
- Th17 Cells
    - Induce inflammatory response and stimulates antimicrobial production → defensins
    - Destroys extracellular bacteria and fungi
    - Important cytokines: IL-17 and IL-22
        - IL-17: recruits leukocytes (neutrophils)
        - IL-22: important in epithelial barrier integrity
    - Defects: leads to bacterial abscesses and chronic mucocutaneous candidiasis
- Memory T-Cells
    - Lymphoid organs: central memory cells → rapid clonal expansion
    - Peripheral tissues: effector memory cells → rapid effector cell
    - Requires IL-7 to stay alive; normally remains in G0 phase (quiescent)
    - Responsible for the secondary immune response
- Immune System Decline
    - Homeostasis: return to steady-state of the immune response
        - During response: survival and proliferation is maintained by antigen, CD28 costimulatory signal, and cytokines (IL-2)
        - Once infection is cleared, survival signal is turned off
        - Cells that are produced are apoptosed
            - Only survivors: memory lymphocytes
        - Response subsides after 1-2 weeks
    - Activation and Exhaustion
        - In an acute viral infection, virus-specific CD8+ T-cells proliferate, differentiate into effector Tc cells and memory cells, and clear the virus
        - In a chronic viral infection, CD8+ T-cells mount an initial response, but begin to express inhibitory receptors (i.e., PD01 and CTLA-4) and are inactivated, leading to a persistence of the virus
            - T-cell anergy/clonal anergy
                - Inability of cells to proliferate in response to engagement with the MHC-peptide complex; due to absence of appropriate costimulatory signals or by inhibition of CTLA-4 or another inhibitory signal

	Th1 Cells	**Th2 Cells**
Differentiation from Th0 mediated by	IL-12	IL-4
Secretes	IL-2: stimulates both Th and Tc proliferation IFN-γ: activates macrophages and NK cells and IL-12	IL-4: stimulates B-cells to produce IgE and stimulates further Th2 response and inhibits Th1 response IL-5: promotes eosinophil growth IL-10: inhibits Th1 response
Mediates	Cell-mediated immunity	Humoral immunity
Infection: *Leishmania major*	Recovery	Disseminated infection

Infection: *Mycobacterium leprae*	Tuberculoid	Lepromatous

Effector T-Cells	Cytokines	Target Cells	Immune Response	Host Defense	Role in Disease
Th1	IL-2 IFN-γ	Macrophages	Macrophage activation	Intracellular pathogens	Autoimmunity; chronic inflammation
Th2	IL-4 IL-5 IL-13	Eosinophils	Eosinophil and mast cell activation; alternative macrophage activation	Helminths	Allergy
Th17	IL-17 IL-22	Neutrophils	Neutrophil recruitment and activation	Extracellular bacteria and fungi	Autoimmunity; inflammation
Tfh	IL-21 IFN-γ/IL-4	B-Cells	Antibody production	Extracellular pathogens	Autoimmunity (antibodies)

Humoral Immunity

- General Principles
    - Arm of the adaptive immune system that functions to neutralize and eliminate extracellular microbes and microbial toxins; mediated by antibodies and produced by B-cells
    - Principal defense mechanism against microbes with capsules rich in polysaccharides and lipids
    - Processing sites
        - Antigens from tissue spaces → filtered by lymph nodes
        - Antigens that are blood-borne → filtered by the spleen
        - Antigens from lumen → mucosa-associated lymphoid tissue (MALT)
    - CD4+ T-cell Th1 will deal with antigens once phagocytized (endosomal system) and presented in MHC II → cell-mediated immunity
    - CD8+ T-cells will deal with intracellular organisms that are in the cytosolic cell-mediated immunity
    - Antibodies, complements, and innate immunity will deal with extracellular component → humoral-mediated immunity (Th2)
    - Innate immunity deals with everyone
- B-Cell Terminology
    - Follicular B-cells (B-2 cells): most common type of B-cell; found mainly in the lymphoid follicles of secondary lymphoid organs and circulating in the blood. Responsible for generating the majority of high-affinity antibodies during an infection
    - Marginal zone B-cells: found mainly in the marginal zone of the spleen and serves as a first line of defense against blood-borne pathogens; undergo mainly T-cell-independent (TI) activation
    - B-1 cells: predominantly populate the peritoneal and pleural cavities; generate natural antibodies (produced without infection) against mucosal pathogens; undergo mainly T-cell-independent (TI) activation
    - Plasmablast: a short-lived, proliferating antibody-secreting cell; results from T-cell-independent (TI) or T-cell-dependent (TD) activation of B-cells
    - Plasma cell: a long-lived, non-proliferating antibody-secreting cell; results from the germinal center reaction from TD or TI activation of B-cells
    - Memory B-cell: dormant B-cell; their function is to circulate through the body and initiate a stronger, more rapid antibody response (i.e., secondary antibody response)
    - Regulatory B-cell: immunosuppressive cell type that stops the expansion of pathogenic, pro-inflammatory lymphocytes through the secretion of IL-10, IL-35, and TGF-β
    - Lymphoplasmacytic cell: cell with a mixture of B-cell and plasma cell morphological features; pre-malignant and malignant IgM-producing cells → IgM monoclonal gammopathy of undetermined significance and Waldenstrom's macroglobulinemia
- Antibody Responses
    - Thymus-independent antigens (TI): non-protein antigens; consists of marginal-zone B-cells (spleen) and B-1 cells (mucosal tissue & peritoneum)
        - Activate B-cells by pattern-recognition receptors
            - Type 1 (TI-1): lipopolysaccharide

        - Immunoglobulin produced: IgM
    - Type 2 (TI-2): highly repetitious molecules → bacterial flagella
        - TI-2 antigens alone can signal B-cells to produce IgM
        - Activated dendritic cells release the cytokine BAFF that augments production of the antibody against TI-2 antigens and induces class switching → produces IgG
  - No isotype switching
  - No affinity maturation
  - Short-lived plasma cells
  - Location: spleen, other lymphoid organs, mucosal tissues, peritoneal cavity
  - Activation
    - Microbes become coated with C3d (product of Factor I on C3b)
    - B-cells have C3d receptor CR2 or CD21
    - Engagement enhances antigen-dependent activation of B-cells
    - B-cells also express toll-like receptors (TLR's) that can recognize microbial products directly
- Thymus-dependent antigens (TD): protein antigens → processed in APC's and recognized by Th cells (B-cells require direct contact with Th cells); majority responders → follicular B-cells (B-2 cells)
  - Help B-cell activation: memory B-cell & long-lived plasma cells
  - Heavy chain isotype switching
  - Affinity maturation
  - Protein antigens with no Th cell: weak or absent antibody response
  - Location: spleen, other lymphoid organs
  - Immunoglobulins produced: IgM, IgG, IgE, IgA
  - Activation
    - Membrane-bound antibodies have short cytoplasmic tails
    - Membrane immunoglobulins must be associated with B-cell receptors → Igα/Igβ
    - Interaction of CD40 (B-cell) and CD40L (Th cell) provides signal 2
      - Interaction of B7 (B-cell) and CD28 (Th cell) provides costimulation to the Th cell
    - Tfh cells
      - Generation and function of follicular helper T-cells (Tfh cells) is dependent on costimulatory ICOS
        - ICOS (T-cells; CD28 family) interaction with ICOS-L (APC's; B7 family) → activation: follicular helper T-cells (Tfh) in antibody responses
      - Secretes IL-21 → antibody production/proliferation
      - In the germinal center of the lymph node: class switching and affinity maturation
  - Induction of humoral response
    - Germinal centers arise within 7-10 days after the initial exposure to thymus-dependent (TD) antigen in the lymph node; 3 events take place:
      - Affinity maturation → result of somatic hypermutation of Ig genes (point mutations)
        - Affinity of antibodies produced in response to a protein antigen increases with prolonged/repeated exposure
        - Followed by selection of high affinity B-cells by antigen
        - Enzyme activation-induced cytidine deaminase (AID) is required → creates mutations in DNA by deamination of cytosine base
      - Class switching
        - Same variable domains as generated during VDJ recombination; different constant domains in the heavy chains
        - Enzyme AID is required
        - Dependent on cytokines to switch from IgM to other isotypes
          - Thymus-dependent (TD) antigens
          - Interaction of CD40 on B-cells and CD40L on T-cells
        - X-Linked Hyper-IgM Syndrome: Th cells do not express CD40L → patients only produce IgM; no memory cell populations, no germinal centers of lymph nodes
      - Formation of plasma cells and memory B-cells
- Antibody Effector Functions

- Neutralization: antibody prevents bacterial adherence; Fab
- Opsonization: antibody promotes phagocytosis via antibody-dependent cellular phagocytosis (ADCP) and antibody-dependent cellular cytotoxicity (ADCC); Fab & Fc
    - Fc Receptors ($\gamma \rightarrow$ IgG; $\varepsilon \rightarrow$ IgE)
        - FcγRI (CD64): macrophages; phagocytosis, activation of phagocytes (ADCP)
            - Phagocytosis of microbe → killing of ingested microbe
        - FcγRIIA (CD32): macrophages; phagocytosis, cell activation (inefficient)
        - FcγRIIB (CD32): B-cells; feedback inhibition of B-cells, attenuation of inflammation
        - FcγRIIIA (CD16): NK cells; antibody-dependent cellular cytotoxicity (ADCC)
            - NK cells → killing of antibody-coated cell
        - FcεRI: mast cells, basophils, eosinophils; activation/degranulation of mast cells and basophils
            - Eosinophil activation → helminth death
- Complement activation: antibody activates complement which enhances opsonization and lyses some bacteria → complement-dependent cytotoxicity (CDC) via the classical pathway
- Main Functions
    - Neutralization: IgA
    - Opsonization (ADCP): IgG1
    - Sensitization for killing by NK cells: IgG1 & IgG3
    - Sensitization of mast cells: IgE
    - Activation of complement system: IgM (best) & IgG3
- Main Distributions
    - Transport across epithelium: IgA
    - Transport across placenta: IgG1
    - Diffusion to extravascular sites: IgG1, IgG2, IgG3, IgG4
- Mucosal Immunity
    - IgG is produced in mucosal lymphoid tissue and is transported across epithelia; neutralizes microbes in the lumen of mucosal organs
- Neonatal Immunity
    - Maternal antibodies are actively transported across the placenta to the fetus and are also transported across the gut epithelium of neonates
    - Incomplete immune systems → passive immunity from the mother
    - Acquires IgG via two routes that rely on neonatal Fc receptor (FcRn): placenta while in uterus or gut after ingestion of mother's colostrum
        - Note: IgG half-life expands in neonates via contribution of neonatal FcRn

- B-Cell Activation
    - Co-receptors
        - Complex of three proteins
            - CD19: cytoplasmic tail for signal transduction
            - CR2: receptor for C3d (from complement system)
            - TAPA-1: transmembrane molecule
        - Inhibitory co-receptors
            - CD22: cytoplasmic tail for signal transduction → contains immunoreceptor tyrosine-based inhibitory motif (ITIM)
                - Functions to deactivate B-cells (negative regulation) → important in preventing autoimmunity
    - Signal 1: Igα/Igβ, CD19, CR2, TAPA-1, CD22
    - Signal 2: CD40/CD40L
- Regulation
    - Humoral and cell-mediated branches must be heavily regulated → cytokines play an important role in antibody feedback (antigen-antibody complex forms)
        - Fc tail binds to FcγRIIB receptors on B-cells
        - Inhibition of receptor-mediated B-cell activation

	Primary Response	**Secondary Response**
Lag After Immunization	5-10 days	1-3 days
Peak Response	Smaller	Larger

Antibody Isotype	IgM > IgG	IgM < IgG; presence of IgA & IgE
Antibody Affinity	Lower	Higher
Antigen	TD or TI	TD

Immune Response to Pathogens

- General Principles
    - Intracellular organisms are targeted mainly by NK cells and cytotoxic T-cells
    - Intracellular antigens are presented in MHC I molecules to cytotoxic T-cells
    - Phagocytized organisms are presented in MHC II molecules to Th1 cells to initiate antibody production
    - Extracellular organisms are targeted mainly by the innate system and by antibodies
    - Immune Evasion
        - Antigenic variation: many viruses (i.e., HIV, Influenza) and bacteria (i.e., *Neisseria gonorrhoeae, E. coli*)
        - Inhibition of complement activation: many bacteria
        - Blocking of hyaluronic acid capsule: *Streptococcus* Group A
- Viral Infections
    - Cell-Mediated Immunity
        - Natural Killer (NK) cells: cytotoxic for virus-infected cells and participates in antibody-dependent cell-mediated cytotoxicity (ADCC)
            - ADCC: antibodies bind to membrane-surface antigens on the target cell → NK92 cells expressing CD16 Fc receptors recognize cell-bound antibodies → crosslinking of CD16 triggers degranulation into a lytic synapse → target cell dies via apoptosis
        - Cytotoxic CD8+ T-cells: destroys cells with viral peptides presented on cell surface in association with MHC I; enzymes utilized include perforin and granzymes
        - Resistance
            - Many viruses inhibit MHC I-associated antigen presentation
                - NK cells recognize and act on low MHC I molecules on the cell surface
            - Other viruses produce inhibitory cytokines or decoy soluble cytokine receptors that neutralize cytokines (i.e., INF-γ); other viruses directly infect T-cells (HIV CD4+ T-cells)
            - Examples
                - Herpes Simplex Virus (HSV): inhibition of antigen presentation; HSV peptide interferes with TAP transporter
                - Cytomegalovirus (CMV): inhibition of antigen presentation; inhibition of proteasomal activity; removal of MHC I molecules from the endoplasmic reticulum (ER)
                - Epstein-Barr Virus (EBV): inhibition of antigen presentation; inhibition of proteasomal activity; production of IL-10 → inhibition of macrophage and dendritic cell activation
                - Pox Virus: inhibition of effector cell activation → production of soluble cytokine receptors
    - Humoral Immunity
        - Antibodies: when in transit and as opsonizing elements
        - IFN-α & IFN-β: produced by virally-infected cells → inhibits transcription and translation in neighboring cells
        - IFN-γ: activates macrophages and NK cells and enhances the adaptive immune system by upregulating expression of MHC I
    - Strategies to Avoid Immunity
        - Antigenic shift & drift: mechanism of antigenic variation (i.e., Influenza)
        - Polymorphism: circumvents immunologic memory by expressing different immunologic targets (i.e., Adenovirus, Rhinovirus)
        - Latent virus: Herpes Simplex Virus (HSV), Varicella-Zoster Virus (VZV)
        - Modulation of MHC expression: Adenovirus, Epstein-Barr Virus, Cytomegalovirus (CMV), HSV, VZV
        - Infection of lymphocytes: leads to their destruction (i.e., HIV, Measles, CMV)
        - Prevention of complement activation: HSV
- Bacterial Infections

- Intracellular Bacteria
    - Cell-Mediated Immunity
        - Infected cells can activate NK cells → cytotoxicity, activation of macrophages
        - CD8+ T-cells: recognize antigens/MHC I molecules and lyse these cells
- Phagocytized Bacteria
    - Cell-Mediated Immunity
        - Antigen-presenting cells (APC's)
        - Th1 cells: bacterial antigens are processed and presented with MHC II molecules on the surface of APC's to CD4+ T-cells
- Extracellular Bacteria
    - Humoral Immunity
        - Complement: activated via the mannose-binding lectin (MBL) pathway or the alternative pathway
            - C3b: opsonin
            - C3a & C5a: recruit leukocytes
            - C5-C9 (MAC): perforate outer lipid bilayer of Gram-negative bacteria
        - Lysozyme: antibacterial that attacks N-acetyl-muramic acid-N-acetylglucosamine linkages (peptidoglycan) in the bacterial cell wall → lysis
        - Antibodies: principal defense against extracellular bacteria
            - Neutralization
            - Activation of complement
            - Opsonization
    - X-Linked Agammaglobulinemia: low count of all immunoglobulins; increased risk for extracellular bacterial infections
- Strategies to Avoid Immunity
    - Prevent phagocytosis
        - Capsules: *Streptococcus pneumoniae*, *Haemophilus*
        - Killing of phagocytes by toxins: *Staphylococcus*
        - Neutralization of IgG opsonin: *Staphylococcus*
        - Intracellular bacteria such as *Mycobacterium*, *Legionella*, and *Listeria*, inhibit the fusion of phagosomes with lysosomes → escape into the cytosol and escape phagocyte killing
    - Allow survival within phagocytes
        - *M. tuberculosis*, *M. leprae*, *Toxoplasmosis*
        - Elicits Type IV hypersensitivity response
    - Prevent complement activation
        - *Streptococcus*, *Staphylococcus*, *Haemophilus*, *Pseudomonas*
    - Avoid recognition by the immune system
        - By polymorphism: *Streptococcus pneumoniae*, *Salmonella typhi*

- Protozoal Infections
    - General Characteristics
        - Protozoa are microscopic, single-celled organisms (fewer than 20 types can infect humans) → Malaria, Trypanosomes, and *Leishmania* cause significant morbidity and mortality
        - Protozoa cause intracellular infections, have marked antigenic variation, and are often immunosuppressive; they have a complex life cycle which poses a variety of challenges to the immune system
        - Protozoal infections are often chronic since the immune system is not efficient at dealing with these organisms; most of the pathology of protozoal infections is caused by the body's immune response to them
    - Cell-Mediated Immunity
        - Important during the intracellular stage of infection
        - Phagocytosis: macrophages, monocytes, neutrophils
        - CD4+ T-cells are activated
        - Cytotoxic CD8+ T-cells: important in the intracellular stage of infection (i.e., sporozoite stage of *Plasmodium falciparum* → causative agent of Malaria)
        - NK cells
    - Humoral Immunity
        - Complement and antibodies are important during the extracellular stage of infection → opsonizes the protozoa and causes lysis
    - Strategies to Avoid Immunity
        - Escape into the cytoplasm following phagocytosis: *Trypanosoma cruzi*

        - Prevention of complement action: *Leishmania*
        - Gene-switching to create antigen variation: Trypanosomes
- Parasitic Infections
    - Helminths
        - General Characteristics
            - Helminths are more accessible to the immune system than protozoa, but they are fewer in number; therefore, the level of immunity generated is very poor
            - Processed as extracellular organisms: antibodies (IgE) and the complement systems are the most important attack mechanisms
                - Th2 cell-mediated production of IgE
- Fungal Infections
    - Innate Immunity
        - Controls most fungal infections (i.e., natural flora against *Candida albicans*)
        - Phagocytosis by neutrophils provides strong defense against fungi
        - The alternative pathway and the MBL pathway of the complement system are triggered by components present in most fungal cell walls
    - Adaptive Immunity
        - Role of IL-7 producing T-cells in protection vs. pathology in fungal infections is controversial
        - Defective Th17 cell differentiation has been linked to recurrent filamentous fungi and the occurrence of mucocutaneous candidiasis in patients with primary immunodeficiencies

<u>Immunization</u>
- Terminology
    - Natural immunity: acquired as a part of normal life experiences; active or passive
    - Artificial immunity: acquired through a medical procedure (i.e., vaccination); active or passive
    - Active immunity: results when a person is challenged with an antigen that stimulates the production of an antibody → creates memory, takes time, is long-lasting
    - Passive immunity: pre-formed antibodies are donated to an individual → no memory, acts immediately, is short-term
- Vaccination
    - Challenge: development of vaccines that stimulate cell-mediated immunity (CMI) against intracellular microbes
    - Types
        - Whole vaccines: killed whole bacterial cells or inactivated virus (stimulates a weaker immune response → humoral only); live-attenuated bacterial cells or viruses (stimulates a strong immune response → humoral + CMI; possibility of reversion back to wild-type)
            - Killed: Cholera, Plague
            - Inactivated: Polio (Salk), Rabies, Influenza, Hepatitis A
            - Live-attenuated: MMR, VZV, Rotavirus, Influenza (flu mist), Polio (Sabin), Smallpox, Yellow Fever
        - Acellular/subunit vaccines: selected antigens (1-20; immunodominant protein) derived from bacterial cells or virus particles; surface molecules of neutralized toxins (toxoids); fewer side effects
            - Toxoid: DTap
            - Acellular: *Haemophilus influenzae* Type B, *Streptococcus pneumonia*, *Neisseria meningitidis*
            - Subunit: Hepatitis B, HPV
        - Recombinant vaccines: genetically engineered; selected genes for microbial antigens are cloned into a vector so that cloning host will produce the antigen for use in a vaccine; DNA/RNA vaccines express foreign proteins in host cells; humoral + CMI
            - Recombinant vector: Vaccina, Canarypox, attenuated Polio, Adenovirus, attenuated *Salmonella*, *Mycobacterium bovis* (BCG)
            - DNA/RNA: COVID-19
                - mRNA: Moderna, Pfizer
    - Fetus and Neonate Vaccination
        - Maternal antibodies affect vaccination
        - Life-attenuated vaccines must be given to neonates > 12 months (residual maternal antibodies will inactivate them if given when neonates < 12 months)
            - If given < 6-9 months, they will need repeated boosters
        - IgM is only useful isotype in diagnosing infections in neonates
        - Infants have 20% of adult IgA at 12 months → colostrum is important

Hypersensitivity
- General Principles
  - Hypersensitivity: an exaggerated or inappropriate immune response that causes tissue damage; includes chronic reactions, allergies, and autoimmunity
    - Takes place in response to an infection that cannot be cleared (i.e., tuberculosis), a normally harmless exogenous substance (i.e., pollen), or to an auto-antigen (i.e., DNA in systemic lupus erythematosus)
    - Immediate hypersensitivity: anaphylactic, antibody-antigen complexes, manifests in minutes
    - Delayed-type hypersensitivity: may occur in days
  - Hypersensitivity reactions only occur in sensitized individuals (i.e., at least one prior contact with the offending agent)
    - Sensitization can be long-lived in the absence of re-exposure (> 10 years) due to immunologic memory
    - Antigen is a protein or is capable of complexing with protein → hapten
- Classifications
  - Type I: antibody-mediated
    - Immediate hypersensitivity mediated by IgE in response to harmless environmental antigens (i.e., pet dander or pollen) → atopic or allergic responses
    - IgE binds to high affinity Fcε receptors on mast cells and basophils
    - Upon subsequent exposures, cross-linking of membrane-bound IgE induces release of mast cell granules
    - Mast cell degranulation
      - Primary mediators
        - Made before and stored in granules → histamine, proteases, eosinophil chemotactic factor, heparin
        - Note: histamine and heparin increase vascular permeability and smooth muscle contraction
      - Secondary mediators
        - Made 2-4 hours after immediate response → platelet-activating factor, leukotrienes, prostaglandins, bradykinins, some cytokines and chemokines
    - Etiology
      - Determine whether an individual manifests atopic (Th2 predominant → IgE) or immune (Th1 predominant → IgG) reaction to an antigen
        - Th2: urban environments, widespread use of antibiotics, western diet
        - Th1: hygiene hypothesis
        - Note: Th1 when exposed to pathogen regularly, Th2 when not
      - Atopy: genetic predisposition to produce IgE in response to many common, naturally-occurring allergens
    - Localized reactions: allergic rhinitis (hay fever), asthma, food allergies, wheals, atopic eczema
    - Systemic reactions: insect stings, drug reactions
    - Note: positive skin-prick test → IgE reaction
    - Desensitization
      - Convert individual from Th2 → Th1
      - Introduce IgG "blocking antibodies" to bind allergens so that there is no reaction → no mast cell degranulation
  - Type II: antibody-mediated
    - Involves IgG or IgM induced damage to cell surface or matrix antigen
    - Cytotoxic processes
      - Classical complement pathway
      - Phagocytosis via Fc receptor and complement receptor
      - ADCC via NK cells or eosinophils
      - Examples
        - Transfusion reactions: antibody attaches to RBC and initiates the complement system to lyse RBC
          - After lysis, hemoglobin is detected in plasma and starts to filter through the kidneys and is found in urine (hemoglobinuria)
          - Hemoglobin converted to bilirubin → toxic at high levels
          - Fever, chills, blood clotting
        - Hemolytic diseases: erythroblastosis fetalis (i.e., Rh+ fetus, Rh- mother)

- IgG antibodies cross placenta → some antibodies may be anti-Rh antibodies
- RhoGAM shot: given to mother; anti-Rh antibodies bind to fetal cells that might have entered the mother's system during the birthing process, facilitates clearing before there is a B-cell response
- Non-cytotoxic processes
  - Interference with receptors
    - Examples
      - Graves disease: stimulating antibodies are directed against the receptor for thyroid-stimulating hormone (overactive thyroid)
      - Myasthenia gravis: blocking antibodies are directed against the acetylcholine receptor
- Type III: antibody-mediated
  - Immune-complex mediated hypersensitivity → antibody combines with a soluble antigen which results in a phagocytosis of the complex; if the complex persists, can lead to immune complex diseases
  - The immune complexes can be deposited in tissue near the site of antigen entry → localized Type III reaction
  - If the complexes form in blood, they can be deposited in blood vessels, kidneys, and joints
    - Serum sickness: after receiving antiserum/serum from another animal (i.e., horse serum → tetanus antitoxin) → fever, vasculitis, arthritis, nephritis
    - Use of monoclonal antibodies for use of cancer treatment → patient developed antibody against mouse monoclonal antibody
    - Autoimmune disease: lupus, rheumatoid arthritis
    - Drug reactions: penicillins, sulfonamides
    - Some infectious diseases
  - Examples
    - Systemic lupus erythematosus: attack of DNA, nucleoproteins, and others → nephritis, arthritis, vasculitis
    - Poststreptococcal glomerulonephritis: attack of streptococcal cell wall antigens → nephritis
    - Serum sickness: attack of various protein antigens → systemic vasculitis, nephritis, arthritis
    - Arthus reaction: attack of various protein antigens → cutaneous vasculitis
- Type IV: cell-mediated
  - Upon first contact with an antigen, a subset of CD4+ Th cells is activated and clonally expanded within 1-2 weeks
  - Upon subsequent encounter with the antigen, sensitized Th cells secrete cytokines → attracts and activates macrophages
  - Reaction peaks at 48-72 hours after contact with the antigen → delayed-type hypersensitivity; frustrated clearance of intracellular pathogens
  - If antigen persists, the lytic products of the activated macrophages can damage healthy tissues → delayed-type hypersensitivity can lead to granuloma formation (i.e., mechanism of TB skin test)
  - Examples
    - *M. tuberculosis*, *Leishmania major*
    - Contact antigens such as nickel and poison ivy

	Type I	**Type II**	**Type III**	**Type IV**
Immune Reactant	IgE	IgG/IgM	IgG + IgM	T-Cells
Antigen Form	Soluble	Cell-Bound	Soluble	Soluble/Cell-Bound
Mechanism	Allergen-specific IgE antibodies bind to mast cells via their Fc receptor. When the specific allergen binds to the IgE,	IgG or IgM antibodies bind to cellular antigen leading to complement activation and cell	Antigen-antibody complexes are deposited in tissues. Complement activation provides inflammatory	Th1 cells secrete cytokines which activate macrophages and cytotoxic T-cells

	cross-linking of IgE induces degranulation of mast cells	lysis. IgG can also mediate ADCC with cytotoxic T-cells, NK cells, macrophages, and neutrophils	mediators and recruits neutrophils. Enzymes released from damaged neutrophil tissue	
Example	Local + systemic anaphylaxis, seasonal hay fever, food allergies, drug allergies	RBC destruction after transfusion with mismatched blood types, during hemolytic disease of the newborn, goodpasture syndrome, myasthenia gravis, graves disease	Post-streptococcal glomerulonephritis, rheumatoid arthritis, systemic lupus erythematosus	Contact dermatitis, type 1 diabetes mellitus, multiple sclerosis, rheumatoid arthritis, hashimoto's thyroiditis

Autoimmunity and Transplantation

- Tolerance
    - Immunological tolerance: non-reactivity to an antigen due to a previous exposure to the same antigen; can exist in T-cells & B-cells
        - Most important form of tolerance is non-reactivity to self-antigens → tolerogen
        - Breakdown of self-tolerance → autoimmunity
    - Central tolerance: deleting T-clones or B-clones before maturity if they have receptors that recognize self-antigens with great affinity (occurs in the thymus or bone marrow)
    - Peripheral tolerance: delete lymphocytes after they leave the bone marrow or thymus
    - Positive selection: the process by which immature double-positive thymocytes expressing T-cell receptors with intermediate affinity and avidity for MHC complexes are induced to differentiate into mature single-positive thymocytes
    - Negative selection/clonal deletion: the intrathymic elimination of double-positive or single-positive thymocytes that express T-cell receptors with high affinity for self-antigens
        - AIRE: autoimmune regulator that is a putative transcription factor that stimulates the expression of many self-antigens in the medullary epithelial cells of the thymus (mTEC) required for the deletion of self-reactive thymocytes
    - Note: APC's include neutrophils, macrophages, dendritic cells, mTEC's
    - Tolerance development: high antigen dosages, persistence of antigen in host, IV or oral introduction, absence of adjuvants, low levels of co-stimulators (CD28/B7 → CTLA-4 will bind to B7 and inhibit)
    - Hypersensitivity: harmless antigens are considered to be dangerous
    - Autoimmunity: self-antigens are considered foreign
- Autoimmunity
    - Causes
        - Tolerance can break down in the thymus for genetic reasons, or can break down in the periphery as a result of environmental factors (i.e., infection)
            - Human leukocyte antigen (HLA) → genetic predisposition
            - Release of sequestered antigens → environmental factors
                - Infection – mimicry: sequestered cardiac antigens and *Streptococcus* that can induce rheumatic fever → short-lived and reversible
        - Inappropriate MHC expression: i.e., type 1 diabetes express high levels of MHC I and MHC II
        - Inappropriate CTLA-4 expression/mutation
        - Immune privileged sites: sites in the body where foreign antigens or tissue grafts do not elicit immune responses → induces tolerance
            - Immunosuppressive cytokines (i.e., TGF-β) seem to be responsible for these responses
            - Sites: brain, eye, testis, placenta
        - CTLA-4: expressed by Treg cells (constitutively) and by activated T-cells; immunomodulatory molecule that transmits an inhibitory signal when bound to CD80 & CD86 (B7-1 & B7-2) on APC's

- Variants in this gene have been associated with type 1 diabetes, Grave's disease, Hashimoto's thyroiditis, celiac disease, systemic lupus erythematosus, and other autoimmune diseases
- Mechanisms
    - Type II: antibody-mediated; organ-specific autoimmune diseases are usually associated with antibody-mediated hypersensitivity mechanisms
        - Non-cytotoxic
            - Grave's disease: auto-antibodies bind to receptor for thyroid stimulating hormone (TSH) resulting in over-stimulation of the thyroid
            - Myasthenia gravis: autoantibodies bind acetylcholine receptors on motor end-plates of muscles progressively weakened skeletal muscles (face, eyes, swallowing reflex)
            - Pernicious anemia: antibodies against membrane-bound intestinal protein (i.e., intrinsic factor, parietal cells) that uptakes vitamin B12 needed for hematopoiesis
        - Cytotoxic
            - Goodpasture syndrome: antibodies directed against type IV collagen (anti-basement membrane) in glomeruli and alveoli → IgG accumulation, which results in basement membrane disruption and leads to kidney damage and pulmonary hemorrhage
    - Type III: immune-complex mediated
        - Systemic lupus erythematosus: produces auto-antibodies to DNA, histones, platelets, leukocytes, and clotting factors; antigen involves dsDNA, Sm, other nucleoproteins → leads to excessive complement activation (C3 down); fever, weakness, arthritis, butterfly skin rash, kidney problems
            - Note: screen with indirect immunofluorescence for antigens that bind to the nucleus (antinuclear antigen → ANA)
        - Rheumatoid arthritis: chronic inflammation of joints; produce auto-antibodies that 1) bind to Fc portion of IgG (circulating in blood that creates immune complexes rheumatoid factor) and 2) anti-CCP against citrulline residues (post-translational modification of arginine)
            - HLA: DR1, DR4
            - Associated with gingivitis
    - Type IV: cell-mediated
        - Hashimoto's thyroiditis: targets thyroid antigens (thyroid peroxidase → TPO); decreased hypothyroidism, can form goiter
        - Type 1 diabetes: antibodies against beta cells that produce insulin
        - Multiple sclerosis: inflammatory lesions in the myelin sheath caused by T-cells; numbness, paralysis, vision loss
- Autoimmune Polyglandular Syndrome I/Autoimmune Polyendocrinopathy-Candidiasis-Ectodermal Dystrophy (APS-1/APECED)
    - Defect in AIRE: decreased expression of self-antigen in thymus → defective negative selection
    - AR or XLR inheritance
    - Symptoms: adrenal insufficiency (Addison's disease), hypoparathyroidism, generalized candidiasis of mucous membranes

<u>Immunodeficiencies</u>

- Primary immunodeficiency: resulting from genetic or developmental defects in the immune system; can involve innate or adaptive immunity
- Secondary immunodeficiency: resulting from an acquired defect (not congenital)
- Forms
    - Primary Immune Deficiency (PID)
        - Results from a genetically-determined abnormality; 65% antibody deficiency
    - B-cell deficiencies: absent or reduced follicles and germinal centers in lymphoid organs; reduced serum immunoglobulin levels → pyogenic bacterial infections, enteric bacterial and viral infections
        - X-Linked Agammaglobulinemia: most common clinical syndrome associated with B-cell maturation (maturation only to pre-B-cell) → mutation in Bruton Tyrosine Kinase (BTK) affects survival signaling of B-cells
    - T-cell deficiencies: may be reduced T-cell zones in lymphoid organs, reduced delayed-type hypersensitivity (DTH) reactions to common antigens, defective T-cell proliferative responses to mitogens in vitro → viral and other intracellular microbial infections, viral-associated malignancies

- DiGeorge Syndrome: autosomal dominant inheritance of congenital thymic aplasia/hypoplasia
    - Deletion of a region on chromosome 22
    - Depressed T-cell numbers
    - Absence of T-cell response; failure to respond to intracellular pathogens
    - Humoral response to T-independent (TI) antigens only
    - Facial abnormalities, congenital heart disease, repeated infections with intracellular bacteria, viruses, and fungi
- Severe combined immunodeficiency (SCID): disorder manifesting as defects in both B-cells and T-cells
  - X-linked: caused by mutation the common γ-chain subunit of several cytokine receptors
    - Affects: IL-2, IL-4, IL-7 (growth factor for pro-T-cells), IL-9, IL-15 (NK cell proliferation), and IL-21
    - Immature lymphocytes cannot proliferate
  - Autosomal: caused by mutation in adenosine deaminase (ADA) → involved in breakdown of adenosine
    - Accumulation of toxic metabolites during proliferation; particularly affects Th-cells
- Immunoglobulin deficiencies
  - IgA deficiency: most common → recurrent respiratory and urinary tract infections, intestinal problems
  - IgG deficiency: rare → treat with IV immunoglobulins
  - X-Linked Hyper-IgM Syndrome: defective isotype switching and cell-mediated immunity
    - Mutation in CD40L (costimulatory)
    - Highly susceptible to *Pneumocystis jirovecii* (intracellular fungus)
- Wiskott-Aldrich Syndrome: X-linked recessive mutation in the *WAS* gene (cellular cytoskeleton)
  - Platelets and leukocytes are small and fail to migrate
  - Diagnosis: eczema, reduced platelets, and immunodeficiency with recurrent infections
- Ataxia Telangiectasia: mutation in a DNA repair gene → defective lymphocyte maturation
  - Gait abnormalities, vascular malformations, immunodeficiency
  - Abnormal repair during recombination of antigen receptor gene segments

Severe Combined Immunodeficiency (SCID)		
Disease	**Deficiencies**	**Mechanism**
X-Linked SCID	Decreased T-cells, normal/increased B-cells, reduced serum immunoglobulins	Cytokine receptor common γ-chain gene mutations, defective T-cell maturation due to lack of IL-7 signals
AR SCID via ADA/PNP Defect	Progressive decrease in T-cells (mainly) and B-cells	ADA or PNP deficiency leads to accumulation of toxic metabolites in lymphocytes
AR SCID via Other Defect	Decreased T-cells and B-cells, reduced serum immunoglobulins	Defective maturation of T-cells and B-cells; may be mutations in *RAG* genes and other genes involved in VDJ recombination or IL-7R signaling
B-Cell Immunodeficiencies		
Disease	**Deficiencies**	**Mechanism**
X-Linked Agammaglobulinemia	Reduction in all serum immunoglobulin isotypes, decreased B-cells	Block in maturation beyond pre-B-cells because of mutation in Bruton tyrosine kinase (BTK)
Ig-Heavy Chain Deficiencies	Deficiency of IgG subclasses (sometimes associated with absent IgA and IgE)	Chromosomal deletion involving immunoglobulin heavy-chain locus at 14q32

T-Cell Maturation Disorders		
Disease	**Deficiencies**	**Mechanism**
DiGeorge Syndrome	Decreased T-cells, normal B-cells, normal/decreased serum immunoglobulins	Anomalous development of 3rd and 4th branchial pouches leading to thymic hypoplasia

Transplantation Immunology

- Types of Transplants
    - Autograft: grafts from yourself
    - Isograft: grafts from an identical twin
    - Allograft: grafts from the human species
    - Xenograft: grafts from a different species
- Types of Donor Transplants
    - Living Donor (completely healthy)
        - Kidney (entire organ) → most commonly-done procedure
        - Liver (segment)
        - Lung (lobe)
        - Intestine (portion)
        - Pancreas (portion)
    - Living Donor (after brain death)
        - Kidney
        - Heart
        - Liver
        - Lungs
        - Pancreas
        - Intestine
        - Heart valves
        - Connective tissue
    - Cadaver (after natural death)
        - Cornea
        - Bone
        - Skin
        - Blood vessels
- Barriers to Transplantation
    - Supply: insufficient number of organ donors
    - Immune Barriers
        - Blood group (ABO) incompatibility
        - Adaptive immune response to foreign HLA or MHC antigens (the goal of immunosuppression for transplantation is to prevent cell-mediated and humoral responses from occurring against the foreign HLA or MHC antigens)
- Histocompatibility
    - Tissues that are antigenically similar → generally indicates allograft acceptance
    - Histocompatibility in humans has > 40 genetic loci; most vigorous rejection is linked to MHC locus/HLA locus
        - HLA: consists of 2 alleles from each loci
            - MHC I genes: A, B, and C loci
            - MHC II genes: DR, DQ, and DP loci
            - Note: A, B, and DR loci are the most important in transplantation
    - Testing Goals
        - Determine ABO and HLA types of patients and donors
        - Selection of donor tissues with minimal differences between the donor and the recipient
        - Look for evidence of prior sensitization
        - Identify incompatible transplants by in-vitro crossmatching
        - Note: even with a perfect HLA match, life-long immunosuppressive therapy is generally required (other minor histocompatibility antigens will differ and can be the basis of transplant rejection)
- Graft Rejection
    - Allograft Rejection

- First-set rejection: primary immune response; leukocytes arrive by 3-7 days (revascularization); full rejection by 10-14 days
- Second-set rejection: secondary immune response (immunologic memory); rejection begins by 3-4 days; full rejection by 5-6 days
- Rejection Stages
    - Sensitization Stage: lymphocytes (T-cells) of the host proliferate in response to the antigens on the graft
        - Both CD8+ and CD4+ cells react
        - Donor APCs may migrate to lymph nodes and stimulate host T-cells (direct allorecognition)
            - T-cells recognize unprocessed allogeneic MHC molecule on graft APCs
        - Host APCs (dendritic cells) can migrate into the graft and endocytose foreign antigens (indirect allorecognition); dendritic cells can cross-present endocytic antigens in the context of MHC I
            - T-cell recognizes processed peptide of allogeneic MHC molecule bound to self-MHC molecule on host APC
    - Effector Stage
        - Host immune system attacks the graft
            - Delayed-type hypersensitivity (DTH) reaction → Type IV
            - CTL-mediated cytotoxicity
            - Complement-mediated antibody response
            - ADCC
        - Th cells play a key role
            - IL-2: T-cell proliferation and CTL response
            - IFN-γ: DTH response, increased MHC II expression
            - TNF-β: increased cytotoxicity
- Clinical Types of Rejection
    - Hyperacute rejection: < 24 hours; involves pre-existing antibodies (i.e., xenotransplants, ABO-incompatible transplants, transplants with positive crossmatch)
        - Sensitizing events
            - Pregnancies: exposure to fetal paternal antigens
            - Transfusion: exposure to foreign leukocytes in blood
            - Example: prior transplant → kidney becomes cyanotic, mottled, and flaccid
    - Acute rejection: weeks (10-14 days post transplant full rejection); involves T-cells and cytokines
    - Chronic rejection: months or longer; involves humoral and cell-mediated immunity
        - Humoral immunity: rejection vasculitis associated with development of donor-specific HLA antibodies
        - Cell-mediated immunity: lymphocytic infiltration of graft

- Organ vs. Stem Cell Transplants
    - Solid organ transplants (i.e., kidney, lung, heart, liver, etc.) almost always have to be ABO-identical
        - ABO-incompatibility can cause hyperacute rejection of graft
        - HLA matching is a secondary priority
    - Stem cell or bone marrow transplants almost always have to be HLA-matched
        - Failure to match can lead to simultaneous rejection of graft and rejection of patient by graft (i.e., Graft-vs-Host Disease → GVHD) → bi-directional rejection
        - ABO compatibility is a secondary priority
            - Note: ABO antigens are not found on stem cells, therefore no hyperacute rejection of graft can occur
- Graft-vs-Host Disease (GVHD): rejection of the host by the graft
    - Host → graft: rejection of non-matched HLA antigens by the host; graft is damaged/lost
    - Graft → host: rejection of non-matched HLA antigens by the graft; host may die
    - Example: bone marrow transplant recipient → immunocompromised due to complete destruction of bone marrow; however, lymphocytes from graft attack allogeneic antigens of the host
        - Symptoms: skin reactions, gastrointestinal hemorrhage, liver failure, splenomegaly
        - Treatment: cyclosporin A, methotrexate
        - Note: 50-70% of bone marrow transplants are affected by GVHD

<u>Principles of Immunodiagnostics</u>

- Immunoassays are based on antigen/antibody reactions

- Factors affecting measurements of antigen/antibody reactions include:
    - Affinity: the strength of a single antigen/antibody interaction; each IgG antigen-binding site typically has a high affinity for its target
    - Avidity: the strength of all interactions combined; IgM typically has a low affinity for antigen-binding sites - however, there are ten binding sites so the avidity is high
    - Polyclonal vs. monoclonal antibodies
        - Polyclonal: often recognize multiple epitopes → more tolerant of small changes in the nature of the antigen, more robust detection
        - Monoclonal: high specificity, highly reproducible, extremely efficient binding of an antigen within a mixture of related molecules (i.e., affinity purification)
    - Antigen/antibody ratio
        - Precipitation Curve
            - Prozone: no precipitation occurs due to the presence of excess antibodies (forming soluble immune complexes)
            - Equivalence zone: precipitation occurs due to the presence of an optimal ratio of antibodies and antigens (forming insoluble immune complexes)
            - Postzone: no precipitation occurs due to the presence of excess antigens (forming soluble immune complexes)
    - Physical form of the antigen
        - Particulate (agglutination): combination of an insoluble particulate antigen with its soluble antibody → forms an antigen-antibody complex → particles agglutinate; used for antigen detection
            - Passive: use carrier particles to improve the visibility of the agglutination reaction
                - Latex agglutination
                - Hemagglutination
                    - Inhibition: antibodies combine with pathogen leading to their neutralization → red blood cells (RBCs) do not hemagglutinate
        - Soluble (precipitation): soluble antigen combines with its specific antibody → antigen-antibody complex is too large to stay in solution and precipitates
        - Enzyme-linked immunosorbent assay (ELISA): use of enzyme-labeled immunoglobulins to detect antigens or antibodies; enzymes convert a colorless substrate to a colored product
            - Direct: detects presence of antigen in the patient sample
            - Indirect: detects presence of antibody in the patient sample
            - Sandwich: detects presence of antigen in the patient sample; antigen is "sandwiched" between the capture antibody and the primary antibody conjugate
        - Immunoblot (Western blot): antigens are separated by polyacrylamide gel electrophoresis (PAGE) and trans-blotted onto nitrocellulose/nylon membranes; antibodies in serum react with specific antigens and those with numerous cross-reacting antibodies are identified more specifically
        - Immunochromatography: rapid tests; dye-labeled antibody that is specific for a target antigen is present on the strip
            - Advantages: fast, inexpensive, minimal infrastructure needed
            - Disadvantages: limited sensitivity (high concentrations needed), limited specificity
        - Immunofluorescence: detect the presence of antigens in specimens using antibodies labeled with a fluorescent molecule
            - Direct (DFA): fluorophore is conjugated to the primary antibody; faster, less sensitive
            - Indirect (IFA): fluorophore is conjugated to the secondary antibody; "sandwich technique", more sensitive
        - Immunoelectrophoresis: combines electrophoresis and double-immunodiffusion
        - Flow cytometry: automated analysis and separation of cells stained with fluorescent antibody techniques → yields quantitative data
    - Seroconversion: the development of specific antibodies in the blood serum as a result of infection or immunization, including vaccination; determines when a test can detect the antibodies
- ABO Blood Typing
    - Note: individuals possess antibodies to ABO antigens that are absent from their own cells
    - Forward grouping: tests for the presence of A and B antigens on RBCs
    - Reverse grouping: tests for the presence of A and B antibodies
    - Examples
        - Blood type A: anti-B antibodies, A antigen
        - Blood type B: anti-A antibodies, B antigen
        - Blood type AB: no antibodies, A and B antigens

- Blood type O: anti-A and anti-B antibodies, no antigens
- Note: blood type AB → universal recipient of RBCs & universal donor of plasma
- Note: blood type O → universal donor of RBCs & universal recipient of plasma

Public Health

Preventative Medicine

- Core Aspects
    - Screening: use of tests, examinations, or other procedures to identify disease, disease precursors, or susceptibility to disease in persons without evidence or suspicion of disease to identify early disease or susceptibility when it is more easily/successfully treated
    - Counseling
    - Immunization/chemoprophylaxis

Public Health Ethics

- Goals
    - Balance competing interests
    - Provide justification for public health interventions and policies
    - Note: the focus on populations contrasts with the duties and goals of medical and research ethics
- Social Responsibility Criteria
    - Expertise: knowledge makes actions more likely to be beneficial than acts by others
    - Proximity: being closer to the event or dilemma increases the obligation to act
    - Effectiveness: greater when the likelihood of making a difference is greater
    - Lower risk/cost: greater when acting doesn't jeopardize one's safety
    - Unique: greater if no other help or outcome is available
    - Severity: greater if failing to act might cause a much worse outcome
    - Public trust: greater for professionals and professional organizations → protect public trust

Death Certificates

- Information collected: identifying information, demographics, place/time/cause of death
    - Completed by physicians, funeral home directors, medical examiners/coroners
        - Pronouncing physician: items 24-31
        - Certifying physician: items 32-37, 45-49 (38-44*)
        - Pronouncing & certifying physician (if same): 24-25, 29-37, 45-49 (38-44*)
        - Medical examiner/coroner: 24-25, 29-30, 32-49
            - Note: * = if medical examiner/coroner does not assume jurisdiction
- Importance
    - Individually: burial permit, settlement of deceased's estate, life insurance claim, obtain death benefits, termination of government services/obligations, closure
    - Public health: characterize and assess general health of population, target and measure results of interventions, follow up on other deaths (infant, maternal, etc.), calculation of life expectancy
        - Flow of information: physician → funeral home → local health department → state health department (vital statistics; keeps certificate on file, provides copies to executor) → national center for health statistics (analyzes and publishes death certificate data)
- Physician Responsibilities
    - Be familiar with state and local regulations regarding medical certifications
    - Complete relevant portions of the death certificate
    - Deliver signed certificate promptly to the funeral director
    - Answer questions from state or local registrars
    - Provide supplemental report if autopsy reveals different cause of death
- Sections of the Death Certificate
    - Part I: Cause of Death
        - Sequence of events leading to death, proceeding backwards from final disease or condition resulting in death
        - Underlying cause (disease or injury that initiated events resulting in death) should be listed last (progress from simple → complicated/complex → death)
            - WHO recommends that all countries use underlying cause of death for basic mortality statistics
            - Only one cause per line but may split lines of needed
            - Duration listed on far right
        - Immediate cause: final disease, injury, or complication directly causing death (not the mode of dying or the terminal event → i.e., cardiac arrest, respiratory arrest, etc.); cannot be left blank

        - If immediate cause of death arose as a complication of, or an error in medical/surgical treatment, report complication/error, procedure, and condition being treated
    - Due to (consequence of)
        - Report any disease, injury, or complication that either gave rise to, had etiologic/pathologic basis, or is believed to have prepared the way for subsequent cause of death by damage to tissues or impairment of function
- Part II: Other Significant Items for Medical Certification
    - Report all other important diseases present at the time of death, even if unrelated; can list more than one condition even though only one line is provided

Mortality Statistics

- Mortality rate: frequency of death in a defined population during a specified period; includes crude death rate, cause-specific death rate, age-specific death rate, infant mortality rate, maternal mortality rate, maternal mortality ratio, case fatality rate, proportionate mortality rate

$$Mortality = \frac{Deaths}{Population}$$

- Crude death rate (CDR): indicates the current mortality impact on population growth; significantly affected by age distribution

$$CDR = \frac{All\text{–}Cause\ Deaths\ in\ 1\ Year}{Midpoint\ Population} x10^n$$

- Cause-specific death rate (CSDR)

$$CSDR = \frac{Specific\ Cause\ Deaths\ in\ 1\ Year}{Midpoint\ Population} x10^n$$

- Age-specific death rate (ASDR)

$$ASDR = \frac{All\text{–}Cause\ Deaths\ for\ Specific\ Age\ Group\ in\ 1\ Year}{Midpoint\ Population\ of\ Specific\ Age\ Group} x10^n$$

- Neonatal mortality rate (NNMR): number of deaths during the first 28 days of life per 1,000 live births

$$NNMR = \frac{Deaths\ in\ Children < 28\ Days\ Old\ in\ 1\ Year}{Live\ Births\ in\ 1\ Year} x10^n$$

- Infant mortality rate (IMR): number of deaths during the first year of life per 1,000 live births

$$IMR = \frac{Deaths\ in\ Children < 1\ Year\ Old\ in\ 1\ Year}{Live\ Births\ in\ 1\ Year} x10^n$$

- Under-five mortality rate (UFMR): number of deaths during the first 5 years of life per 1,000 live births

$$UFMR = \frac{Deaths\ in\ Children < 5\ Years\ Old\ in\ 1\ Year}{Live\ Births\ in\ 1\ Year} x10^n$$

- Maternal mortality rate (MMRate): the number of maternal deaths in a given period per population of women who are of reproductive age

$$MMRate = \frac{Maternal\ Deaths\ in\ 1\ Year}{Women\ of\ Reproductive\ Age}$$

- Maternal mortality ratio (MMRatio): the number of maternal deaths per live births

$$MMRatio = \frac{Maternal\ Deaths\ in\ 1\ Year}{Live\ Births}$$

- Case fatality rate (CFR): the percentage of people who have a certain disease and die within a certain time after their disease was diagnosed; a measure of disease severity

$$CFR = \frac{Deaths\ in\ 1\ Year\ after\ Disease\ Onset/Diagnosis}{Individuals\ with\ Disease\ at\ the\ time} x100$$

- Proportionate mortality rate (PMR): deaths from a certain cause expressed as a percentage of the total deaths in the same year and locality; reflects the burden of disease in the locality being studied

$$PMR = \frac{Deaths\ from\ a\ Disease\ in\ a\ Given\ Year\ and\ Locality}{All-Cause\ Deaths\ in\ the\ Same\ Year\ and\ Locality} x100$$

Family Violence

- Family Violence: the intentional intimidation or abuse of children (< 18 years old), adults, or elders (> 60 years old) by a family member, intimate partner (current or former), or caretaker in an effort to gain power and control over the victim
    - Includes: neglect, physical assault, sexual assault, emotional or psychological mistreatment, threats and intimidation, economic abuse and financial exploitation, and violation of individual rights
- Screening Tools
    - Intimate partner violence: the Hurt, Insult, Threaten, and Scream (HITS) Instrument, the Partner Abuse Interview, and the Women's Experience with Battering (WEB) Scale
    - Caretaker violence: the Caregiver Abuse Screen (CASE) and the Hwalek-Sengstock Elder Brief Abuse Screen Test (HSEAST)
- Physician Responsibilities
    - Approach to Family Violence
        - Initial diagnosis: battered person syndrome; adult physical abuse, confirmed; etc.
        - Treat for physical injuries, be alert to possible severe depression
        - If abuse is suspected, interview the patient alone
        - Provide resource information
        - Refer to a safe haven, if needed
    - Reporting
        - The American Medical Association (AMA) states that physicians have an ethical obligation to identify victims of partner violence and intervene appropriately
            - Mandatory reporting of family/domestic violence is law in some U.S. states
            - Some states have exceptions for reporting injuries due to domestic violence → NH, OK, PA; physicians must know the laws that pertain to their states
            - Mandatory reporting of suspected child or elder abuse
            - Good-faith reports of suspected abuse are a shield to lawsuits
            - Failure to report can result in legal action against the physician
        - Note: when seeing a suspected abuse victim, always consider the possibility that there could be other abuse victims in the household

Environmental & Occupational Health

- Terminology
    - Environment: a system of interdependent living and nonliving components in a given area over a given period of time, including all physical, chemical, and biological interactions
    - Environmental health: the segment of public health that is concerned with assessing, understanding, and controlling the impacts of people on their environment and the impacts of the environment on them
    - Hazard: a source of danger (factor) that has the potential to adversely affect health; a qualitative term expressing the potential of an environmental or occupational agent to harm the health of certain individuals if the exposure level is high enough and/or other conditions apply
        - Results in a risk only if there has been exposure; no risk if the hazard is contained, or if there is no opportunity for exposure
        - Biological hazards can be transmitted
            - Direct: physical contact between reservoir and host
            - Indirect: transfer of an infectious agent from a reservoir to a host by air particles, vehicles, or vectors
    - Risk: the probability of an unfavorable outcome (i.e., that an individual will become ill or die within a stated period or before a given age); a quantitative probability that a health effect will occur after an individual has been exposed to a specified amount of a hazard
        - The likelihood (probability) that a hazard will cause harm
    - Environmental determinants of health: any external agent (biological, chemical, physical) that can be causally linked to an involuntary change in health status
        - Note: air pollution is 4th in global ranking of risk factors by total deaths from all causes

- Human health: a state of complete physical, mental, social, and ecological well-being
- Occupation: what a person does for a living
- Occupational health: the science of protecting the health of working individuals through control of the work environment; the ability of a working individual to function at an optimum level of productivity in a work environment while enjoying good health and longevity
    - Note: if an occupational environment, event, or exposure either causes or contributes to the resulting condition, or significantly aggravates a pre-existing condition, it is classifiable as work-related by OSHA

- Vulnerability
    - Inherently more sensitive: genetic predisposition, pregnant persons, unborn/young children
    - Increased sensitivity: old age, immunocompromised, previous respiratory disease
    - Exposed to unusually large amounts of air pollutants: workers in certain industries, children, inhabitants of polluted areas
    - Note: Air Quality Index (AQI) → low score = better air quality; high score → worse air quality
- Hierarchy of Hazard Control
    - Most effective → least effective
        - Elimination: physically remove the hazard → most effective
        - Substitution: replace the hazard
        - Engineering: isolate people from the hazard
        - Administrative: change the way people work
        - PPE: protect the individual worker → least effective

Bioethics

Introduction to Bioethics

- Bioethics Principles
    - Respect for Persons: show fidelity to patient; show respect for patient dignity and autonomy
        - Fidelity: be loyal to the patient and keep your promises (social contract of medicine, confidentiality)
        - Autonomy: individual capacity to choose (non-interference in the life plans of others); based on capacity to determine own self-interest, unless those actions impinge on others' rights
            - Limits to patient autonomy
                - Harm principle: the only purpose for which power can be rightfully exercised over any member of a civilized community against their will is to prevent harm to others
                - Diminished autonomy/capacity
                - Paternalism: the view that the 'doctor knows best'; sometimes justifiable when used to withhold information, but is not a justifiable exemption from informed consent
                - Public health protections: reportable conditions
        - Veracity: be truthful to the patient
        - Avoidance of killing: *primum non nocere* → differs from nonmaleficence (i.e., withholding vs. withdrawing treatment)
        - Note: non-Hippocratic codes of ethics emphasize patient benefits and their duties, rights, and social welfare; implies that individuals have justifiable moral or legal claims to entitlements or liberties (i.e., human rights)
            - Positive rights: the right to a good or service; requires actions
                - Be truthful (veracity), be fair in providing care (justice), explore patient preferences
            - Negative rights: the right to be left alone; does not require actions
                - Protect privacy, avoid killing, avoid harming
    - Beneficence: act in patient's interest
    - Nonmaleficence: avoid harm to patient
    - Utility: balance benefit and harm (beneficence + nonmaleficence); physicians should act to maximize the net benefit to the patient and must consider probable outcomes, probabilities, severity of harm, value of benefit from the patient's perspective, and reasonable alternatives to the action
    - Justice: be fair in distribution, procedures, respect, and equity
        - Procedural justice: fairness of law and policy
        - Distributive justice: criteria for distributing goods, harms, rights, etc.
        - Social justice: improve conditions for those least well-off (solidarity)

        - Justice upholds public trust in medicine; prioritize patient and community health over personal interests - treats equals equally and unequals unequally (Aristotle)
- Bioethics Theories
    - Duty-based (deontological): act in ways that fulfill your moral obligations
    - Outcome-based (consequential): act in ways that maximize outcomes
    - Virtue-based: act from virtuous character and intention
- General Principles
    - Universal Declaration of Human Rights (UDHR)
        - Everyone has the right to a standard of living adequate for the health and well-being of themselves and their family (i.e., food, clothing, housing, medical care, social services, and security in the event of negative impacts)
        - Motherhood and childhood are entitled to special care and assistance
    - Universal Declaration on Bioethics and Human Rights (UDBHR)
        - Recognizes the interrelation between ethics and human rights in the specific field of bioethics
        - Describes human rights relative to health care, non-discrimination, government responsibility, future generations, protecting the environment/biosphere/biodiversity
    - Goals of Medicine
        - Prevention of disease and injury, and the promotion and maintenance of health
        - Relief of pain and suffering caused by maladies
        - Care and cure of those with a malady, and the care of those who cannot be cured
        - Avoidance of premature death, and the pursuit of a peaceful death
    - Physician Expertise and Judgment
        - Hippocratic: subjective judgment of what will benefit the patient
        - Non-Hippocratic
            - Modern medicine: objective judgment based on scientific evidence
            - Post-modern (contemporary) medicine: as objective as possible and prioritizes patient preferences when plausible
        - Patients are experts in their overall well-being while physicians are experts in biomedicine (implications → humility, communication, competencies, respect for persons)
- Cognitive Biases and Fallacies
    - Availability heuristic: tend to overestimate the likelihood of events that have occurred recently
    - Bandwagon effect: tend to do/believe things because many other people in your proximity do/believe it
    - Base-rate fallacy: tend to ignore some information while focusing on other information
    - Belief bias: tend to evaluate the strength of an argument based on past experience or knowledge
    - Confirmation bias: tend to search for, interpret, focus on, and remember information that corresponds with one's own beliefs
    - Conjunction fallacy: tend to assume a specific condition is more probable than a general one
    - Gambler's fallacy: tend to believe that future probabilities are altered by past events
    - Hindsight bias: tend to see past events as more predictable than they actually were
    - Fundamental attribution error: tend to emphasize personality-based explanations in others

<u>Research</u>
- Goal of Research: truth, understand disease mechanisms/treatments/prevention, help populations and future patients
- Goal of Clinical Practice: application of truth to patient care (improve patient care and outcomes)
- Conflict of Priorities: deals with the questions of which goals to pursue and whose interests are served
- Responsible Conduct of Research
    - Use established professional norms and ethical principles to conduct research
    - Critical for integrity of findings and to uphold public trust
    - Must be taught to future researchers through a commitment to provide training and make incremental impacts across generations
    - Minimize conflicts of interest
- Research Ethics
    - Ethical Requirements of Studies
        - Provide accurate and meaningful data (scientific validity)
        - Utilize fair subject selection
        - Maintain a favorable benefit:harm ratio for participants
        - Uphold respect for persons
        - Obtain each element of informed consent
        - Ensure voluntariness of participants

        - Undergo ethical review and receive approval by an Institutional Review Board (IRB) before starting (social value, method)
        - Clinical equipoise (for clinical trials)
            - Ethical, reliable, and effective; can and should be done during emergencies
            - Recruiting patients to randomized controlled trials (RTC's) offers better outcomes than giving unproven treatments
    - Basic Ethical Principles
        - Respect for Persons
        - Beneficence
        - Justice
    - Applications of Ethical Principles
        - Informed Consent
        - Assessment of Risk and Benefits
        - Selection of Subjects
    - U.S. Public Health Service Syphilis Study
        - A philanthropic activity that evolved into a research study where the method used was not a clinical trial → observational; aims and methods shifted in three historical phases
            - Participants were routinely deceived about the aims of the study and were withheld treatment even after it became the standard treatment for syphilis; led to the drafting of the Belmont Report → enacted legal mechanisms to protect research participants from harm
    - U.S. Common Rule
        - Created by the U.S. Department of Health and Human Services; for all participating departments and agencies, the Common Rule outlines the basic provisions for IRBs, informed consent, and Assurances of Compliance. Human subject research conducted or supported by each federal department/agency is governed by the regulations of that department/agency
        - The main elements of the Common Rule includes:
            - Requirements for ensuring compliance by research institutions
            - Requirements for obtaining, waiving, and documenting informed consent
            - Requirements for IRB membership, function, operations, review of research, and record keeping
        - The Common Rule includes additional protections pertaining to certain vulnerable research subjects:
            - Subpart B provides additional protections for pregnant women, in vitro fertilization, and fetuses
            - Subpart C contains additional protections for prisoners
            - Subpart D does the same for children

<u>Informed Consent</u>

- General Overview
    - Has 5 elements that all must be met to support respect for persons
        - Capacity, disclosure, understanding, voluntariness, authorization
    - Required by law in much of the world in terms of diagnosis, treatment, and research
    - Patients have the right to refuse treatment (capacity, disclosure)
    - Physicians have a duty to disclose all risks and present the medical facts accurately and to sensitively and respectfully disclose all relevant medical information to patients
    - Informed consent is a right, not a duty, and can be waived (autonomy)
        - May justify withholding information that poses a serious psychological threat; must be based on danger and/or patient incompetence and not on beneficence → does not justify withholding bad news
- Medical Malpractice
    - Negligence (often unintentional), existing duty of care
    - Breach of care (failed to act according to accepted standard of care for their specialty and jurisdiction)
        - Breach of standard of care: caused the patient's injury and patient suffered damages because of the injury
    - Note: a physician can perform the correct procedure perfectly, but can still be considered malpractice if appropriate disclosures are not made to the patient, or if inappropriate disclosures are made to others who are not the patient
        - Common inpatient reasons: surgical errors accounted for about 34% of claims
        - Common outpatient reasons: diagnostic errors accounted for about 46% of claims

        - Note: increased age of physicians is correlated with increased risk of malpractice suits
        - Note: most malpractice cases are resolved before a trial and favor the patient, and in those that go to trial, the majority of them favor the physicians/hospitals
    - Litigation
        - Duty: patient must prove that the physician owed a legal duty of care
        - Breach of duty: patient must prove that the physician failed to abide by the standard of care
            - Had disclosure been made, the patient would not have consented
            - Had disclosure been made, a reasonable person in the patient's circumstances would not have consented
            - The materialized risk must have been caused by the intervention
        - Causation: physician's breach of duty caused the patient harm; but for the physician's actions, would this injury have occurred? (i.e., event is the result of the occurrence of the preceding event)
            - ≥ 50% chance that the harm came from physician negligence → 100% damages
            - < 50% chance that the harm came from physician negligence → 0% damages
        - Damages: physician's breach of the standard of care resulted in some kind of harm (i.e., physical, psychological, financial, etc.)
    - Good Samaritan Laws
        - Protects volunteers from civil liability for acts or omissions that cause injury to another while providing assistance in an emergency; provides immunity from civil damages for personal injuries that result from ordinary negligence → no protection against gross negligence (i.e., willful, malicious)
        - Requirements
            - Outside medical setting (i.e., accidents, natural disasters, etc.)
            - No pre-existing duty to provide care
            - No expectation of remuneration of any sort
            - Recipient does not object
- Medical Battery
    - Physician intentionally touches or treats a patient against the patient's will; physician acted with intent
    - Patient does not agree to the harmful or offensive action
    - Note: no need to prove injury or negligence
- Disclosure
    - Information that must be disclosed
        - Alternative risks (including no action or treatment)
        - Inherent risks associated with each alternative (probability and severity)
        - Who will provide treatment and their roles
        - Physician experience
- Patient Comprehension
    - Reasonable Person Standard: obligation to ensure the patient understands and is able to use information to weigh their treatment options
    - Shows respect for patient autonomy by treating in accordance with patient priorities and values (consent is continually re-evaluated and may change)
    - Reasonable Patient
        - Disclose any major risks: includes rare occurrences, death, permanent damage to major organs or mobility
        - Disclose all common risks: includes those that are not major and may be reversible (i.e., hair loss, nausea, change in taste sensation, insomnia, fatigue, etc.)
    - Reasonable Physician
        - Disclose only risks that are widely agreed among physicians and the medical profession → the professional standard
            - Note: if the custom is to not disclose, then there is no duty to disclose
    - Competence: legal determination by a court that applies to all decisions
    - Insanity: legal determination by a court that applies to criminal responsibility
    - Capacity: clinical determination that is decision-specific and has potential legal consequences
        - Ability to understand and make a decision, and the ability to communicate that decision; this can fluctuate over time
        - Note: all patients are presumed to have capacity until the presumption is revoked (i.e., dementia, minors, mental disability)
            - Minors: incompetent if < 18 years old but are presumed competent if they are emancipated, if they are seeking certain types of treatments (i.e., contraception,

STDs, mental health, alcohol/substance abuse, etc.), or if they are a mature minor (legal decision; physician opinion)
            - Note: parents are shared-decision makers (SDM) for minors (consent of 1 parent is sufficient); parent must act in the child's best interest and cannot refuse life-saving treatment unless there is a low chance of effectiveness and heavy burdens
    - Detaining Patients
        - Detained patients lose the right to leave but have other rights intact → informed consent, refusal of treatment; can be detained up to 48 hours pending a court hearing
        - Justified when a patient has an infectious disease of great civil peril (i.e., Ebola, Tuberculosis, Measles, etc.) and when a patient has mental health issues (i.e., poses a danger to self or others)

<u>Confidentiality and Malpractice</u>
- Confidentiality
    - Duty of Confidentiality: patient's right to confidentiality arises when a physician-patient relationship is formed
        - Exceptions: gunshot/knife wounds, abuse or neglect, communicable diseases, neurologic impairment (affects driving), patient poses danger to self or others
    - Privacy: a human right possessed by every person
    - Privilege: applies only in the context of court proceedings

<u>Liability & Treatment Relationships</u>
- History of Treatment Relationship
    - Duty is based on consent and contract
    - Treatment relationships are specific to each episode of illness → prior treatment history is irrelevant
    - The act of agreeing to give or actually giving advice to a patient, or a person on the patient's behalf, in person or by telephone establishes a treatment relationship
    - Once a relationship is formed, physicians must uphold informed consent, non-abandonment, confidentiality, and standard of care
- Duty to Treat
    - Starting point: no duty to treat
    - Providers may refuse treatment for any/no reason
    - Duty to treat is created by the physician's own voluntary consent
    - If there is no treatment relationship, and no physician consent, there is no duty to treat
    - Limits to refusal
        - Discrimination by race, religion, disability, national origin, gender, etc.
        - You have already agreed (i.e., managed care organization contract → BCBS network)
        - Another type of prior agreement (i.e., on-call, ER patients)
    - Contraindications to treatment
        - When common treatments serve no purpose (i.e., blind patient and cataracts surgery)
        - When a patient may react badly (i.e., invasive procedure on psychotic patient)
        - Involves those who cannot make their autonomous wishes known
        - Terminal illness
    - Morally expendable treatment
        - Uselessness: treatment does not help the patient
        - Grave burden: even if the treatment serves a useful purpose, it represents a grave burden to the patient or another
        - Proportionality: harms outweigh benefits
- Standard of Care
    - A diagnostic and treatment process that a clinician should follow for a certain type of patient, illness, or clinical circumstance
    - Legal definition: the level at which the average, prudent provider in a given community would practice → how a similarly qualified practitioner would have managed the patient's care under the same/similar circumstances
    - Reasonable physician: how to determine what is reasonable → expert witness
        - No expert witness testimony → no standard of care → no breach of duty → no case
- Exceptions to Duty: no duty of informed consent if any of the following 6 exceptions apply
    - Information is already known to that particular patient, or is commonly known
    - It is an emergency (must meet all four criteria)
        - Urgent, immediate need
        - Patient lacks capacity

    - No opportunity for consent from surrogate
    - No known objection
  - Therapeutic privilege: disclosing risk information would make the patient so upset that they could not make a rational choice or that it would materially affect their medical condition
  - Waiver: the patient does not want to know → defers to the physician
  - Public health: the physician must treat the patient to protect the community (i.e., serious infectious disease)
  - Conscious-based objections: clinicians can sometimes avoid duty for moral/religious reasons
- Termination of Relationship
  - No treatment relationship: you may refuse to treat for any reason (unless discrimination or prior agreement to treat)
  - Existing relationship: you must continue to treat until relationship terminates via one of four valid ways; otherwise → tortuous abandonment (purposeful, deliberate decision for a non-medical reason)
    - Mutual consent
    - Patient dismisses physician
    - Treatment is no longer needed
    - Firing a patient: noncompliance, failure to pay, verbal abuse/threats, drug seeking, failure to keep appointments, violating policies, lack skills for adequate treatment, lacks resources
      - Permitted with sufficient notice: amount of time required for patient to get another physician → use "certified mail"

<u>Empathy, Morality, and Burnout</u>

- General Overview
  - Too much empathy → moral distress → burnout
  - Burnout erodes empathy and professionalism
    - Highest levels of burnout are seen among emergency medicine practitioners
    - Highest levels of work satisfaction are seen among practitioners in the preventative, occupational, and environmental medicine fields
    - Lowest levels of work satisfaction are seen among general surgery practitioners
  - Medicine is demanding → moral maturity, self-care, compassion, and self-compassion are essential
  - Empathy: feeling the same emotions as the other person
  - Sympathy: feeling sorrow or concern for the other person
  - Compassion: feeling care and warmth for the other person
- Empathy
  - Requires a capacity for down-regulation
  - Empathy is a gateway to compassion, and compassion is a gateway to altruistic behavior
    - Altruism: unselfish concern for others → can be motivated by fear
  - Mature empathy: reaffirming how the other person is feeling and being actively aware and attentive to their emotions/situation
  - Immature empathy: sharing similar experiences, unsolicited "fix-it" advice, pity, humor as a distraction, poorly-placed reassurance
- Morality
  - Stages of Development
    - Avoid punishment
    - Reciprocity and payoff
    - Interpersonal conformity
    - Respect for authority
    - Social contract and individual rights
    - Morality of care and morality of justice
  - Moral Culture of Medicine
    - Respect for persons
    - Compassionate caring (beneficence and nonmaleficence)
    - Competent caring
    - Compassion, altruism, empathy
    - Social justice
  - Moral Distress: the emotional state that arises from a situation when a person feels that the ethically correct action to take is different from policies or procedures, or from what the person is currently tasked with doing (i.e., the anguish experienced when one's integrity is compromised; feeling compromised) → often defines critical incidents in professionalism

- Integrity: acting in accordance with one's own beliefs and values
- Moral distress is an occupational hazard in health care professions and can lead to health problems, workplace bullying, employee absenteeism, and other negative consequences

- Burnout
    - Causes: excessive workload, inefficient environment, inadequate support, loss of autonomy/flexibility, problems with work-life balance, loss of meaning in work
    - Effects: medical errors, depression and other mental health problems, poorer patient care
    - Solutions: maintaining passion for one's work, improving emotional intelligence, proper exercise/sleep/nutrition, supportive relationships (especially professional relationships), ongoing professional development, hobbies outside of medicine, time away from work, professional help
        - Compassion training: meet suffering directly and commit to being moved by it enough to do something helpful (i.e., mindfulness)

<u>Start of Life</u>

- Terminology
    - Necessary condition: something which must be present in order for another thing to be possible (i.e., eyes open is necessary to watch TV)
    - Sufficient condition: something which, if present, guarantees that another thing will occur (i.e., drinking a fifth of alcohol is sufficient to become drunk)
    - Prevention view: preventing the birth of children with severe genetic disorders
    - Dominant view: achieving societal goals in reproductive choices (i.e., cost to society, distributive justice; must uphold reproductive autonomy)
    - Gradualism: the idea of increased moral status as the fetus continues developing
- Considerations in Decision Making
    - The action itself must be either morally good, or at least morally neutral
    - The bad consequences must not be intended
    - The good consequences cannot be the direct causal result of the bad consequences
    - The good consequences must be proportionate to the bad consequences
- Contraception
    - Kantian view: reproduction is not the sole purpose of intercourse, the purpose of marriage is not procreation → teaches categorical imperative (i.e., not to use people as a means to an end)
        - Contraception might increase promiscuity; does not address the emotional side of relationships (i.e., private reasons for using contraception)
    - Utilitarianism (Singer) view: reasons for having sex is pleasure, it provides the greatest happiness for the greatest number of people; Singer argued for using contraception to stop population growth
        - Promote the greatest happiness for greatest number of people, but argues that contraception leading to sexual freedom is not necessarily the highest good as the consequences cannot always be predicted
    - Christianity: contraception is a sin (Pope John Paul II), abstinence is preferred, condom use is acceptable for disease prevention (Pope Benedict), issue is not important for the Church to focus on (Pope Francis)
- Abortion: USMLE stance → termination of pregnancy is wrong

<u>End-of-Life Ethical and Legal Issues</u>

- Terminology
    - Physician-assisted dying (passive): patient refusing treatment or intervention, physician sanctions refusal (i.e., chemotherapy, CPR, artificial ventilation, clinically-assisted nutrition and hydration, antibiotics, voluntary stopping eating and drinking)
    - Physician-assisted dying (active): intent must be to relieve suffering and not to end life (i.e., high-dose opioids → double effect; palliative sedation to unconsciousness)
    - Physician-assisted suicide: physician provides the means for death (most often with a prescription); the patient, not the physician, will ultimately administer the lethal medication
    - Euthanasia: the painless killing of a patient suffering from an incurable and painful disease or who is in an irreversible coma; physician prescribes and administers the method of death
    - Palliative Care: interdisciplinary care focusing on improving quality of life for patients with any serious illness and for their families
    - Hospice Care: palliative care that is delivered under the Medicare hospice benefit (6-month prognosis)
- Quality of Life
    - Control 5 key symptoms: pain, nausea/vomiting, agitation, dyspnea, retained respiratory secretions (RTS)
- Pain Management

- Algorithm
    - No pain (score: 0)
        - Suggested route & type of analgesic: none
        - Initial assessment: within 20 minutes of arrival
        - Re-evaluation: within 60 minutes of initial assessment
    - Mild pain (score: 1-3)
        - Suggested route & type of analgesic: oral analgesic → aspirin, acetaminophen, NSAIDS
        - Initial assessment: within 20 minutes of arrival
        - Re-evaluation: within 60 minutes of analgesia
    - Moderate pain (score: 4-6)
        - Suggested route & type of analgesic: oral analgesia → aspirin, acetaminophen, opioids (codeine, oxycodone, tramadol)
        - Initial assessment: within 20 minutes of arrival
        - Re-evaluation: within 60 minutes of analgesia
    - Severe pain (score: 7-10)
        - Suggested route & type of analgesic: IV opiates or PR NSAID → morphine, methadone, fentanyl, non-opioid analgesics
        - Initial assessment: within 20 minutes of arrival
        - Re-evaluation: within 30 minutes of analgesia
- Analgesic Ladder
    - Pain → step 1 (+/- nonopioid, +/- adjuvant)
    - Pain persisting or increasing → step 2 (opioid for mild to moderate pain, +/- nonopioid, +/- adjuvant)
    - Pain persisting or increasing → step 3 (opioid for moderate to severe pain, +/- nonopioid, +/- adjuvant
    - Freedom from pain
- Oligoanalgesia: the failure to provide analgesia to patients with acute pain

- Surrogate Decision-Makers (SDM)
    - Appropriate entities
        - Patient's guardian, patient's spouse, patient's adult children, either parent of the patient, any adult siblings of the patient, any grandchildren of the patient, an adult relative (must have showed special care and close contact and is familiar with the patient's activities and their health, religious, and moral beliefs), a close friend of the patient, the patient's guardian of estate
    - Inappropriate entities
        - A patient's healthcare provider, an employee of a patient's healthcare provider (unless the patient is related to that employee), the owner/operator/administrator of a healthcare facility, an employee of an owner/operator/administrator of a healthcare facility (unless the patient is related to that employee)
    - Decision-making standards
        - Expressed Interest Standard: refers all healthcare decisions to the explicit instructions that the patient provided prior to losing their decision-making capacity; preferred situation
            - Note: requires a previously competent patient and requires an advance directive/prior statement addressing the situation at hand
        - Substituted Judgment: refers to the patient's values, beliefs, character traits, and past decisions in order to make an educated guess with regard to what the patient would have decided in this particular instance; advance directive must either not exist or must not cover the situation at hand (no advance directive is the ideal standard in this instance)
            - Note: requires a previously competent patient, the surrogate must know the patient well enough to accurately estimate their wishes
        - Best Interests Standard: used when a surrogate has no indication of what the patient's values are or what they would have wanted; used for children, never-competent adults, and patients who cannot be identified (least preferred standard since the surrogate uses their own values to interpret what is in the patient's best interests → places heavy burden on the surrogate)
            - Never-Competent Patients
                - Agents: surrogate decision-makers that are formally appointed by the patient (have the same level of authority to make medical decisions as patients)
                - Representatives: surrogate decision-makers that are informally appointed when no agents are available (have a limited scope of

decision-making authority and cannot withhold or withdraw life-sustaining medical treatment without first petitioning the court unless the patient is in an end-stage condition or is permanently unconscious - even if the representative is a court-appointed guardian)
            - If a representative chooses to petition the court, they must show by clear and convincing evidence that death is in the patient's best interests (i.e., intense suffering)
        - Note: difficult to attempt to define the proper course of action → should often involve second or third opinions and possibly an ethics consultation

<u>Physician Impairment</u>

- Terminology
    - Physician impairment: a physical, mental, or substance-abuse related disorder that interferes with a physician's ability to undertake professional activities competently and safely
    - Note: the diagnosis of an illness does not equate to impairment → impairment is a functional classification; individuals with an illness may or may not have evidence of impairment
- General Notes
    - Most abused substance: alcohol (39.5%)
    - Least abused substance: amphetamines (1.8%)
    - Emergency medicine physicians: abuse cocaine and marijuana most
    - Anesthesiologists: abuse opioids the most
    - Psychiatry: abuse benzodiazepines the most (x3 higher than other groups)
    - Female doctors: higher rates of alcohol abuse than the public
    - Female surgeons: higher rates of alcohol abuse than other female physicians
    - 8-12% of physicians will experience a substance-related problem in their lifetimes
    - Substance abuse is the most common reason for disciplinary action by state boards
    - Substance abuse frequently pre-dates entry into the profession
    - Physicians disciplined by their regulatory boards were x3 as likely to have demonstrated unprofessional behavior in medical school → largest number related to alcohol and drugs
    - Female physicians do not appear to be referred for treatment as frequently as male physicians
    - 81% of physicians with substance-use problems successfully complete treatment and return to practice under monitoring
- Reporting Policy
    - AMA: physicians have an ethical obligation to report impaired, incompetent, and unethical colleagues
    - How to intervene: avoid confronting the physician alone, express positive regard for their abilities, describe specific observable behaviors, avoid accusation or blame, avoid arguing or negotiating, have a specific plan of action, clearly indicate the consequences, insist upon immediate action, provide for safe transition and transportation to the next step in the plan

<u>Substance-Use Disorder</u>

- General Notes
    - All forms of drug dependence and disease burden were highest in men between 20-29 years old
    - Global and US deaths highest due to tobacco
    - JUUL vaping use dropped significantly among older grade-levels YOY
    - Marijuana vaping dropped significantly among 10th graders YOY
- Substance-Use Disorder: maladaptive substance use that disrupts or impairs functioning at work, school, home, or leisure
    - Diagnosis: individual must display ≥ 2 of 11 symptoms within 12 months
        - Mild: 2-3 symptoms
        - Moderate: 4-5 symptoms
        - Severe: ≥ 6 symptoms
    - Symptoms
        - Consuming more of a substance over a longer period of time than initially intended
        - Worrying about stopping or consistently failed efforts to stop
        - Spending significant time obtaining, using, or recovering from the effects of a substance
        - Use of the substance results in failure to fulfill major role obligations
        - Craving the substance
        - Continuing the use of a substance despite the health problems caused/worsened by it
        - Continued substance use despite it having negative effects on other relationships
        - Repeated use in a dangerous situation (i.e., heavy machinery)
        - Giving up or reducing social, occupational, or recreational activities due to substance use

        - Building up a tolerance to the substance
        - Experiencing withdrawal symptoms after stopping use
    - Patterns
        - Experimental: may cease, intensify, or turn into other types of use
        - Opportunistic: may intensify or become compulsive
        - Social-recreational: may remain as such, intensify, or become compulsive
        - Compulsive: obsessive, uncontrollable habit that is maladaptive
        - Intensified: often a quality of the other patterns
- Substance intoxication: develops soon after ingestion of a drug and reverses once the drug is metabolized and excreted
- Tolerance: a state of need for increased amounts of the substance to achieve a desired effect
- Withdrawal: the physical and psychological signs and symptoms that occur when a person that is dependent on a substance stops using it for a period of time
- Prevention
    - Theories
        - Transtheoretical Model (Stages of Change Theory): focuses on the decision-making of the individual and is a model of intentional change; operates on the assumption that people do not change behaviors quickly and decisively but rather change occurs continuously through a cyclical process
            - Six stages: precontemplation → contemplation → preparation → action → maintenance → termination
            - Note: ignores social context in which change occurs
        - Health Belief Model: suggests that a person's belief in a personal threat of an illness or disease, together with a person's belief in the effectiveness of the recommended health behavior or action, will predict the likelihood that the person will adopt the behavior
            - Six constructs: perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cue to action, self-efficacy
        - Community Organization Theory
    - Continuum
        - Universal programs: reach the general population such as all students in a school or all parents in a community
        - Selective programs: target groups such as children of substance abuse or by virtue of their membership in a particular segment of the population and have an above-average risk of developing substance-use issues
        - Indicated programs: for those whose actions put them at high risk for substance-use issues
- Screening
    - Screening, Brief Intervention, and Referral to Treatment (SBIRT)
        - (S) Screening quickly assesses the severity of substance use and identifies the appropriate level of treatment
        - (BI) Brief intervention focuses on increasing insight and awareness regarding substance use and motivation toward behavioral change
        - (RT) Referral to treatment provides those identified as needing more extensive treatment with access to specialty care
        - Note: this approach is recommended when addressing substance-use disorders
        - Screening Forms
            - Pre-Screening Form: administered to all adult patients; rules out patients who are at low or no-risk using one pre-screening question for alcohol and one pre-screening question for drugs
            - The Alcohol Use Disorders Identification Test (AUDIT): administered to adult patients who screen positive on the pre-screening question for alcohol use
            - The Drug Abuse Screening Test (DAST): administered to adult patients who screen positive on the pre-screening question for drug use
            - CRAFFT: used to screen for alcohol and drug use in patients under 21 years old
                - (C) Have you ever ridden in a car by someone who was using a substance?
                - (R) Do you ever use a substance to relax, feel better, or fit in?
                - (A) Do you ever use a substance when you are alone?
                - (F) Do you ever forget things you did while using a substance?
                - (F) Do your family/friends ever tell you that you should cut down on your substance use?
                - (T) Have you ever gotten into trouble as a result of your substance use?

- Treatment
    - Hazardous use: brief counseling and ongoing assessment by a physician
    - Substance abuse: brief counseling, intensive ongoing follow-up, re-evaluation
    - Substance dependence: counseling, referral to specialty treatment, pharmacotherapy
    - Goals of treatment
        - Reduce substance use and craving for substance use
        - Improve health, well-being, and social function of the affected individual
        - Prevent future harm by decreasing the risk of complications and relapse
- Harm Reduction: programs which attempt to minimize the consequences of any psychoactive substance misuse, and encompasses a broad range of activities and interventions designed to improve the health and quality of life of individuals and communities